

This is a part of SFQED-Loops script collection
developed for calculating loop and tree-level processes in
Strong-Field Quantum Electrodynamics.
The scripts are available on <https://github.com/ArsenyMironov/SFQED-Loops>

If you use this script in your research, please, consider
citing our papers:

- A. A. Mironov, S. Meuren, and A. M. Fedotov, PRD 102, 053005 (2020); arXiv:2003.06909
<https://doi.org/10.1103/PhysRevD.102.053005>
- A. A. Mironov, A. M. Fedotov, 105 (3), 033005 (2022); arXiv:2109.00634
<https://doi.org/10.1103/PhysRevD.105.033005>
- Y. V. Selivanov, A. A. Mironov, A. I. Alexeenko, A. M. Fedotov arXiv:2605.12479 (2026)

If you have any questions, please, don't hesitate to contact:
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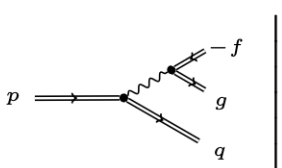
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This notebook is a supplement for our work
[Y. V. Selivanov, A. A. Mironov, A. I. Alexeenko, A. M. Fedotov arXiv:2605.12479 (2026)]

The notebook was prepared by A. A. Mironov and Y. V. Selivanov

The notebook presents a tree-level calculation of the trident process rate (the direct term contribution) initiated by a relativistic electron propagating in a strong constant crossed electromagnetic field, W_{dir}

The corresponding diagram considered in this notebook reads

$$\sum_{\text{spin}} \int d\Pi(q, g, f) \left| \begin{array}{c} \text{diagram} \\ \text{"direct" diagram} \end{array} \right|^2$$


The notebook depends on two additional files
 definitions.nb
 TridentDiracSimplify.nb
 which should be contained in the same directory

```
In[1]:= NotebookEvaluate[NotebookOpen[NotebookDirectory[] <> "definitions.nb"]]
```

FeynCalc 10.1.0 (stable version). For help, use the
 online documentation, visit the forum and have a look at the supplied
 examples. The PDF-version of the manual can be downloaded [here](#).

If you use FeynCalc in your research, please
 evaluate FeynCalcHowToCite[] to learn how to cite this software.

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 of our work is crucial to ensure the future development of this package!*

Trident process ($e > 0$)

$$S_{fi}^{\text{trident, full}} = S_{fi}^{\text{dir}} + S_{fi}^{\text{ex}},$$

where S_{fi}^{dir} is the direct term and S_{fi}^{ex}
 is the exchange term. We will consider only S_{fi}^{dir}

$$\text{Note that } \int d\Pi \left| S_{fi}^{\text{dir}} \right|^2 = \int d\Pi \left| S_{fi}^{\text{ex}} \right|^2,$$

where $\int d\Pi$ is the integral over all the outgoing particle momenta

The direct term matrix element is defined by*

$$S_{fi}^{\text{dir}} = (ie)^2 \int d^4x_1 d^4x_2 \bar{\psi}_q(x_1) \gamma^\mu \psi_p(x_1) D_{\mu\nu}^c(x_1 - x_2) \bar{\psi}_f(x_2) \gamma^\nu \psi_g^+(x_2);$$

x_1^μ, x_2^μ – position of the left and right vertices of the diagram;

p^μ, q^μ – initial and final electron momenta;

$S^c, D_{\mu\nu}^c$ – electron and photon casual propagators;

Superscript + stands for Hermitian conjugation

The Ritus E_p – function

$$E_p(x_1) = \left[1 - \frac{e(\gamma k)(\gamma a)}{2(kp)} (kx_2) \right]$$

$$\text{Exp}\left[-i(\mathbf{p} \cdot \mathbf{x}_1) + i \frac{e(a \cdot \mathbf{p})}{2(k \cdot \mathbf{p})} (k \cdot \mathbf{x}_1)^2 + i \frac{e^2 a^2}{6(k \cdot \mathbf{p})} (k \cdot \mathbf{x}_1)^3\right];$$

$$\bar{E}_q(x_2) = \gamma^0 E_q(x_2) \gamma^0;$$

Volkov states

$$\psi_{p\lambda}(x_1) = E_p(x_1) u_{p\lambda}$$

$$\bar{\psi}_{p\lambda}(x_1) = \bar{u}_{p\lambda} \bar{E}_p(x_1)$$

$$\psi_{p\lambda}^+(x_1) = E_{-p}(x_1) v_{p\lambda}$$

$$\bar{u}_{p\lambda} = u_{p\lambda} \gamma^0$$

Photon propagator

$$D_{\mu\nu}^c(x) = \int \frac{d^4 l}{(2\pi)^4} e^{ilx} \frac{-i}{l^2 + i0} g_{\mu\nu}$$

* Note some differences in notations used in this

Notebook and in section 6.1 of [arXiv 2605.12479]. Here, $e > 0$ denotes the absolute value of the electron charge, and momenta f and g are interchanged. While this slightly changes some intermediate steps, this does not affect the final result.

Main

Preliminary definitions

We assume that

$\mathbf{F}$ is the crossed field tensor,

$\mathbf{k}$ is the 'wave' 4 - vector, normalised such that $k^0 = m$,

$\mathbf{a}$ is the transverse (space - like) constant 4 - vector,

χp denotes the χ parameter of a particle with 4 - momentum p ,

All objects such as vectors and tensors introduced via commands from `definitions.nb` are assumed to be D - dimensional. However, since there is no regularization involved in this calculation, we will set $D \rightarrow 4$ along the Notebook

Command **NewMomentum** is defined in `definitions.nb`. It introduces a set of 4 - vectors and scalars induced by a 4 - momenta and tensor F , see `definitions.nb` for more details

```

In[2]:= NewMomentum["p"] (*Initial electron*)
NewMomentum["q"] (*Scattered electron*)
NewMomentum["g"] (*New positron, created by a photon*)
NewMomentum["f"] (*New electron, created by a photon*)
NewMomentum["l"] (*Intermediate photon*)
NewMomentum["l1"] (*Also intermediate photon,
l1 is used when mod-sqaring the amplitude*)
NewMomentum["l2"] (*Same as above*)
NewCoordinate["x1"] (*Coordinate at which emission happens*)
NewCoordinate["x2"] (*Coordinate at which pair creation happens*)
ScalarProduct[p, p] = m^2;
ScalarProduct[q, q] = m^2;
ScalarProduct[f, f] = m^2;
ScalarProduct[g, g] = m^2;

{p^α, p^2, k · p, Fp^α, FFp^α, FDp^α, a · p, 0, 0, 0, -a^2 (k · p), 0, 0, - $\frac{m^6 \chi p^2}{e^2}$ , - $\frac{m^6 \chi p^2}{e^2}$ ,  $\frac{m^6 \chi p^2}{e^2}$ , 0, 0, 0, 0, 0, 0}

{q^α, q^2, k · q, Fq^α, FFq^α, FDq^α, a · q, 0, 0, 0, -a^2 (k · q), 0, 0, - $\frac{m^6 \chi q^2}{e^2}$ , - $\frac{m^6 \chi q^2}{e^2}$ ,  $\frac{m^6 \chi q^2}{e^2}$ , 0, 0, 0, 0, 0, 0}

{g^α, g^2, g · k, Fg^α, FFg^α, FDg^α, a · g, 0, 0, 0, -a^2 (g · k), 0, 0, - $\frac{m^6 \chi g^2}{e^2}$ , - $\frac{m^6 \chi g^2}{e^2}$ ,  $\frac{m^6 \chi g^2}{e^2}$ , 0, 0, 0, 0, 0, 0}

{f^α, f^2, f · k, Ff^α, FFf^α, FDf^α, a · f, 0, 0, 0, -a^2 (f · k), 0, 0, - $\frac{m^6 \chi f^2}{e^2}$ , - $\frac{m^6 \chi f^2}{e^2}$ ,  $\frac{m^6 \chi f^2}{e^2}$ , 0, 0, 0, 0, 0, 0}

{l^α, l^2, k · l, Fl^α, FFl^α, FDll^α, a · l, 0, 0, 0, -a^2 (k · l), 0, 0, - $\frac{m^6 \chi l^2}{e^2}$ , - $\frac{m^6 \chi l^2}{e^2}$ ,  $\frac{m^6 \chi l^2}{e^2}$ , 0, 0, 0, 0, 0, 0}

{l1^α, l1^2, k · l1, Fl1^α, FFl1^α, FDll1^α, a · l1, 0, 0, 0, -a^2 (k · l1), 0, 0, - $\frac{m^6 \chi l1^2}{e^2}$ , - $\frac{m^6 \chi l1^2}{e^2}$ ,  $\frac{m^6 \chi l1^2}{e^2}$ , 0, 0, 0, 0, 0, 0}

{l2^α, l2^2, k · l2, Fl2^α, FFl2^α, FDll2^α, a · l2, 0, 0, 0, -a^2 (k · l2), 0, 0, - $\frac{m^6 \chi l2^2}{e^2}$ , - $\frac{m^6 \chi l2^2}{e^2}$ ,  $\frac{m^6 \chi l2^2}{e^2}$ , 0, 0, 0, 0, 0, 0}

{x1^α, x1^2, k · x1, a · x1, Fx1^α, FFx1^α, FDx1^α, k · x1, 0, 0, 0,
-a^2 (k · x1), 0, 0, - $\frac{m^2 \xi^2 (k \cdot x1)^2}{e^2}$ , - $\frac{m^2 \xi^2 (k \cdot x1)^2}{e^2}$ ,  $\frac{m^2 \xi^2 (k \cdot x1)^2}{e^2}$ , 0, 0, 0, 0, 0, 0}

{x2^α, x2^2, k · x2, a · x2, Fx2^α, FFx2^α, FDx2^α, k · x2, 0, 0, 0,
-a^2 (k · x2), 0, 0, - $\frac{m^2 \xi^2 (k \cdot x2)^2}{e^2}$ , - $\frac{m^2 \xi^2 (k \cdot x2)^2}{e^2}$ ,  $\frac{m^2 \xi^2 (k \cdot x2)^2}{e^2}$ , 0, 0, 0, 0, 0, 0}

```

Commands **Ep** and **EpC** introduce E_p –

functions in the form of an array {prefactor, phase},
see definitions.nb for details

In[15]:= **Ep**x1 = **Ep**[x1, p]

EqBarx1 = **EpC**[x1, q]

EfBarx2 = **EpC**[x2, f]

Emingx2 = **Ep**[x2, -g]

$$\text{Out[15]} = \left\{ 1 - \frac{e(k \cdot x1)(\gamma \cdot k)(\gamma \cdot a)}{2(k \cdot p)}, \frac{a^2 e^2 (k \cdot x1)^3}{6(k \cdot p)} + \frac{e(a \cdot p)(k \cdot x1)^2}{2(k \cdot p)} - p \cdot x1 \right\}$$

$$\text{Out[16]} = \left\{ 1 - \frac{e(k \cdot x1)(\gamma \cdot a)(\gamma \cdot k)}{2(k \cdot q)}, -\frac{a^2 e^2 (k \cdot x1)^3}{6(k \cdot q)} - \frac{e(a \cdot q)(k \cdot x1)^2}{2(k \cdot q)} + q \cdot x1 \right\}$$

$$\text{Out[17]} = \left\{ 1 - \frac{e(k \cdot x2)(\gamma \cdot k)(\gamma \cdot k)}{2(f \cdot k)}, -\frac{a^2 e^2 (k \cdot x2)^3}{6(f \cdot k)} - \frac{e(a \cdot f)(k \cdot x2)^2}{2(f \cdot k)} + f \cdot x2 \right\}$$

$$\text{Out[18]} = \left\{ \frac{e(k \cdot x2)(\gamma \cdot k)(\gamma \cdot a)}{2(g \cdot k)} + 1, -\frac{a^2 e^2 (k \cdot x2)^3}{6(g \cdot k)} + \frac{e(a \cdot g)(k \cdot x2)^2}{2(g \cdot k)} + g \cdot x2 \right\}$$

The dressed vertex (with an outgoing photon line)

$$\Gamma^\mu(l, q, p) = \int d^4x \bar{E}_q(x) \gamma^\mu E_p(x) e^{i l x};$$

x^μ – position of the vertex;

p^μ, q^μ – initial and final electron momenta;

l^μ, q^μ – momentum of the outgoing photon

$-\infty < s < \infty$ – measure for the momentum exchanged

with the field (may be viewed as the 'photon number')

$$\Gamma^\mu(l, q, p) = \int ds (2\pi)^4 \delta(p + sk - q - l) \tilde{\Gamma}^\mu(s, q, p)$$

$$\tilde{\Gamma}^\mu(s, q, p) = \gamma^\mu A_0(s, q, p) -$$

$$\frac{e}{2} \left(\frac{(\gamma a)(\gamma k) \gamma^\mu}{kq} + \frac{\gamma^\mu (\gamma k)(\gamma a)}{kp} \right) A_1(s, q, p) - \frac{e^2 a^2 k^\mu (\gamma k)}{2 kq kp} A_2(s, q, p)$$

$$A_0(s, q, p) = \int \frac{d\phi}{2\pi} e^{is\phi} e^{-i\left(\frac{\alpha}{2}\phi^2 - \frac{4\beta}{3}\phi^3\right)} = \frac{\text{Exp}\left[i\left(\frac{\alpha s}{8\beta} - \frac{\alpha^3}{192\beta^2}\right)\right]}{(4\beta)^{1/3}} \text{Ai}(y)$$

$$A_1(s, q, p) = -i \frac{dA_0}{ds} = \int \frac{d\phi}{2\pi} e^{is\phi} \phi e^{-i\left(\frac{\alpha}{2}\phi^2 - \frac{4\beta}{3}\phi^3\right)} =$$

$$\frac{\text{Exp}\left[i\left(\frac{\alpha s}{8\beta} - \frac{\alpha^3}{192\beta^2}\right)\right]}{(4\beta)^{1/3}} \left\{ \frac{\alpha}{8\beta} \text{Ai}(y) - i \frac{1}{(4\beta)^{1/3}} \text{Ai}'(y) \right\}$$

$$A_2(s, q, p) = (-i)^2 \frac{d^2 A_0}{ds^2} = \int \frac{d\phi}{2\pi} e^{is\phi} \phi^2 e^{-i\left(\frac{\alpha}{2}\phi^2 - \frac{4\beta}{3}\phi^3\right)} =$$

$$\frac{\text{Exp}\left[i\left(\frac{\alpha s}{8\beta} - \frac{\alpha^3}{192\beta^2}\right)\right]}{(4\beta)^{1/3}} \left\{ \left[2\left(\frac{\alpha}{8\beta}\right)^2 - \frac{s}{4\beta}\right] \text{Ai}(y) - i \frac{\alpha}{(4\beta)^{4/3}} \text{Ai}'(y) \right\}$$

where

$$\alpha = \alpha(q, p) = -e \left(\frac{ap}{kp} - \frac{aq}{kq} \right)$$

$$\beta = \beta(q, p) = \frac{e^2 a^2}{8} \left(\frac{1}{kp} - \frac{1}{kq} \right)$$

$$y = y(s, q, p) = (4\beta(q, p))^{2/3} \left[\frac{s}{4\beta(q, p)} - \left(\frac{\alpha(q, p)}{8\beta(q, p)} \right)^2 \right]$$

The amplitude expressed via $\tilde{\Gamma}^\mu$

$$S_{fi}^{\text{dir}} = (ie)^2 \int d^4x_1 d^4x_2 \bar{\psi}_q(x_1) \gamma^\mu \psi_p(x_1) D_{\mu\nu}^c(x_1 - x_2) \bar{\psi}_f(x_2) \gamma^\nu \psi_g(x_2)$$

$$= (ie)^2 \int d^4x_1 d^4x_2 \frac{d^4l}{(2\pi)^4} \frac{-i}{l^2 + i0} g_{\mu\nu}$$

$$\left(\bar{u}_{q\lambda_2} \bar{E}_q(x_1) \gamma^\mu E_p(x_1) u_{p\lambda_1} e^{ilx_1} \right) \left(\bar{u}_{f\sigma_2} \bar{E}_f(x_2) \gamma^\mu E_g(x_2) v_{g\sigma_1} e^{-ilx_2} \right)$$

$$= ie^2 \int \frac{d^4l}{(2\pi)^4} \frac{g_{\mu\nu}}{l^2 + i0} \left(\bar{u}_{q\lambda_2} \int d^4s_1 (2\pi)^4 \delta(p + s_1 k - q - l) \tilde{\Gamma}^\mu(s_1, q, p) u_{p\lambda_1} \right)$$

$$\left(\bar{u}_{f\sigma_2} \int d^4s_2 (2\pi)^4 \delta(l + s_2 k - f - g) \tilde{\Gamma}^\mu(s_2, g, -f) v_{g\sigma_1} \right)$$

$$= ie^2 \int d^4s_1 d^4s_2 (2\pi)^4 \delta(p + (s_1 + s_2)k - q - f - g)$$

$$\frac{g_{\mu\nu}}{(p + s_1 k - q)^2 + i0} \left(\bar{u}_{q\lambda_2} \tilde{\Gamma}^\mu(s_1, q, p) u_{p\lambda_1} \right) \left(\bar{u}_{f\sigma_2} \tilde{\Gamma}^\mu(s_2, g, -f) v_{g\sigma_1} \right)$$

Let us introduce partial amplitude $m(s_1, s_2)$

(curly M in [arXiv 2605.12479], see Eq. (6.1), note that here $f \leftrightarrow g$)

$$S_{fi}^{\text{dir}} = \int ds_1 ds_2 i m(s_1, s_2) (2\pi)^4 \delta(g + f + q - p - (s_1 + s_2)k)$$

$$i m(s_1, s_2) = ie^2 \left[\bar{u}_{q\lambda_2} \tilde{\Gamma}^\mu(s_1, q, p) u_{p\lambda_1} \right] \frac{g_{\mu\nu}}{l^2 + i0} \left[\bar{u}_{f\sigma_2} \tilde{\Gamma}^\nu(s_2, f, -g) v_{g\sigma_1} \right]$$

$$m(s_1, s_2) = e^2 \frac{1}{l^2 + i0} \left[\bar{u}_{q\lambda_2} \tilde{\Gamma}^\mu(s_1, q, p) u_{p\lambda_1} \right] \left[\bar{u}_{g\sigma_2} \tilde{\Gamma}_\mu(s_2, f, -g) v_{g\sigma_1} \right]$$

$$m^+(t_1, t_2) =$$

$$e^2 \frac{1}{l'^2 - i0} \left[\bar{u}_{p\lambda_1} \left(\tilde{\Gamma}^\mu(t_1, q, p) \right)^+ u_{q\lambda_2} \right] \left[\bar{v}_{g\sigma_1} \left(\tilde{\Gamma}_\mu(t_2, f, -g) \right)^+ u_{f\sigma_2} \right]$$

$$\mathbf{l} = \mathbf{p} + \mathbf{s}_1 \mathbf{k} - \mathbf{q}$$

$$\mathbf{l}' = \mathbf{p} + \mathbf{t}_1 \mathbf{k} - \mathbf{q}$$

$$\sum u_{\mathbf{p}\lambda}^{\alpha} \bar{u}_{\mathbf{p}\lambda}^{\beta} = (\mathbf{p}\gamma + m)^{\alpha\beta}$$

$$\sum v_{\mathbf{p}\lambda}^{\alpha} \bar{v}_{\mathbf{p}\lambda}^{\beta} = (\mathbf{p}\gamma - m)^{\alpha\beta}$$

$$dW = \frac{dw}{VT 2 p^0};$$

$$\frac{1}{2} \sum_{\lambda\sigma} - \text{averaging over the initial electron spin states}$$

and summation over the final particle spin projections

$$\frac{1}{2} - \text{a symmetry factor to account for the two identical final electrons}$$

$$dw = \frac{1}{2} \times \frac{1}{2} \sum_{\lambda\sigma} \frac{d^3 \mathbf{f}}{(2\pi)^3 2 f^0} \frac{d^3 \mathbf{g}}{(2\pi)^3 2 g^0} \frac{d^3 \mathbf{q}}{(2\pi)^3 2 q^0} (|S_{fi}^{dir}|)^2;$$

$$(|S_{fi}^{dir}|)^2 = \frac{VT}{L_{\phi}} \int d\mathbf{s}_1 d\mathbf{s}_2 dt_1 dt_2 M(\mathbf{s}_1, \mathbf{s}_2, \mathbf{t}_1, \mathbf{t}_2)$$

$$(2\pi)^4 \delta(\mathbf{g} + \mathbf{f} + \mathbf{q} - \mathbf{p} - (\mathbf{s}_1 + \mathbf{s}_2) \mathbf{k}) (2\pi) \delta(\mathbf{s}_1 + \mathbf{s}_2 - \mathbf{t}_1 - \mathbf{t}_2)$$

$$M(\mathbf{s}_1, \mathbf{s}_2, \mathbf{t}_1, \mathbf{t}_2) = m(\mathbf{s}_1, \mathbf{s}_2) m^+(\mathbf{t}_1, \mathbf{t}_2)$$

Steps 1-2: Integrations using δ -functions

Step 1 : integrate over $d^3 \mathbf{f}$ using δ - function

$$\int \frac{d^3 \mathbf{f}}{(2\pi)^3 2 f^0} (2\pi)^4 \delta(\mathbf{g} + \mathbf{f} + \mathbf{q} - \mathbf{p} - (\mathbf{s}_1 + \mathbf{s}_2) \mathbf{k}) =$$

$$\frac{2\pi}{2 f^0} \delta(g^0 + f^0 + q^0 - p^0 - (\mathbf{s}_1 + \mathbf{s}_2) k^0)$$

$\delta(\mathbf{g} + \mathbf{f} + \mathbf{q} - \mathbf{p} - (\mathbf{s}_1 + \mathbf{s}_2) \mathbf{k})$ also implies the conservation of :

$$\mathbf{k}\mathbf{g} + \mathbf{k}\mathbf{f} + \mathbf{k}\mathbf{q} - \mathbf{k}\mathbf{p} = 0, \quad \mathbf{g}_{\perp} + \mathbf{f}_{\perp} + \mathbf{q}_{\perp} - \mathbf{p}_{\perp} = 0$$

$$\text{Consider } g^0 + f^0 + q^0 - p^0 - (\mathbf{s}_1 + \mathbf{s}_2) k^0 = 0$$

$$\text{By substituting } f^0 = \sqrt{m^2 + \left(\vec{\mathbf{f}}\right)^2} = \sqrt{m^2 + \left(\vec{\mathbf{p}} + \tilde{\mathbf{s}} \vec{\mathbf{k}} - \vec{\mathbf{g}} - \vec{\mathbf{q}}\right)^2}, \text{ we get}$$

$$\sqrt{m^2 + \left(\vec{\mathbf{p}} + (\mathbf{s}_1 + \mathbf{s}_2) \vec{\mathbf{k}} - \vec{\mathbf{g}} - \vec{\mathbf{q}}\right)^2} = p^0 + (\mathbf{s}_1 + \mathbf{s}_2) k^0 - g^0 - q^0$$

If we square this expression and rearrange terms, we will arrive at

$$s_1 + s_2 = \tilde{s} = \frac{(pq + pg - qg - m^2)}{kp - kq - kg} = \frac{(pq + pg - qg - m^2)}{kf}$$

Hence,

$$\delta (g^0 + f^0 + q^0 - p^0 - (s_1 + s_2) k^0) = \delta (F(s_1 + s_2)) = \frac{\delta (s_1 + s_2 - \tilde{s})}{|F'(\tilde{s})|}$$

$$\text{where } F(s) = p^0 + s k^0 - g^0 - q^0 - \sqrt{m^2 + (\vec{p} + s \vec{k} - \vec{g} - \vec{q})^2}$$

By taking the derivative and substituting

$$\text{back } \sqrt{m^2 + (\vec{p} + \tilde{s} \vec{k} - \vec{g} - \vec{q})^2} = (p^0 + \tilde{s} k^0 - g^0 - q^0) = f^0,$$

it is straightforward to show that

$$F'(\tilde{s}) = \frac{(kf)}{f^0} = \frac{m^2 \chi_f}{\xi f^0}$$

hence

$$\delta (g^0 + f^0 + q^0 - p^0 - \tilde{s} k^0) = \frac{\xi f^0}{m^2 \chi_f} \delta (s_1 + s_2 - \tilde{s})$$

$$\int \frac{d^3 f}{(2\pi)^3 2 f^0} (2\pi)^4 \delta (g + f + q - p - (s_1 + s_2) k) = \frac{\pi \xi}{m^2 \chi_f} \delta (s_1 + s_2 - \tilde{s})$$

Step 2 : intergate over s_2, t_2

$$\int ds_1 ds_2 dt_1 dt_2 (2\pi) \delta$$

$$(s_1 + s_2 - t_1 - t_2) \frac{\pi \xi}{m^2 \chi_f} \delta (s_1 + s_2 - \tilde{s}) M(s_1, s_2, t_1, t_2)$$

$$= \frac{2\pi^2 \xi}{m^2 \chi_f} \int ds_1 dt_1 dt_2 \delta (\tilde{s} - t_1 - t_2) M(s_1, \tilde{s} - s_1, t_1, t_2)$$

$$= \frac{2\pi^2 \xi}{m^2 \chi_f} \int ds dt M(s_1, \tilde{s} - s, t, \tilde{s} - t), \text{ where } s = s_1, t = t_1$$

In the end, we are calculating :

$$dW =$$

$$\frac{1}{2} \times \frac{1}{2} \frac{1}{2 p^0 L_\phi} \frac{2\pi^2 \xi}{(2\pi)^6 2 g^0 2 q^0 m^2 \chi_f} \sum_{\lambda\sigma} d^3 g d^3 q ds dt M(s, \tilde{s} - s, t, \tilde{s} - t)$$

$$M(s, \tilde{s} - s, t, \tilde{s} - t) = m(s, \tilde{s} - s) m^+(t, \tilde{s} - t)$$

$$m(s, \tilde{s} - s) = e^2 \frac{1}{l^2 + i0} [\bar{u}_{q\lambda_2} \tilde{\Gamma}^\mu(s, q, p) u_{p\lambda_1}] [\bar{u}_{g\sigma_2} \tilde{\Gamma}_\mu(\tilde{s} - s, f, -g) v_{g\sigma_1}]$$

$$m^+(t, \tilde{s} - t) =$$

$$e^2 \frac{1}{l'^2 - i0} [\bar{u}_{p\lambda_1} (\tilde{\Gamma}^{\mu 1} (t, q, p))^+ u_{q\lambda_2}] [\bar{v}_{g\sigma_1} (\tilde{\Gamma}_{\mu 1} (\tilde{s} - t, f, -g))^+ u_{f\sigma_2}]$$

$$l = p + s - k - q$$

$$l' = p + t - k - q$$

Step 3: Spin sums

For technical convenience,
we first define the operation for a dummy matrix element mod squared
 $[\bar{u}_{q\lambda_2} \gamma^\mu u_{p\lambda_1}] [\bar{u}_{g\sigma_2} \gamma^\nu v_{g\sigma_1}] g_{\mu\nu} [\bar{u}_{p\lambda_1} \gamma^{\mu 1} u_{q\lambda_2}] [\bar{v}_{g\sigma_1} \gamma^{\nu 1} u_{f\sigma_2}] g_{\mu 1 \nu 1},$
then perform the formal spin summation, and after this we substitute
 $\gamma^\mu \rightarrow \tilde{\Gamma}^\mu (s, q, p)$
 $\gamma^\nu \rightarrow \tilde{\Gamma}^\nu (\tilde{s} - s, f, -g)$
 $\gamma^{\mu 1} \rightarrow (\tilde{\Gamma}^{\mu 1} (t, q, p))^+$
 $\gamma^{\nu 1} \rightarrow (\tilde{\Gamma}^{\nu 1} (\tilde{s} - t, f, -g))^+$

```
In[19]:= DummyExpr = (SpinorUBarD[q, m].GAD[μ].SpinorUD[p, m])
              (SpinorUBarD[f, m].GAD[ν].SpinorVD[g, m])
DummyExprC =
  ComplexConjugate[DummyExpr, FCRenameDummyIndices → False] /. {μ → μ1, ν → ν1}

Out[19]= u(q, m).γμ.u(p, m) u(f, m).γν.v(g, m)

Out[20]= (φ(-g, m)).γν1.(φ(f, m)) (φ(p, m)).γμ1.(φ(q, m))

In[21]:= DummyExpr1 = FermionSpinSum[DummyExpr DummyExprC]

Out[21]= tr((γ.f + m).γν.(γ.g - m).γν1) tr((m + γ.p).γμ1.(m + γ.q).γμ)
```

We define

$$\tilde{\Gamma}^\mu (s, q, p) = \gamma^\mu A_0 (s, q, p) - \frac{e}{2} \left(\frac{(\gamma a) (\gamma k) \gamma^\mu}{kq} + \frac{\gamma^\mu (\gamma k) (\gamma a)}{kp} \right) A_1 (s, q, p) - \frac{e^2 a^2 k^\mu (\gamma k)}{2 kq kp} A_2 (s, q, p)$$

$$(\tilde{\Gamma}^{\mu 1} (t, q, p))^+ = \gamma^\mu B_0 (t, q, p) - \frac{e}{2} \left(\frac{\gamma^\mu (\gamma k) (\gamma a)}{kq} + \frac{(\gamma a) (\gamma k) \gamma^\mu}{kp} \right) B_1 (t, q, p) - \frac{e^2 a^2 k^\mu (\gamma k)}{2 kq kp} B_2 (t, q, p)$$

$$B_n = (A_n)^*$$

It what follows,
our goal is to integrate the expression for the trident rate;

This involves some variable changes
in the integrand and algebraic simplifications;

To track the calculation, we split the expression into variables

- **Expr** - the integrand
- **Coeff** -
the overall coefficient (also dependent on the integration variables)
- **Diffs** - the integration measure to track which integrations are left,
namely

$$W = \int [\text{Diffs}] * \text{Coeff} * \text{Expr}$$

We will be modifying there three variables,
in each step adding a number to the name (DiffsN, CoeffN, ExprN)

Commands **rtilde** and **rtildeC** are introduced
in definitions.nb in accordance with formulas for

$\tilde{\Gamma}^{\mu}(s, q, p)$ and $(\tilde{\Gamma}^{\mu}(t, q, p))^+$, respectively;

chif[p] is also a handy command introduced in definitions.nb,
it returns symbol ' χ_p ' for a corresponding argument 'p'

```
In[22]:= Diffs = ds dt d3q d3g
Coeff = (1 / 2) (1 / 2) (1 / Lφ / 2 / p0) 2 π^2 ξ / ((2 π) ^6 2 g0 2 q0 m^2 chif[f])
e^4 / (l1v2 + iε) / (l2v2 - iε) /. {e^4 → (4 π α) ^2}
Expr = DummyExpr1 /.
{DiracGamma[LorentzIndex[μ, D], D] → rtilde[μ, s, q, p],
DiracGamma[LorentzIndex[ν, D], D] → rtilde[μ, stilde - s, f, -g],
DiracGamma[LorentzIndex[μ1, D], D] → rtildeC[μ1, t, q, p],
DiracGamma[LorentzIndex[ν1, D], D] → rtildeC[μ1, stilde - t, f, -g]}
```

```
Out[22]= d3g d3q ds dt
```

$$\text{Out[23]} = \frac{\alpha^2 \xi}{64 \pi^2 g_0 L \phi m^2 p_0 q_0 \chi f (i\epsilon + l_1^2) (l_2^2 - i\epsilon)}$$

$$\begin{aligned} \text{Out[24]} = & \text{tr} \left((m + \gamma \cdot p) \left(-\frac{a^2 e^2 k^{\mu 1} \gamma \cdot k B_2(t, q, p)}{2 (k \cdot p) (k \cdot q)} - \frac{1}{2} e B_1(t, q, p) \left(\frac{(\gamma \cdot a) \cdot (\gamma \cdot k) \cdot \gamma^{\mu 1}}{k \cdot p} + \frac{\gamma^{\mu 1} \cdot (\gamma \cdot k) \cdot (\gamma \cdot a)}{k \cdot q} \right) + \gamma^{\mu 1} B_0(t, q, p) \right) (m + \gamma \cdot q) \cdot \right. \\ & \left. \left(-\frac{a^2 e^2 k^{\mu} \gamma \cdot k A_2(s, q, p)}{2 (k \cdot p) (k \cdot q)} - \frac{1}{2} e A_1(s, q, p) \left(\frac{\gamma^{\mu} \cdot (\gamma \cdot k) \cdot (\gamma \cdot a)}{k \cdot p} + \frac{(\gamma \cdot a) \cdot (\gamma \cdot k) \cdot \gamma^{\mu}}{k \cdot q} \right) + \gamma^{\mu} A_0(s, q, p) \right) \right) \\ & \text{tr} \left((\gamma \cdot f + m) \left(\frac{a^2 e^2 k^{\mu} \gamma \cdot k A_2(\text{stilde} - s, f, -g)}{2 (f \cdot k) (g \cdot k)} - \frac{1}{2} e A_1(\text{stilde} - s, f, -g) \left(\frac{(\gamma \cdot a) \cdot (\gamma \cdot k) \cdot \gamma^{\mu}}{f \cdot k} - \frac{\gamma^{\mu} \cdot (\gamma \cdot k) \cdot (\gamma \cdot a)}{g \cdot k} \right) + \right. \right. \\ & \left. \left. \gamma^{\mu} A_0(\text{stilde} - s, f, -g) \right) (\gamma \cdot g - m) \left(\frac{a^2 e^2 k^{\mu 1} \gamma \cdot k B_2(\text{stilde} - t, f, -g)}{2 (f \cdot k) (g \cdot k)} - \right. \right. \\ & \left. \left. \frac{1}{2} e B_1(\text{stilde} - t, f, -g) \left(\frac{\gamma^{\mu 1} \cdot (\gamma \cdot k) \cdot (\gamma \cdot a)}{f \cdot k} - \frac{(\gamma \cdot a) \cdot (\gamma \cdot k) \cdot \gamma^{\mu 1}}{g \cdot k} \right) + \gamma^{\mu 1} B_0(\text{stilde} - t, f, -g) \right) \right) \end{aligned}$$

```
In[25]:= Diffs1 = Diffs;
Coeff1 = Coeff;
```

The following step is saved as a pre - evaluated expression stored in TridentDiracSimplify.nb. To perform the actual computation, uncomment commented line. The calculation takes several minutes

```
In[27]:= (*Expr1=DiracSimplify[Expr];*)
NotebookEvaluate[
NotebookOpen[NotebookDirectory[] <> "TridentDiracSimplify.nb"]];
```

Step 4: Rearranging terms in Expr and further simplifications

SortTerms and **SortTermsSimplify** are two lists, which we use as an argument in command Collect to collect terms in Expr

```
In[28]:= SortTerms[X_] := Collect[X, {A0 B0, A0 B1, A0 B2, A1 B0, A1 B1, A1 B2, A2 B0,
A2 B1, A2 B2, C0 D0, C0 D1, C0 C2, C1 D0, C1 D1, C1 C2, C2 D0, C2 D1, C2 C2}]
SortTermsSimplify[X_] :=
Collect[X, {A0 B0, A0 B1, A0 B2, A1 B0, A1 B1, A1 B2, A2 B0, A2 B1, A2 B2,
C0 D0, C0 D1, C0 C2, C1 D0, C1 D1, C1 C2, C2 D0, C2 D1, C2 C2}, Simplify]
```

For brevity, we substitute A_n and B_n functions with symbols :
 $A_n[s, q, p] \rightarrow A_n$

$$B_n[t, q, p] \rightarrow B_n$$

$$A_n[\tilde{s} - s, f, -g] \rightarrow C_n$$

$$B_n[\tilde{s} - t, f, -g] \rightarrow D_n$$

Then terms in the expressions are collected according to the rules introduced in the previous input cell

```
In[30]:= DiffS2 = DiffS1;
Coeff2 = Coeff1;
Expr1 /. {A0[s, q, p] → A0, A1[s, q, p] → A1, A2[s, q, p] → A2,
  B0[t, q, p] → B0, B1[t, q, p] → B1, B2[t, q, p] → B2,
  A0[stilde - s, f, -g] → C0,
  A1[stilde - s, f, -g] → C1, A2[stilde - s, f, -g] → C2,
  B0[stilde - t, f, -g] → D0, B1[stilde - t, f, -g] → D1, B2[stilde - t, f, -g] → D2};
Expr2 = SortTerms[%];
Expr2 /. {D → 4}
```

$$\begin{aligned} \text{Out[34]} = & \frac{16 A_2 B_2 C_0 D_0 \bar{a}^4 (\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{k}) e^4}{(\bar{k} \cdot \bar{p}) (\bar{k} \cdot \bar{q})} + A_1 B_2 \left(C_1 D_0 \left(\frac{8 \bar{a}^4 (\bar{f} \cdot \bar{k}) e^4}{\bar{k} \cdot \bar{p}} - \frac{8 \bar{a}^4 (\bar{g} \cdot \bar{k}) e^4}{\bar{k} \cdot \bar{p}} + \frac{8 \bar{a}^4 (\bar{f} \cdot \bar{k}) e^4}{\bar{k} \cdot \bar{q}} - \frac{8 \bar{a}^4 (\bar{g} \cdot \bar{k}) e^4}{\bar{k} \cdot \bar{q}} \right) + \right. \\ & C_0 D_0 \left(-\frac{8 \bar{a}^2 (\bar{a} \cdot \bar{g}) (\bar{f} \cdot \bar{k}) e^3}{\bar{k} \cdot \bar{p}} - \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{f}) (\bar{g} \cdot \bar{k}) e^3}{\bar{k} \cdot \bar{p}} - \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{g}) (\bar{f} \cdot \bar{k}) e^3}{\bar{k} \cdot \bar{q}} - \right. \\ & \left. \left. \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{f}) (\bar{g} \cdot \bar{k}) e^3}{\bar{k} \cdot \bar{q}} + \frac{16 \bar{a}^2 (\bar{a} \cdot \bar{p}) (\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{k}) e^3}{(\bar{k} \cdot \bar{p}) (\bar{k} \cdot \bar{q})} + \frac{16 \bar{a}^2 (\bar{a} \cdot \bar{q}) (\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{k}) e^3}{(\bar{k} \cdot \bar{p}) (\bar{k} \cdot \bar{q})} \right) \right) + \\ & A_2 B_1 \left(C_0 D_1 \left(\frac{8 \bar{a}^4 (\bar{f} \cdot \bar{k}) e^4}{\bar{k} \cdot \bar{p}} - \frac{8 \bar{a}^4 (\bar{g} \cdot \bar{k}) e^4}{\bar{k} \cdot \bar{p}} + \frac{8 \bar{a}^4 (\bar{f} \cdot \bar{k}) e^4}{\bar{k} \cdot \bar{q}} - \frac{8 \bar{a}^4 (\bar{g} \cdot \bar{k}) e^4}{\bar{k} \cdot \bar{q}} \right) + \right. \\ & C_0 D_0 \left(-\frac{8 \bar{a}^2 (\bar{a} \cdot \bar{g}) (\bar{f} \cdot \bar{k}) e^3}{\bar{k} \cdot \bar{p}} - \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{f}) (\bar{g} \cdot \bar{k}) e^3}{\bar{k} \cdot \bar{p}} - \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{g}) (\bar{f} \cdot \bar{k}) e^3}{\bar{k} \cdot \bar{q}} - \right. \\ & \left. \left. \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{f}) (\bar{g} \cdot \bar{k}) e^3}{\bar{k} \cdot \bar{q}} + \frac{16 \bar{a}^2 (\bar{a} \cdot \bar{p}) (\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{k}) e^3}{(\bar{k} \cdot \bar{p}) (\bar{k} \cdot \bar{q})} + \frac{16 \bar{a}^2 (\bar{a} \cdot \bar{q}) (\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{k}) e^3}{(\bar{k} \cdot \bar{p}) (\bar{k} \cdot \bar{q})} \right) \right) + \\ & A_0 B_0 \left(\frac{16 C_2 D_2 \bar{a}^4 (\bar{k} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^4}{(\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{k})} + \frac{8 C_1 D_2 \bar{a}^2 (\bar{a} \cdot \bar{q}) (\bar{k} \cdot \bar{p}) e^3}{\bar{f} \cdot \bar{k}} - \frac{8 C_1 D_2 \bar{a}^2 (\bar{a} \cdot \bar{q}) (\bar{k} \cdot \bar{p}) e^3}{\bar{g} \cdot \bar{k}} + \right. \\ & \frac{16 C_1 D_2 \bar{a}^2 (\bar{a} \cdot \bar{f}) (\bar{k} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^3}{(\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{k})} - \frac{16 C_1 D_2 \bar{a}^2 (\bar{a} \cdot \bar{g}) (\bar{k} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^3}{(\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{k})} + \frac{8 C_1 D_2 \bar{a}^2 (\bar{a} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^3}{\bar{f} \cdot \bar{k}} - \\ & \frac{8 C_1 D_2 \bar{a}^2 (\bar{a} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^3}{\bar{g} \cdot \bar{k}} + 16 C_0 D_2 m^2 \bar{a}^2 e^2 + \frac{8 C_0 D_2 \bar{a}^2 (\bar{f} \cdot \bar{q}) (\bar{k} \cdot \bar{p}) e^2}{\bar{f} \cdot \bar{k}} + \\ & \frac{8 C_0 D_2 \bar{a}^2 (\bar{g} \cdot \bar{q}) (\bar{k} \cdot \bar{p}) e^2}{\bar{g} \cdot \bar{k}} + \frac{8 C_0 D_2 \bar{a}^2 (\bar{f} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^2}{\bar{f} \cdot \bar{k}} + \frac{8 C_0 D_2 \bar{a}^2 (\bar{g} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^2}{\bar{g} \cdot \bar{k}} - \\ & \left. \frac{16 C_0 D_2 m^2 \bar{a}^2 (\bar{k} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^2}{(\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{k})} - \frac{16 C_0 D_2 \bar{a}^2 (\bar{f} \cdot \bar{g}) (\bar{k} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^2}{(\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{k})} - 16 C_0 D_2 \bar{a}^2 (\bar{p} \cdot \bar{q}) e^2 + \right. \end{aligned}$$

$$\begin{aligned}
& \text{C1 D0} \left(16 e (\bar{a} \cdot \bar{f}) m^2 - 16 e (\bar{a} \cdot \bar{g}) m^2 + \frac{16 e (\bar{a} \cdot \bar{f}) (\bar{g} \cdot \bar{k}) m^2}{\bar{f} \cdot \bar{k}} - \frac{16 e (\bar{a} \cdot \bar{g}) (\bar{f} \cdot \bar{k}) m^2}{\bar{g} \cdot \bar{k}} - 16 e (\bar{a} \cdot \bar{q}) (\bar{f} \cdot \bar{p}) - \right. \\
& \quad 16 e (\bar{a} \cdot \bar{p}) (\bar{f} \cdot \bar{q}) + 16 e (\bar{a} \cdot \bar{q}) (\bar{g} \cdot \bar{p}) + 16 e (\bar{a} \cdot \bar{p}) (\bar{g} \cdot \bar{q}) - \frac{16 e (\bar{a} \cdot \bar{f}) (\bar{g} \cdot \bar{q}) (\bar{k} \cdot \bar{p})}{\bar{f} \cdot \bar{k}} + \\
& \quad \left. \frac{16 e (\bar{a} \cdot \bar{g}) (\bar{f} \cdot \bar{q}) (\bar{k} \cdot \bar{p})}{\bar{g} \cdot \bar{k}} - \frac{16 e (\bar{a} \cdot \bar{f}) (\bar{g} \cdot \bar{p}) (\bar{k} \cdot \bar{q})}{\bar{f} \cdot \bar{k}} + \frac{16 e (\bar{a} \cdot \bar{g}) (\bar{f} \cdot \bar{p}) (\bar{k} \cdot \bar{q})}{\bar{g} \cdot \bar{k}} \right) + \\
& \text{C0 D1} \left(16 e (\bar{a} \cdot \bar{f}) m^2 - 16 e (\bar{a} \cdot \bar{g}) m^2 + \frac{16 e (\bar{a} \cdot \bar{f}) (\bar{g} \cdot \bar{k}) m^2}{\bar{f} \cdot \bar{k}} - \frac{16 e (\bar{a} \cdot \bar{g}) (\bar{f} \cdot \bar{k}) m^2}{\bar{g} \cdot \bar{k}} - 16 e (\bar{a} \cdot \bar{q}) (\bar{f} \cdot \bar{p}) - \right. \\
& \quad 16 e (\bar{a} \cdot \bar{p}) (\bar{f} \cdot \bar{q}) + 16 e (\bar{a} \cdot \bar{q}) (\bar{g} \cdot \bar{p}) + 16 e (\bar{a} \cdot \bar{p}) (\bar{g} \cdot \bar{q}) - \frac{16 e (\bar{a} \cdot \bar{f}) (\bar{g} \cdot \bar{q}) (\bar{k} \cdot \bar{p})}{\bar{f} \cdot \bar{k}} + \\
& \quad \left. \frac{16 e (\bar{a} \cdot \bar{g}) (\bar{f} \cdot \bar{q}) (\bar{k} \cdot \bar{p})}{\bar{g} \cdot \bar{k}} - \frac{16 e (\bar{a} \cdot \bar{f}) (\bar{g} \cdot \bar{p}) (\bar{k} \cdot \bar{q})}{\bar{f} \cdot \bar{k}} + \frac{16 e (\bar{a} \cdot \bar{g}) (\bar{f} \cdot \bar{p}) (\bar{k} \cdot \bar{q})}{\bar{g} \cdot \bar{k}} \right) + \\
& \text{C2 D1} \left(\frac{8 \bar{a}^2 (\bar{a} \cdot \bar{q}) (\bar{k} \cdot \bar{p}) e^3}{\bar{f} \cdot \bar{k}} - \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{q}) (\bar{k} \cdot \bar{p}) e^3}{\bar{g} \cdot \bar{k}} + \frac{16 \bar{a}^2 (\bar{a} \cdot \bar{f}) (\bar{k} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^3}{(\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{k})} - \right. \\
& \quad \left. \frac{16 \bar{a}^2 (\bar{a} \cdot \bar{g}) (\bar{k} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^3}{(\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{k})} + \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^3}{\bar{f} \cdot \bar{k}} - \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^3}{\bar{g} \cdot \bar{k}} \right) + \\
& \text{C0 D0} (-64 m^4 - 32 (\bar{f} \cdot \bar{g}) m^2 + 32 (\bar{p} \cdot \bar{q}) m^2 + 32 (\bar{f} \cdot \bar{q}) (\bar{g} \cdot \bar{p}) + 32 (\bar{f} \cdot \bar{p}) (\bar{g} \cdot \bar{q})) + \\
& \text{C2 D0} \left(16 m^2 \bar{a}^2 e^2 + \frac{8 \bar{a}^2 (\bar{f} \cdot \bar{q}) (\bar{k} \cdot \bar{p}) e^2}{\bar{f} \cdot \bar{k}} + \frac{8 \bar{a}^2 (\bar{g} \cdot \bar{q}) (\bar{k} \cdot \bar{p}) e^2}{\bar{g} \cdot \bar{k}} + \frac{8 \bar{a}^2 (\bar{f} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^2}{\bar{f} \cdot \bar{k}} + \right. \\
& \quad \left. \frac{8 \bar{a}^2 (\bar{g} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^2}{\bar{g} \cdot \bar{k}} - \frac{16 m^2 \bar{a}^2 (\bar{k} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^2}{(\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{k})} - \frac{16 \bar{a}^2 (\bar{f} \cdot \bar{g}) (\bar{k} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^2}{(\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{k})} - 16 \bar{a}^2 (\bar{p} \cdot \bar{q}) e^2 \right) + \\
& \text{C1 D1} \left(-64 (\bar{a} \cdot \bar{p}) (\bar{a} \cdot \bar{q}) e^2 + \frac{16 m^2 \bar{a}^2 (\bar{g} \cdot \bar{k}) e^2}{\bar{f} \cdot \bar{k}} - \frac{16 \bar{a}^2 (\bar{f} \cdot \bar{q}) (\bar{k} \cdot \bar{p}) e^2}{\bar{f} \cdot \bar{k}} - \frac{16 \bar{a}^2 (\bar{g} \cdot \bar{q}) (\bar{k} \cdot \bar{p}) e^2}{\bar{f} \cdot \bar{k}} - \right. \\
& \quad \frac{16 \bar{a}^2 (\bar{g} \cdot \bar{q}) (\bar{k} \cdot \bar{p}) e^2}{\bar{g} \cdot \bar{k}} + \frac{32 (\bar{a} \cdot \bar{f}) (\bar{a} \cdot \bar{q}) (\bar{k} \cdot \bar{p}) e^2}{\bar{f} \cdot \bar{k}} + \frac{32 (\bar{a} \cdot \bar{g}) (\bar{a} \cdot \bar{q}) (\bar{k} \cdot \bar{p}) e^2}{\bar{g} \cdot \bar{k}} - \frac{16 \bar{a}^2 (\bar{f} \cdot \bar{q}) (\bar{k} \cdot \bar{p}) e^2}{\bar{g} \cdot \bar{k}} - \\
& \quad \frac{16 \bar{a}^2 (\bar{f} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^2}{\bar{f} \cdot \bar{k}} - \frac{16 \bar{a}^2 (\bar{g} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^2}{\bar{f} \cdot \bar{k}} - \frac{16 \bar{a}^2 (\bar{g} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^2}{\bar{g} \cdot \bar{k}} + \frac{32 m^2 \bar{a}^2 (\bar{k} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^2}{(\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{k})} - \\
& \quad \frac{64 (\bar{a} \cdot \bar{f}) (\bar{a} \cdot \bar{g}) (\bar{k} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^2}{(\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{k})} + \frac{32 \bar{a}^2 (\bar{f} \cdot \bar{g}) (\bar{k} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^2}{(\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{k})} + \frac{32 (\bar{a} \cdot \bar{f}) (\bar{a} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^2}{\bar{f} \cdot \bar{k}} + \\
& \quad \left. \frac{32 (\bar{a} \cdot \bar{g}) (\bar{a} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^2}{\bar{g} \cdot \bar{k}} - \frac{16 \bar{a}^2 (\bar{f} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^2}{\bar{g} \cdot \bar{k}} + 32 \bar{a}^2 (\bar{p} \cdot \bar{q}) e^2 + \frac{16 m^2 \bar{a}^2 (\bar{f} \cdot \bar{k}) e^2}{\bar{g} \cdot \bar{k}} \right) + \\
& \text{A0 B1} \left(\text{C2 D1} \left(\frac{8 \bar{a}^4 (\bar{k} \cdot \bar{p}) e^4}{\bar{f} \cdot \bar{k}} - \frac{8 \bar{a}^4 (\bar{k} \cdot \bar{p}) e^4}{\bar{g} \cdot \bar{k}} + \frac{8 \bar{a}^4 (\bar{k} \cdot \bar{q}) e^4}{\bar{f} \cdot \bar{k}} - \frac{8 \bar{a}^4 (\bar{k} \cdot \bar{q}) e^4}{\bar{g} \cdot \bar{k}} \right) + \right. \\
& \quad \left. \text{C2 D0} \left(-16 \bar{a}^2 (\bar{a} \cdot \bar{p}) e^3 - 16 \bar{a}^2 (\bar{a} \cdot \bar{q}) e^3 + \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{f}) (\bar{k} \cdot \bar{p}) e^3}{\bar{f} \cdot \bar{k}} + \right. \right.
\end{aligned}$$

$$\begin{aligned}
& \frac{8 \bar{a}^2 (\bar{f} \cdot \bar{p}) (\bar{g} \cdot \bar{k}) (\bar{k} \cdot \bar{q}) e^2}{(\bar{f} \cdot \bar{k}) (\bar{k} \cdot \bar{p})} + \frac{16 (\bar{a} \cdot \bar{f}) (\bar{a} \cdot \bar{p}) (\bar{g} \cdot \bar{k}) (\bar{k} \cdot \bar{q}) e^2}{(\bar{f} \cdot \bar{k}) (\bar{k} \cdot \bar{p})} + \frac{8 \bar{a}^2 (\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) e^2}{(\bar{g} \cdot \bar{k}) (\bar{k} \cdot \bar{p})} - \\
& \frac{16 (\bar{a} \cdot \bar{g}) (\bar{a} \cdot \bar{p}) (\bar{f} \cdot \bar{k}) (\bar{k} \cdot \bar{q}) e^2}{(\bar{g} \cdot \bar{k}) (\bar{k} \cdot \bar{p})} - \frac{8 \bar{a}^2 (\bar{f} \cdot \bar{k}) (\bar{p} \cdot \bar{q}) e^2}{\bar{k} \cdot \bar{p}} + \frac{8 \bar{a}^2 (\bar{g} \cdot \bar{k}) (\bar{p} \cdot \bar{q}) e^2}{\bar{k} \cdot \bar{p}} - \\
& \frac{8 \bar{a}^2 (\bar{f} \cdot \bar{k}) (\bar{p} \cdot \bar{q}) e^2}{\bar{k} \cdot \bar{q}} + \frac{8 \bar{a}^2 (\bar{g} \cdot \bar{k}) (\bar{p} \cdot \bar{q}) e^2}{\bar{k} \cdot \bar{q}} + \frac{8 m^2 \bar{a}^2 (\bar{f} \cdot \bar{k}) e^2}{\bar{k} \cdot \bar{p}} + \frac{16 (\bar{a} \cdot \bar{p}) (\bar{a} \cdot \bar{q}) (\bar{f} \cdot \bar{k}) e^2}{\bar{k} \cdot \bar{p}} - \\
& \frac{8 m^2 \bar{a}^2 (\bar{g} \cdot \bar{k}) e^2}{\bar{k} \cdot \bar{p}} - \frac{16 (\bar{a} \cdot \bar{p}) (\bar{a} \cdot \bar{q}) (\bar{g} \cdot \bar{k}) e^2}{\bar{k} \cdot \bar{p}} + \frac{8 m^2 \bar{a}^2 (\bar{f} \cdot \bar{k}) e^2}{\bar{k} \cdot \bar{q}} + \frac{16 (\bar{a} \cdot \bar{p}) (\bar{a} \cdot \bar{q}) (\bar{f} \cdot \bar{k}) e^2}{\bar{k} \cdot \bar{q}} - \\
& \frac{8 m^2 \bar{a}^2 (\bar{g} \cdot \bar{k}) e^2}{\bar{k} \cdot \bar{q}} - \frac{16 (\bar{a} \cdot \bar{p}) (\bar{a} \cdot \bar{q}) (\bar{g} \cdot \bar{k}) e^2}{\bar{k} \cdot \bar{q}} - \frac{8 \bar{a}^2 (\bar{f} \cdot \bar{q}) (\bar{g} \cdot \bar{k}) (\bar{k} \cdot \bar{p}) e^2}{(\bar{f} \cdot \bar{k}) (\bar{k} \cdot \bar{q})} + \\
& \frac{16 (\bar{a} \cdot \bar{f}) (\bar{a} \cdot \bar{q}) (\bar{g} \cdot \bar{k}) (\bar{k} \cdot \bar{p}) e^2}{(\bar{f} \cdot \bar{k}) (\bar{k} \cdot \bar{q})} + \frac{8 \bar{a}^2 (\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{q}) (\bar{k} \cdot \bar{p}) e^2}{(\bar{g} \cdot \bar{k}) (\bar{k} \cdot \bar{q})} - \frac{16 (\bar{a} \cdot \bar{g}) (\bar{a} \cdot \bar{q}) (\bar{f} \cdot \bar{k}) (\bar{k} \cdot \bar{p}) e^2}{(\bar{g} \cdot \bar{k}) (\bar{k} \cdot \bar{q})} \Bigg) + \\
& \text{A1 B0} \left(\frac{8 \text{C1 D2 } \bar{a}^4 (\bar{k} \cdot \bar{p}) e^4}{\bar{f} \cdot \bar{k}} - \frac{8 \text{C1 D2 } \bar{a}^4 (\bar{k} \cdot \bar{p}) e^4}{\bar{g} \cdot \bar{k}} + \frac{8 \text{C1 D2 } \bar{a}^4 (\bar{k} \cdot \bar{q}) e^4}{\bar{f} \cdot \bar{k}} - \frac{8 \text{C1 D2 } \bar{a}^4 (\bar{k} \cdot \bar{q}) e^4}{\bar{g} \cdot \bar{k}} - \right. \\
& \frac{16 \text{C0 D2 } \bar{a}^2 (\bar{a} \cdot \bar{p}) e^3}{\bar{f} \cdot \bar{k}} - \frac{16 \text{C0 D2 } \bar{a}^2 (\bar{a} \cdot \bar{q}) e^3}{\bar{g} \cdot \bar{k}} + \\
& \frac{8 \text{C0 D2 } \bar{a}^2 (\bar{a} \cdot \bar{f}) (\bar{k} \cdot \bar{p}) e^3}{\bar{f} \cdot \bar{k}} + \frac{8 \text{C0 D2 } \bar{a}^2 (\bar{a} \cdot \bar{g}) (\bar{k} \cdot \bar{p}) e^3}{\bar{g} \cdot \bar{k}} + \\
& \frac{8 \text{C0 D2 } \bar{a}^2 (\bar{a} \cdot \bar{f}) (\bar{k} \cdot \bar{q}) e^3}{\bar{f} \cdot \bar{k}} + \frac{8 \text{C0 D2 } \bar{a}^2 (\bar{a} \cdot \bar{g}) (\bar{k} \cdot \bar{q}) e^3}{\bar{g} \cdot \bar{k}} + \\
& \text{C0 D0} \left(16 e (\bar{a} \cdot \bar{p}) m^2 + 16 e (\bar{a} \cdot \bar{q}) m^2 - \frac{16 e (\bar{a} \cdot \bar{p}) (\bar{k} \cdot \bar{q}) m^2}{\bar{k} \cdot \bar{p}} - \frac{16 e (\bar{a} \cdot \bar{q}) (\bar{k} \cdot \bar{p}) m^2}{\bar{k} \cdot \bar{q}} + 16 e (\bar{a} \cdot \bar{g}) (\bar{f} \cdot \bar{p}) + \right. \\
& 16 e (\bar{a} \cdot \bar{g}) (\bar{f} \cdot \bar{q}) + 16 e (\bar{a} \cdot \bar{f}) (\bar{g} \cdot \bar{p}) + 16 e (\bar{a} \cdot \bar{f}) (\bar{g} \cdot \bar{q}) - \frac{16 e (\bar{a} \cdot \bar{p}) (\bar{f} \cdot \bar{q}) (\bar{g} \cdot \bar{k})}{\bar{k} \cdot \bar{p}} - \\
& \frac{16 e (\bar{a} \cdot \bar{p}) (\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{q})}{\bar{k} \cdot \bar{p}} - \frac{16 e (\bar{a} \cdot \bar{q}) (\bar{f} \cdot \bar{p}) (\bar{g} \cdot \bar{k})}{\bar{k} \cdot \bar{q}} - \frac{16 e (\bar{a} \cdot \bar{q}) (\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{p})}{\bar{k} \cdot \bar{q}} \Bigg) + \\
& \text{C1 D1} \left(-16 \bar{a}^2 (\bar{a} \cdot \bar{p}) e^3 - 16 \bar{a}^2 (\bar{a} \cdot \bar{q}) e^3 + \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{f}) (\bar{k} \cdot \bar{p}) e^3}{\bar{f} \cdot \bar{k}} - \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{g}) (\bar{k} \cdot \bar{p}) e^3}{\bar{f} \cdot \bar{k}} - \right. \\
& \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{f}) (\bar{k} \cdot \bar{p}) e^3}{\bar{g} \cdot \bar{k}} + \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{g}) (\bar{k} \cdot \bar{p}) e^3}{\bar{g} \cdot \bar{k}} + \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{f}) (\bar{k} \cdot \bar{q}) e^3}{\bar{f} \cdot \bar{k}} - \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{g}) (\bar{k} \cdot \bar{q}) e^3}{\bar{f} \cdot \bar{k}} - \\
& \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{f}) (\bar{k} \cdot \bar{q}) e^3}{\bar{g} \cdot \bar{k}} + \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{g}) (\bar{k} \cdot \bar{q}) e^3}{\bar{g} \cdot \bar{k}} + \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{p}) (\bar{g} \cdot \bar{k}) (\bar{k} \cdot \bar{q}) e^3}{(\bar{f} \cdot \bar{k}) (\bar{k} \cdot \bar{p})} + \\
& \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{p}) (\bar{f} \cdot \bar{k}) (\bar{k} \cdot \bar{q}) e^3}{(\bar{g} \cdot \bar{k}) (\bar{k} \cdot \bar{p})} + \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{q}) (\bar{g} \cdot \bar{k}) (\bar{k} \cdot \bar{p}) e^3}{(\bar{f} \cdot \bar{k}) (\bar{k} \cdot \bar{q})} + \frac{8 \bar{a}^2 (\bar{a} \cdot \bar{q}) (\bar{f} \cdot \bar{k}) (\bar{k} \cdot \bar{p}) e^3}{(\bar{g} \cdot \bar{k}) (\bar{k} \cdot \bar{q})} \Bigg) + \text{C1 D0} \\
& \left(-8 \bar{a}^2 (\bar{f} \cdot \bar{p}) e^2 - 8 \bar{a}^2 (\bar{f} \cdot \bar{q}) e^2 + 8 \bar{a}^2 (\bar{g} \cdot \bar{p}) e^2 + 8 \bar{a}^2 (\bar{g} \cdot \bar{q}) e^2 - \frac{8 m^2 \bar{a}^2 (\bar{k} \cdot \bar{p}) e^2}{\bar{f} \cdot \bar{k}} - \frac{8 \bar{a}^2 (\bar{f} \cdot \bar{g}) (\bar{k} \cdot \bar{p}) e^2}{\bar{f} \cdot \bar{k}} + \right.
\end{aligned}$$

$$\left. \frac{8 \bar{a}^2 (\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{p}) e^2}{\bar{k} \cdot \bar{p}} - \frac{8 \bar{a}^2 (\bar{f} \cdot \bar{q}) (\bar{g} \cdot \bar{k}) e^2}{\bar{k} \cdot \bar{q}} - \frac{8 \bar{a}^2 (\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{q}) e^2}{\bar{k} \cdot \bar{q}} - \frac{16 m^2 \bar{a}^2 (\bar{f} \cdot \bar{k}) (\bar{g} \cdot \bar{k}) e^2}{(\bar{k} \cdot \bar{p}) (\bar{k} \cdot \bar{q})} \right)$$

To simplify further, contractions of momenta with k , such as (kp) , are replaced by χ - parameters

$$(kp) = \frac{m^2 \chi_p}{\xi}$$

ξ is defined from

$$\xi^2 = -\frac{e^2 a^2}{m^2}$$

Function **ktochiSubs** is defined in `definitons.nb`, it generates substitutions of scalar products (kp) , (kq) , etc, to pass to the corresponding χ parameters as defined above

In[35]:= **(*example*)**

kp /. ktochiSubs[p, q, f, g]

Out[35]=
$$\frac{m^2 \chi_p}{\xi}$$

In[36]:= **Diffs3 = Diffs2;**

Coeff3 = Coeff2;

Expr3 = Expr2 /. ktochiSubs[p, q, f, g] /. {av2 → -m^2 ξ^2 / e^2};

% /. {D → 4}

Out[39]=
$$\frac{16 A2 B2 C0 D0 m^4 \chi f \chi g \xi^4}{\chi p \chi q} + A1 B2$$

$$\left(C1 D0 \left(\frac{8 m^4 \chi f \xi^4}{\chi p} - \frac{8 m^4 \chi g \xi^4}{\chi p} + \frac{8 m^4 \chi f \xi^4}{\chi q} - \frac{8 m^4 \chi g \xi^4}{\chi q} \right) + C0 D0 \left(\frac{8 e m^2 \chi g (\bar{a} \cdot \bar{f}) \xi^2}{\chi p} + \frac{8 e m^2 \chi g (\bar{a} \cdot \bar{f}) \xi^2}{\chi q} + \frac{8 e m^2 \chi f (\bar{a} \cdot \bar{g}) \xi^2}{\chi p} + \frac{8 e m^2 \chi f (\bar{a} \cdot \bar{g}) \xi^2}{\chi q} - \frac{16 e m^2 \chi f \chi g (\bar{a} \cdot \bar{p}) \xi^2}{\chi p \chi q} - \frac{16 e m^2 \chi f \chi g (\bar{a} \cdot \bar{q}) \xi^2}{\chi p \chi q} \right) \right) + A2 B1$$

$$\left(C0 D1 \left(\frac{8 m^4 \chi f \xi^4}{\chi p} - \frac{8 m^4 \chi g \xi^4}{\chi p} + \frac{8 m^4 \chi f \xi^4}{\chi q} - \frac{8 m^4 \chi g \xi^4}{\chi q} \right) + C0 D0 \left(\frac{8 e m^2 \chi g (\bar{a} \cdot \bar{f}) \xi^2}{\chi p} + \frac{8 e m^2 \chi g (\bar{a} \cdot \bar{f}) \xi^2}{\chi q} + \frac{8 e m^2 \chi f (\bar{a} \cdot \bar{g}) \xi^2}{\chi p} + \frac{8 e m^2 \chi f (\bar{a} \cdot \bar{g}) \xi^2}{\chi q} - \frac{16 e m^2 \chi f \chi g (\bar{a} \cdot \bar{p}) \xi^2}{\chi p \chi q} - \frac{16 e m^2 \chi f \chi g (\bar{a} \cdot \bar{q}) \xi^2}{\chi p \chi q} \right) \right) +$$

$$A0 B0 \left(-16 C0 D2 \xi^2 m^4 + \frac{16 C2 D2 \xi^4 \chi p \chi q m^4}{\chi f \chi g} + \frac{16 C0 D2 \xi^2 \chi p \chi q m^4}{\chi f \chi g} - \frac{16 C1 D2 e \xi^2 \chi p \chi q (\bar{a} \cdot \bar{f}) m^2}{\chi f \chi g} + \frac{16 C1 D2 e \xi^2 \chi p \chi q (\bar{a} \cdot \bar{g}) m^2}{\chi f \chi g} - \frac{8 C1 D2 e \xi^2 \chi q (\bar{a} \cdot \bar{p}) m^2}{\chi f} + \frac{8 C1 D2 e \xi^2 \chi q (\bar{a} \cdot \bar{p}) m^2}{\chi g} - \frac{8 C1 D2 e \xi^2 \chi p (\bar{a} \cdot \bar{q}) m^2}{\chi f} + \frac{8 C1 D2 e \xi^2 \chi p (\bar{a} \cdot \bar{q}) m^2}{\chi g} + \frac{16 C0 D2 \xi^2 \chi p \chi q (\bar{f} \cdot \bar{g}) m^2}{\chi f \chi g} - \right)$$

$$\begin{aligned}
& \frac{8 \text{ C0 D2 } \xi^2 \chi q (\bar{f} \cdot \bar{p}) m^2}{\chi^f} - \frac{8 \text{ C0 D2 } \xi^2 \chi p (\bar{f} \cdot \bar{q}) m^2}{\chi^f} - \frac{8 \text{ C0 D2 } \xi^2 \chi q (\bar{g} \cdot \bar{p}) m^2}{\chi^g} - \frac{8 \text{ C0 D2 } \xi^2 \chi p (\bar{g} \cdot \bar{q}) m^2}{\chi^g} + \\
& 16 \text{ C0 D2 } \xi^2 (\bar{p} \cdot \bar{q}) m^2 + \text{C2 D1} \left(-\frac{16 e m^2 \chi p \chi q (\bar{a} \cdot \bar{f}) \xi^2}{\chi^f \chi^g} + \frac{16 e m^2 \chi p \chi q (\bar{a} \cdot \bar{g}) \xi^2}{\chi^f \chi^g} - \right. \\
& \quad \left. \frac{8 e m^2 \chi q (\bar{a} \cdot \bar{p}) \xi^2}{\chi^f} + \frac{8 e m^2 \chi q (\bar{a} \cdot \bar{p}) \xi^2}{\chi^g} - \frac{8 e m^2 \chi p (\bar{a} \cdot \bar{q}) \xi^2}{\chi^f} + \frac{8 e m^2 \chi p (\bar{a} \cdot \bar{q}) \xi^2}{\chi^g} \right) + \\
& \text{C1 D0} \left(16 e (\bar{a} \cdot \bar{f}) m^2 + \frac{16 e \chi^g (\bar{a} \cdot \bar{f}) m^2}{\chi^f} - 16 e (\bar{a} \cdot \bar{g}) m^2 - \frac{16 e \chi^f (\bar{a} \cdot \bar{g}) m^2}{\chi^g} + \right. \\
& \quad \frac{16 e \chi q (\bar{a} \cdot \bar{g}) (\bar{f} \cdot \bar{p})}{\chi^g} - 16 e (\bar{a} \cdot \bar{q}) (\bar{f} \cdot \bar{p}) + \frac{16 e \chi p (\bar{a} \cdot \bar{g}) (\bar{f} \cdot \bar{q})}{\chi^g} - 16 e (\bar{a} \cdot \bar{p}) (\bar{f} \cdot \bar{q}) - \\
& \quad \left. \frac{16 e \chi q (\bar{a} \cdot \bar{f}) (\bar{g} \cdot \bar{p})}{\chi^f} + 16 e (\bar{a} \cdot \bar{q}) (\bar{g} \cdot \bar{p}) - \frac{16 e \chi p (\bar{a} \cdot \bar{f}) (\bar{g} \cdot \bar{q})}{\chi^f} + 16 e (\bar{a} \cdot \bar{p}) (\bar{g} \cdot \bar{q}) \right) + \\
& \text{C0 D1} \left(16 e (\bar{a} \cdot \bar{f}) m^2 + \frac{16 e \chi^g (\bar{a} \cdot \bar{f}) m^2}{\chi^f} - 16 e (\bar{a} \cdot \bar{g}) m^2 - \frac{16 e \chi^f (\bar{a} \cdot \bar{g}) m^2}{\chi^g} + \right. \\
& \quad \frac{16 e \chi q (\bar{a} \cdot \bar{g}) (\bar{f} \cdot \bar{p})}{\chi^g} - 16 e (\bar{a} \cdot \bar{q}) (\bar{f} \cdot \bar{p}) + \frac{16 e \chi p (\bar{a} \cdot \bar{g}) (\bar{f} \cdot \bar{q})}{\chi^g} - 16 e (\bar{a} \cdot \bar{p}) (\bar{f} \cdot \bar{q}) - \\
& \quad \left. \frac{16 e \chi q (\bar{a} \cdot \bar{f}) (\bar{g} \cdot \bar{p})}{\chi^f} + 16 e (\bar{a} \cdot \bar{q}) (\bar{g} \cdot \bar{p}) - \frac{16 e \chi p (\bar{a} \cdot \bar{f}) (\bar{g} \cdot \bar{q})}{\chi^f} + 16 e (\bar{a} \cdot \bar{p}) (\bar{g} \cdot \bar{q}) \right) + \\
& \text{C0 D0} (-64 m^4 - 32 (\bar{f} \cdot \bar{g}) m^2 + 32 (\bar{p} \cdot \bar{q}) m^2 + 32 (\bar{f} \cdot \bar{q}) (\bar{g} \cdot \bar{p}) + 32 (\bar{f} \cdot \bar{p}) (\bar{g} \cdot \bar{q})) + \\
& \text{C1 D1} \left(-\frac{16 \xi^2 \chi^g m^4}{\chi^f} - \frac{32 \xi^2 \chi p \chi q m^4}{\chi^f \chi^g} - \frac{16 \xi^2 \chi^f m^4}{\chi^g} - \frac{32 \xi^2 \chi p \chi q (\bar{f} \cdot \bar{g}) m^2}{\chi^f \chi^g} + \right. \\
& \quad \frac{16 \xi^2 \chi q (\bar{f} \cdot \bar{p}) m^2}{\chi^f} + \frac{16 \xi^2 \chi q (\bar{f} \cdot \bar{p}) m^2}{\chi^g} + \frac{16 \xi^2 \chi p (\bar{f} \cdot \bar{q}) m^2}{\chi^f} + \frac{16 \xi^2 \chi p (\bar{f} \cdot \bar{q}) m^2}{\chi^g} + \\
& \quad \frac{16 \xi^2 \chi q (\bar{g} \cdot \bar{p}) m^2}{\chi^f} + \frac{16 \xi^2 \chi q (\bar{g} \cdot \bar{p}) m^2}{\chi^g} + \frac{16 \xi^2 \chi p (\bar{g} \cdot \bar{q}) m^2}{\chi^f} + \frac{16 \xi^2 \chi p (\bar{g} \cdot \bar{q}) m^2}{\chi^g} - \\
& \quad 32 \xi^2 (\bar{p} \cdot \bar{q}) m^2 - \frac{64 e^2 \chi p \chi q (\bar{a} \cdot \bar{f}) (\bar{a} \cdot \bar{g})}{\chi^f \chi^g} + \frac{32 e^2 \chi q (\bar{a} \cdot \bar{f}) (\bar{a} \cdot \bar{p})}{\chi^f} + \frac{32 e^2 \chi q (\bar{a} \cdot \bar{g}) (\bar{a} \cdot \bar{p})}{\chi^g} + \\
& \quad \left. \frac{32 e^2 \chi p (\bar{a} \cdot \bar{f}) (\bar{a} \cdot \bar{q})}{\chi^f} + \frac{32 e^2 \chi p (\bar{a} \cdot \bar{g}) (\bar{a} \cdot \bar{q})}{\chi^g} - 64 e^2 (\bar{a} \cdot \bar{p}) (\bar{a} \cdot \bar{q}) \right) + \\
& \text{C2 D0} \left(-16 \xi^2 m^4 + \frac{16 \xi^2 \chi p \chi q m^4}{\chi^f \chi^g} + \frac{16 \xi^2 \chi p \chi q (\bar{f} \cdot \bar{g}) m^2}{\chi^f \chi^g} - \frac{8 \xi^2 \chi q (\bar{f} \cdot \bar{p}) m^2}{\chi^f} - \right. \\
& \quad \left. \frac{8 \xi^2 \chi p (\bar{f} \cdot \bar{q}) m^2}{\chi^f} - \frac{8 \xi^2 \chi q (\bar{g} \cdot \bar{p}) m^2}{\chi^g} - \frac{8 \xi^2 \chi p (\bar{g} \cdot \bar{q}) m^2}{\chi^g} + 16 \xi^2 (\bar{p} \cdot \bar{q}) m^2 \right) + \\
& \text{A1 B0} \left(\frac{8 \text{ C1 D2 } m^4 \chi p \xi^4}{\chi^f} - \frac{8 \text{ C1 D2 } m^4 \chi p \xi^4}{\chi^g} + \frac{8 \text{ C1 D2 } m^4 \chi q \xi^4}{\chi^f} - \frac{8 \text{ C1 D2 } m^4 \chi q \xi^4}{\chi^g} - \right. \\
& \quad \left. \frac{8 \text{ C0 D2 } e m^2 \chi p (\bar{a} \cdot \bar{f}) \xi^2}{\chi^f} - \frac{8 \text{ C0 D2 } e m^2 \chi q (\bar{a} \cdot \bar{f}) \xi^2}{\chi^f} - \frac{8 \text{ C0 D2 } e m^2 \chi p (\bar{a} \cdot \bar{g}) \xi^2}{\chi^g} - \right.
\end{aligned}$$

$$\begin{aligned}
& \frac{8 \text{ C0 D2 } e m^2 \chi q (\bar{a} \cdot \bar{g}) \xi^2}{\chi g} + 16 \text{ C0 D2 } e m^2 (\bar{a} \cdot \bar{p}) \xi^2 + 16 \text{ C0 D2 } e m^2 (\bar{a} \cdot \bar{q}) \xi^2 + \\
& \text{C1 D1} \left(-\frac{8 e m^2 \chi p (\bar{a} \cdot \bar{f}) \xi^2}{\chi f} + \frac{8 e m^2 \chi p (\bar{a} \cdot \bar{f}) \xi^2}{\chi g} - \frac{8 e m^2 \chi q (\bar{a} \cdot \bar{f}) \xi^2}{\chi f} + \right. \\
& \quad \frac{8 e m^2 \chi q (\bar{a} \cdot \bar{f}) \xi^2}{\chi g} + \frac{8 e m^2 \chi p (\bar{a} \cdot \bar{g}) \xi^2}{\chi f} - \frac{8 e m^2 \chi p (\bar{a} \cdot \bar{g}) \xi^2}{\chi g} + \frac{8 e m^2 \chi q (\bar{a} \cdot \bar{g}) \xi^2}{\chi f} - \\
& \quad \frac{8 e m^2 \chi q (\bar{a} \cdot \bar{g}) \xi^2}{\chi g} + 16 e m^2 (\bar{a} \cdot \bar{p}) \xi^2 - \frac{8 e m^2 \chi g \chi q (\bar{a} \cdot \bar{p}) \xi^2}{\chi f \chi p} - \frac{8 e m^2 \chi f \chi q (\bar{a} \cdot \bar{p}) \xi^2}{\chi g \chi p} + \\
& \quad \left. 16 e m^2 (\bar{a} \cdot \bar{q}) \xi^2 - \frac{8 e m^2 \chi g \chi p (\bar{a} \cdot \bar{q}) \xi^2}{\chi f \chi q} - \frac{8 e m^2 \chi f \chi p (\bar{a} \cdot \bar{q}) \xi^2}{\chi g \chi q} \right) + \\
& \text{C0 D0} \left(16 e (\bar{a} \cdot \bar{p}) m^2 - \frac{16 e \chi q (\bar{a} \cdot \bar{p}) m^2}{\chi p} + 16 e (\bar{a} \cdot \bar{q}) m^2 - \frac{16 e \chi p (\bar{a} \cdot \bar{q}) m^2}{\chi q} + \right. \\
& \quad 16 e (\bar{a} \cdot \bar{g}) (\bar{f} \cdot \bar{p}) - \frac{16 e \chi g (\bar{a} \cdot \bar{q}) (\bar{f} \cdot \bar{p})}{\chi q} + 16 e (\bar{a} \cdot \bar{g}) (\bar{f} \cdot \bar{q}) - \frac{16 e \chi g (\bar{a} \cdot \bar{p}) (\bar{f} \cdot \bar{q})}{\chi p} + \\
& \quad \left. 16 e (\bar{a} \cdot \bar{f}) (\bar{g} \cdot \bar{p}) - \frac{16 e \chi f (\bar{a} \cdot \bar{q}) (\bar{g} \cdot \bar{p})}{\chi q} + 16 e (\bar{a} \cdot \bar{f}) (\bar{g} \cdot \bar{q}) - \frac{16 e \chi f (\bar{a} \cdot \bar{p}) (\bar{g} \cdot \bar{q})}{\chi p} \right) + \\
& \text{C0 D1} \left(-\frac{8 \xi^2 \chi p m^4}{\chi f} + \frac{8 \xi^2 \chi p m^4}{\chi g} - \frac{8 \xi^2 \chi q m^4}{\chi f} + \frac{8 \xi^2 \chi q m^4}{\chi g} - \frac{8 \xi^2 \chi f m^4}{\chi p} + \frac{8 \xi^2 \chi g m^4}{\chi p} - \right. \\
& \quad \frac{8 \xi^2 \chi f m^4}{\chi q} + \frac{8 \xi^2 \chi g m^4}{\chi q} - \frac{8 \xi^2 \chi p (\bar{f} \cdot \bar{g}) m^2}{\chi f} + \frac{8 \xi^2 \chi p (\bar{f} \cdot \bar{g}) m^2}{\chi g} - \frac{8 \xi^2 \chi q (\bar{f} \cdot \bar{g}) m^2}{\chi f} + \\
& \quad \frac{8 \xi^2 \chi q (\bar{f} \cdot \bar{g}) m^2}{\chi g} + 8 \xi^2 (\bar{f} \cdot \bar{p}) m^2 + \frac{8 \xi^2 \chi g \chi q (\bar{f} \cdot \bar{p}) m^2}{\chi f \chi p} + 8 \xi^2 (\bar{f} \cdot \bar{q}) m^2 + \frac{8 \xi^2 \chi g \chi p (\bar{f} \cdot \bar{q}) m^2}{\chi f \chi q} - \\
& \quad 8 \xi^2 (\bar{g} \cdot \bar{p}) m^2 - \frac{8 \xi^2 \chi f \chi q (\bar{g} \cdot \bar{p}) m^2}{\chi g \chi p} - 8 \xi^2 (\bar{g} \cdot \bar{q}) m^2 - \frac{8 \xi^2 \chi f \chi p (\bar{g} \cdot \bar{q}) m^2}{\chi g \chi q} + \\
& \quad \frac{8 \xi^2 \chi f (\bar{p} \cdot \bar{q}) m^2}{\chi p} - \frac{8 \xi^2 \chi g (\bar{p} \cdot \bar{q}) m^2}{\chi p} + \frac{8 \xi^2 \chi f (\bar{p} \cdot \bar{q}) m^2}{\chi q} - \frac{8 \xi^2 \chi g (\bar{p} \cdot \bar{q}) m^2}{\chi q} - \\
& \quad \frac{16 e^2 \chi p (\bar{a} \cdot \bar{f}) (\bar{a} \cdot \bar{g})}{\chi f} + \frac{16 e^2 \chi p (\bar{a} \cdot \bar{f}) (\bar{a} \cdot \bar{g})}{\chi g} - \frac{16 e^2 \chi q (\bar{a} \cdot \bar{f}) (\bar{a} \cdot \bar{g})}{\chi f} + \frac{16 e^2 \chi q (\bar{a} \cdot \bar{f}) (\bar{a} \cdot \bar{g})}{\chi g} - \\
& \quad 16 e^2 (\bar{a} \cdot \bar{f}) (\bar{a} \cdot \bar{p}) + \frac{16 e^2 \chi g \chi q (\bar{a} \cdot \bar{f}) (\bar{a} \cdot \bar{p})}{\chi f \chi p} + 16 e^2 (\bar{a} \cdot \bar{g}) (\bar{a} \cdot \bar{p}) - \frac{16 e^2 \chi f \chi q (\bar{a} \cdot \bar{g}) (\bar{a} \cdot \bar{p})}{\chi g \chi p} - \\
& \quad 16 e^2 (\bar{a} \cdot \bar{f}) (\bar{a} \cdot \bar{q}) + \frac{16 e^2 \chi g \chi p (\bar{a} \cdot \bar{f}) (\bar{a} \cdot \bar{q})}{\chi f \chi q} + 16 e^2 (\bar{a} \cdot \bar{g}) (\bar{a} \cdot \bar{q}) - \frac{16 e^2 \chi f \chi p (\bar{a} \cdot \bar{g}) (\bar{a} \cdot \bar{q})}{\chi g \chi q} + \\
& \quad \left. \frac{16 e^2 \chi f (\bar{a} \cdot \bar{p}) (\bar{a} \cdot \bar{q})}{\chi p} - \frac{16 e^2 \chi g (\bar{a} \cdot \bar{p}) (\bar{a} \cdot \bar{q})}{\chi p} + \frac{16 e^2 \chi f (\bar{a} \cdot \bar{p}) (\bar{a} \cdot \bar{q})}{\chi q} - \frac{16 e^2 \chi g (\bar{a} \cdot \bar{p}) (\bar{a} \cdot \bar{q})}{\chi q} \right) + \\
& \text{C1 D0} \left(\frac{8 \xi^2 \chi p m^4}{\chi f} - \frac{8 \xi^2 \chi p m^4}{\chi g} + \frac{8 \xi^2 \chi q m^4}{\chi f} - \frac{8 \xi^2 \chi q m^4}{\chi g} + \frac{8 \xi^2 \chi f m^4}{\chi p} - \frac{8 \xi^2 \chi g m^4}{\chi p} + \right. \\
& \quad \frac{8 \xi^2 \chi f m^4}{\chi q} - \frac{8 \xi^2 \chi g m^4}{\chi q} + \frac{8 \xi^2 \chi p (\bar{f} \cdot \bar{g}) m^2}{\chi f} - \frac{8 \xi^2 \chi p (\bar{f} \cdot \bar{g}) m^2}{\chi g} + \frac{8 \xi^2 \chi q (\bar{f} \cdot \bar{g}) m^2}{\chi f} - \\
& \quad \left. \frac{8 \xi^2 \chi q (\bar{f} \cdot \bar{g}) m^2}{\chi g} \right)
\end{aligned}$$

$$\begin{aligned}
& \frac{8 \xi^2 \chi q (\bar{f} \cdot \bar{g}) m^2}{\chi g} + 8 \xi^2 (\bar{f} \cdot \bar{p}) m^2 - \frac{8 \xi^2 \chi g \chi q (\bar{f} \cdot \bar{p}) m^2}{\chi f \chi p} + 8 \xi^2 (\bar{f} \cdot \bar{q}) m^2 - \frac{8 \xi^2 \chi g \chi p (\bar{f} \cdot \bar{q}) m^2}{\chi f \chi q} - \\
& 8 \xi^2 (\bar{g} \cdot \bar{p}) m^2 + \frac{8 \xi^2 \chi f \chi q (\bar{g} \cdot \bar{p}) m^2}{\chi g \chi p} - 8 \xi^2 (\bar{g} \cdot \bar{q}) m^2 + \frac{8 \xi^2 \chi f \chi p (\bar{g} \cdot \bar{q}) m^2}{\chi g \chi q} - \\
& \left. \frac{8 \xi^2 \chi f (\bar{p} \cdot \bar{q}) m^2}{\chi p} + \frac{8 \xi^2 \chi g (\bar{p} \cdot \bar{q}) m^2}{\chi p} - \frac{8 \xi^2 \chi f (\bar{p} \cdot \bar{q}) m^2}{\chi q} + \frac{8 \xi^2 \chi g (\bar{p} \cdot \bar{q}) m^2}{\chi q} \right) \Bigg) + \\
A0 B1 & \left(C2 D1 \left(\frac{8 m^4 \chi p \xi^4}{\chi f} - \frac{8 m^4 \chi p \xi^4}{\chi g} + \frac{8 m^4 \chi q \xi^4}{\chi f} - \frac{8 m^4 \chi q \xi^4}{\chi g} \right) + C2 D0 \left(-\frac{8 e m^2 \chi p (\bar{a} \cdot \bar{f}) \xi^2}{\chi f} - \right. \right. \\
& \left. \frac{8 e m^2 \chi q (\bar{a} \cdot \bar{f}) \xi^2}{\chi f} - \frac{8 e m^2 \chi p (\bar{a} \cdot \bar{g}) \xi^2}{\chi g} - \frac{8 e m^2 \chi q (\bar{a} \cdot \bar{g}) \xi^2}{\chi g} + 16 e m^2 (\bar{a} \cdot \bar{p}) \xi^2 + 16 e m^2 (\bar{a} \cdot \bar{q}) \xi^2 \right) + \\
C1 D1 & \left(-\frac{8 e m^2 \chi p (\bar{a} \cdot \bar{f}) \xi^2}{\chi f} + \frac{8 e m^2 \chi p (\bar{a} \cdot \bar{f}) \xi^2}{\chi g} - \frac{8 e m^2 \chi q (\bar{a} \cdot \bar{f}) \xi^2}{\chi f} + \frac{8 e m^2 \chi q (\bar{a} \cdot \bar{f}) \xi^2}{\chi g} + \right. \\
& \frac{8 e m^2 \chi p (\bar{a} \cdot \bar{g}) \xi^2}{\chi f} - \frac{8 e m^2 \chi p (\bar{a} \cdot \bar{g}) \xi^2}{\chi g} + \frac{8 e m^2 \chi q (\bar{a} \cdot \bar{g}) \xi^2}{\chi f} - \frac{8 e m^2 \chi q (\bar{a} \cdot \bar{g}) \xi^2}{\chi g} + \\
& 16 e m^2 (\bar{a} \cdot \bar{p}) \xi^2 - \frac{8 e m^2 \chi g \chi q (\bar{a} \cdot \bar{p}) \xi^2}{\chi f \chi p} - \frac{8 e m^2 \chi f \chi q (\bar{a} \cdot \bar{p}) \xi^2}{\chi g \chi p} + \\
& \left. 16 e m^2 (\bar{a} \cdot \bar{q}) \xi^2 - \frac{8 e m^2 \chi g \chi p (\bar{a} \cdot \bar{q}) \xi^2}{\chi f \chi q} - \frac{8 e m^2 \chi f \chi p (\bar{a} \cdot \bar{q}) \xi^2}{\chi g \chi q} \right) + \\
C0 D0 & \left(16 e (\bar{a} \cdot \bar{p}) m^2 - \frac{16 e \chi q (\bar{a} \cdot \bar{p}) m^2}{\chi p} + 16 e (\bar{a} \cdot \bar{q}) m^2 - \frac{16 e \chi p (\bar{a} \cdot \bar{q}) m^2}{\chi q} + \right. \\
& 16 e (\bar{a} \cdot \bar{g}) (\bar{f} \cdot \bar{p}) - \frac{16 e \chi g (\bar{a} \cdot \bar{q}) (\bar{f} \cdot \bar{p})}{\chi q} + 16 e (\bar{a} \cdot \bar{g}) (\bar{f} \cdot \bar{q}) - \frac{16 e \chi g (\bar{a} \cdot \bar{p}) (\bar{f} \cdot \bar{q})}{\chi p} + \\
& \left. 16 e (\bar{a} \cdot \bar{f}) (\bar{g} \cdot \bar{p}) - \frac{16 e \chi f (\bar{a} \cdot \bar{q}) (\bar{g} \cdot \bar{p})}{\chi q} + 16 e (\bar{a} \cdot \bar{f}) (\bar{g} \cdot \bar{q}) - \frac{16 e \chi f (\bar{a} \cdot \bar{p}) (\bar{g} \cdot \bar{q})}{\chi p} \right) + \\
C1 D0 & \left(-\frac{8 \xi^2 \chi p m^4}{\chi f} + \frac{8 \xi^2 \chi p m^4}{\chi g} - \frac{8 \xi^2 \chi q m^4}{\chi f} + \frac{8 \xi^2 \chi q m^4}{\chi g} - \frac{8 \xi^2 \chi f m^4}{\chi p} + \frac{8 \xi^2 \chi g m^4}{\chi p} - \right. \\
& \frac{8 \xi^2 \chi f m^4}{\chi q} + \frac{8 \xi^2 \chi g m^4}{\chi q} - \frac{8 \xi^2 \chi p (\bar{f} \cdot \bar{g}) m^2}{\chi f} + \frac{8 \xi^2 \chi p (\bar{f} \cdot \bar{g}) m^2}{\chi g} - \frac{8 \xi^2 \chi q (\bar{f} \cdot \bar{g}) m^2}{\chi f} + \\
& \frac{8 \xi^2 \chi q (\bar{f} \cdot \bar{g}) m^2}{\chi g} + 8 \xi^2 (\bar{f} \cdot \bar{p}) m^2 + \frac{8 \xi^2 \chi g \chi q (\bar{f} \cdot \bar{p}) m^2}{\chi f \chi p} + 8 \xi^2 (\bar{f} \cdot \bar{q}) m^2 + \frac{8 \xi^2 \chi g \chi p (\bar{f} \cdot \bar{q}) m^2}{\chi f \chi q} - \\
& 8 \xi^2 (\bar{g} \cdot \bar{p}) m^2 - \frac{8 \xi^2 \chi f \chi q (\bar{g} \cdot \bar{p}) m^2}{\chi g \chi p} - 8 \xi^2 (\bar{g} \cdot \bar{q}) m^2 - \frac{8 \xi^2 \chi f \chi p (\bar{g} \cdot \bar{q}) m^2}{\chi g \chi q} + \\
& \frac{8 \xi^2 \chi f (\bar{p} \cdot \bar{q}) m^2}{\chi p} - \frac{8 \xi^2 \chi g (\bar{p} \cdot \bar{q}) m^2}{\chi p} + \frac{8 \xi^2 \chi f (\bar{p} \cdot \bar{q}) m^2}{\chi q} - \frac{8 \xi^2 \chi g (\bar{p} \cdot \bar{q}) m^2}{\chi q} - \\
& \frac{16 e^2 \chi p (\bar{a} \cdot \bar{f}) (\bar{a} \cdot \bar{g})}{\chi f} + \frac{16 e^2 \chi p (\bar{a} \cdot \bar{f}) (\bar{a} \cdot \bar{g})}{\chi g} - \frac{16 e^2 \chi q (\bar{a} \cdot \bar{f}) (\bar{a} \cdot \bar{g})}{\chi f} + \frac{16 e^2 \chi q (\bar{a} \cdot \bar{f}) (\bar{a} \cdot \bar{g})}{\chi g} - \\
& 16 e^2 (\bar{a} \cdot \bar{f}) (\bar{a} \cdot \bar{p}) + \frac{16 e^2 \chi g \chi q (\bar{a} \cdot \bar{f}) (\bar{a} \cdot \bar{p})}{\chi f \chi p} + 16 e^2 (\bar{a} \cdot \bar{g}) (\bar{a} \cdot \bar{p}) - \frac{16 e^2 \chi f \chi q (\bar{a} \cdot \bar{g}) (\bar{a} \cdot \bar{p})}{\chi g \chi p} -
\end{aligned}$$

$$\begin{aligned}
& 16 e^2 (\bar{a} \cdot \bar{f}) (\bar{a} \cdot \bar{q}) + \frac{16 e^2 \chi g \chi p (\bar{a} \cdot \bar{f}) (\bar{a} \cdot \bar{q})}{\chi f \chi q} + 16 e^2 (\bar{a} \cdot \bar{g}) (\bar{a} \cdot \bar{q}) - \frac{16 e^2 \chi f \chi p (\bar{a} \cdot \bar{g}) (\bar{a} \cdot \bar{q})}{\chi g \chi q} + \\
& \left. \frac{16 e^2 \chi f (\bar{a} \cdot \bar{p}) (\bar{a} \cdot \bar{q})}{\chi p} - \frac{16 e^2 \chi g (\bar{a} \cdot \bar{p}) (\bar{a} \cdot \bar{q})}{\chi p} + \frac{16 e^2 \chi f (\bar{a} \cdot \bar{p}) (\bar{a} \cdot \bar{q})}{\chi q} - \frac{16 e^2 \chi g (\bar{a} \cdot \bar{p}) (\bar{a} \cdot \bar{q})}{\chi q} \right) + \\
& \text{C0 D1} \left(\frac{8 \xi^2 \chi p m^4}{\chi f} - \frac{8 \xi^2 \chi p m^4}{\chi g} + \frac{8 \xi^2 \chi q m^4}{\chi f} - \frac{8 \xi^2 \chi q m^4}{\chi g} + \frac{8 \xi^2 \chi f m^4}{\chi p} - \frac{8 \xi^2 \chi g m^4}{\chi p} + \right. \\
& \frac{8 \xi^2 \chi f m^4}{\chi q} - \frac{8 \xi^2 \chi g m^4}{\chi q} + \frac{8 \xi^2 \chi p (\bar{f} \cdot \bar{g}) m^2}{\chi f} - \frac{8 \xi^2 \chi p (\bar{f} \cdot \bar{g}) m^2}{\chi g} + \frac{8 \xi^2 \chi q (\bar{f} \cdot \bar{g}) m^2}{\chi f} - \\
& \frac{8 \xi^2 \chi q (\bar{f} \cdot \bar{g}) m^2}{\chi g} + 8 \xi^2 (\bar{f} \cdot \bar{p}) m^2 - \frac{8 \xi^2 \chi g \chi q (\bar{f} \cdot \bar{p}) m^2}{\chi f \chi p} + 8 \xi^2 (\bar{f} \cdot \bar{q}) m^2 - \frac{8 \xi^2 \chi g \chi p (\bar{f} \cdot \bar{q}) m^2}{\chi f \chi q} - \\
& 8 \xi^2 (\bar{g} \cdot \bar{p}) m^2 + \frac{8 \xi^2 \chi f \chi q (\bar{g} \cdot \bar{p}) m^2}{\chi g \chi p} - 8 \xi^2 (\bar{g} \cdot \bar{q}) m^2 + \frac{8 \xi^2 \chi f \chi p (\bar{g} \cdot \bar{q}) m^2}{\chi g \chi q} - \\
& \left. \frac{8 \xi^2 \chi f (\bar{p} \cdot \bar{q}) m^2}{\chi p} + \frac{8 \xi^2 \chi g (\bar{p} \cdot \bar{q}) m^2}{\chi p} - \frac{8 \xi^2 \chi f (\bar{p} \cdot \bar{q}) m^2}{\chi q} + \frac{8 \xi^2 \chi g (\bar{p} \cdot \bar{q}) m^2}{\chi q} \right) + \\
& \text{A0 B2} \left(-16 \text{C2 D0} m^4 \xi^4 + \text{C1 D0} \left(16 e m^2 (\bar{a} \cdot \bar{f}) \xi^2 - 16 e m^2 (\bar{a} \cdot \bar{g}) \xi^2 - \frac{8 e m^2 \chi f (\bar{a} \cdot \bar{p}) \xi^2}{\chi p} + \right. \right. \\
& \left. \frac{8 e m^2 \chi g (\bar{a} \cdot \bar{p}) \xi^2}{\chi p} - \frac{8 e m^2 \chi f (\bar{a} \cdot \bar{q}) \xi^2}{\chi q} + \frac{8 e m^2 \chi g (\bar{a} \cdot \bar{q}) \xi^2}{\chi q} \right) + \\
& \text{C0 D0} \left(-16 \xi^2 m^4 + \frac{16 \xi^2 \chi f \chi g m^4}{\chi p \chi q} - 16 \xi^2 (\bar{f} \cdot \bar{g}) m^2 + \frac{8 \xi^2 \chi g (\bar{f} \cdot \bar{p}) m^2}{\chi p} + \frac{8 \xi^2 \chi g (\bar{f} \cdot \bar{q}) m^2}{\chi q} + \right. \\
& \left. \frac{8 \xi^2 \chi f (\bar{g} \cdot \bar{p}) m^2}{\chi p} + \frac{8 \xi^2 \chi f (\bar{g} \cdot \bar{q}) m^2}{\chi q} - \frac{16 \xi^2 \chi f \chi g (\bar{p} \cdot \bar{q}) m^2}{\chi p \chi q} \right) + \\
& \text{A2 B0} \left(-16 \text{C0 D2} m^4 \xi^4 + \text{C0 D1} \left(16 e m^2 (\bar{a} \cdot \bar{f}) \xi^2 - 16 e m^2 (\bar{a} \cdot \bar{g}) \xi^2 - \frac{8 e m^2 \chi f (\bar{a} \cdot \bar{p}) \xi^2}{\chi p} + \right. \right. \\
& \left. \frac{8 e m^2 \chi g (\bar{a} \cdot \bar{p}) \xi^2}{\chi p} - \frac{8 e m^2 \chi f (\bar{a} \cdot \bar{q}) \xi^2}{\chi q} + \frac{8 e m^2 \chi g (\bar{a} \cdot \bar{q}) \xi^2}{\chi q} \right) + \\
& \text{C0 D0} \left(-16 \xi^2 m^4 + \frac{16 \xi^2 \chi f \chi g m^4}{\chi p \chi q} - 16 \xi^2 (\bar{f} \cdot \bar{g}) m^2 + \frac{8 \xi^2 \chi g (\bar{f} \cdot \bar{p}) m^2}{\chi p} + \frac{8 \xi^2 \chi g (\bar{f} \cdot \bar{q}) m^2}{\chi q} + \right. \\
& \left. \frac{8 \xi^2 \chi f (\bar{g} \cdot \bar{p}) m^2}{\chi p} + \frac{8 \xi^2 \chi f (\bar{g} \cdot \bar{q}) m^2}{\chi q} - \frac{16 \xi^2 \chi f \chi g (\bar{p} \cdot \bar{q}) m^2}{\chi p \chi q} \right) + \\
& \text{A1 B1} \left(\text{C1 D1} \left(-32 m^4 \xi^4 + \frac{8 m^4 \chi g \chi q \xi^4}{\chi f \chi p} + \frac{8 m^4 \chi f \chi q \xi^4}{\chi g \chi p} + \frac{8 m^4 \chi g \chi p \xi^4}{\chi f \chi q} + \frac{8 m^4 \chi f \chi p \xi^4}{\chi g \chi q} \right) + \right. \\
& \text{C1 D0} \left(16 e m^2 (\bar{a} \cdot \bar{f}) \xi^2 - \frac{8 e m^2 \chi g \chi q (\bar{a} \cdot \bar{f}) \xi^2}{\chi f \chi p} - \frac{8 e m^2 \chi g \chi p (\bar{a} \cdot \bar{f}) \xi^2}{\chi f \chi q} - \right. \\
& 16 e m^2 (\bar{a} \cdot \bar{g}) \xi^2 + \frac{8 e m^2 \chi f \chi q (\bar{a} \cdot \bar{g}) \xi^2}{\chi g \chi p} + \frac{8 e m^2 \chi f \chi p (\bar{a} \cdot \bar{g}) \xi^2}{\chi g \chi q} - \\
& \left. \frac{8 e m^2 \chi f (\bar{a} \cdot \bar{p}) \xi^2}{\chi p} + \frac{8 e m^2 \chi g (\bar{a} \cdot \bar{p}) \xi^2}{\chi p} - \frac{8 e m^2 \chi f (\bar{a} \cdot \bar{p}) \xi^2}{\chi q} + \frac{8 e m^2 \chi g (\bar{a} \cdot \bar{p}) \xi^2}{\chi q} \right) -
\end{aligned}$$

$$\begin{aligned}
& \left. \frac{8 e m^2 \chi f (\bar{a} \cdot \bar{q}) \xi^2}{\chi p} + \frac{8 e m^2 \chi g (\bar{a} \cdot \bar{q}) \xi^2}{\chi p} - \frac{8 e m^2 \chi f (\bar{a} \cdot \bar{q}) \xi^2}{\chi q} + \frac{8 e m^2 \chi g (\bar{a} \cdot \bar{q}) \xi^2}{\chi q} \right) + \\
\text{C0 D1} & \left(16 e m^2 (\bar{a} \cdot \bar{f}) \xi^2 - \frac{8 e m^2 \chi g \chi q (\bar{a} \cdot \bar{f}) \xi^2}{\chi f \chi p} - \frac{8 e m^2 \chi g \chi p (\bar{a} \cdot \bar{f}) \xi^2}{\chi f \chi q} - 16 e m^2 (\bar{a} \cdot \bar{g}) \xi^2 + \right. \\
& \frac{8 e m^2 \chi f \chi q (\bar{a} \cdot \bar{g}) \xi^2}{\chi g \chi p} + \frac{8 e m^2 \chi f \chi p (\bar{a} \cdot \bar{g}) \xi^2}{\chi g \chi q} - \frac{8 e m^2 \chi f (\bar{a} \cdot \bar{p}) \xi^2}{\chi p} + \\
& \frac{8 e m^2 \chi g (\bar{a} \cdot \bar{p}) \xi^2}{\chi p} - \frac{8 e m^2 \chi f (\bar{a} \cdot \bar{p}) \xi^2}{\chi q} + \frac{8 e m^2 \chi g (\bar{a} \cdot \bar{p}) \xi^2}{\chi q} - \frac{8 e m^2 \chi f (\bar{a} \cdot \bar{q}) \xi^2}{\chi p} + \\
& \left. \frac{8 e m^2 \chi g (\bar{a} \cdot \bar{q}) \xi^2}{\chi q} - \frac{8 e m^2 \chi f (\bar{a} \cdot \bar{q}) \xi^2}{\chi q} + \frac{8 e m^2 \chi g (\bar{a} \cdot \bar{q}) \xi^2}{\chi q} \right) + \text{C0 D0} \\
& \left(\frac{16 \xi^2 \chi q m^4}{\chi p} + \frac{16 \xi^2 \chi p m^4}{\chi q} - \frac{32 \xi^2 \chi f \chi g m^4}{\chi p \chi q} + 32 \xi^2 (\bar{f} \cdot \bar{g}) m^2 - \frac{16 \xi^2 \chi g (\bar{f} \cdot \bar{p}) m^2}{\chi p} + \frac{16 \xi^2 \chi g (\bar{f} \cdot \bar{p}) m^2}{\chi q} + \right. \\
& \frac{16 \xi^2 \chi g (\bar{f} \cdot \bar{q}) m^2}{\chi p} - \frac{16 \xi^2 \chi g (\bar{f} \cdot \bar{q}) m^2}{\chi q} - \frac{16 \xi^2 \chi f (\bar{g} \cdot \bar{p}) m^2}{\chi p} + \frac{16 \xi^2 \chi f (\bar{g} \cdot \bar{p}) m^2}{\chi q} + \frac{16 \xi^2 \chi f (\bar{g} \cdot \bar{q}) m^2}{\chi p} - \\
& \frac{16 \xi^2 \chi f (\bar{g} \cdot \bar{q}) m^2}{\chi q} + \frac{32 \xi^2 \chi f \chi g (\bar{p} \cdot \bar{q}) m^2}{\chi p \chi q} + 64 e^2 (\bar{a} \cdot \bar{f}) (\bar{a} \cdot \bar{g}) - \frac{32 e^2 \chi g (\bar{a} \cdot \bar{f}) (\bar{a} \cdot \bar{p})}{\chi p} - \\
& \left. \frac{32 e^2 \chi f (\bar{a} \cdot \bar{g}) (\bar{a} \cdot \bar{p})}{\chi p} - \frac{32 e^2 \chi g (\bar{a} \cdot \bar{f}) (\bar{a} \cdot \bar{q})}{\chi q} - \frac{32 e^2 \chi f (\bar{a} \cdot \bar{g}) (\bar{a} \cdot \bar{q})}{\chi q} + \frac{64 e^2 \chi f \chi g (\bar{a} \cdot \bar{p}) (\bar{a} \cdot \bar{q})}{\chi p \chi q} \right)
\end{aligned}$$

Step 5: Passing to invariant variables ρ_1 , τ_1 , ρ_2 , τ_2 in the integral

In the following step, we consider only the integrand, namely, $\text{Coeff} * \text{Expr}$. We will introduce the Jacobian of the tranformation later

Recall the conservation laws

$$p + sk = q + l$$

$$\chi p = \chi q + \chi l$$

Variables ρ_1 , τ_1 read

$$\begin{aligned}
\rho_1 &= \frac{e(qFp)}{\xi m^4 \chi l} = - \frac{e(lFp)}{\xi m^4 \chi l} = \frac{e(qFl)}{\xi m^4 \chi l} \\
\tau_1 &= - \frac{e(qF^* p)}{m^4 \chi l} = \frac{e(lF^* p)}{m^4 \chi l} = \frac{e(qF^* l)}{m^4 \chi l}
\end{aligned}$$

Similarly, we have ρ_2 and τ_2

$$t k + l = f + g$$

$$\chi l = \chi f + \chi g$$

$$\rho_2 = -\frac{e (gFf)}{\xi m^4 \chi l} = -\frac{e (gFl)}{\xi m^4 \chi l} = \frac{e (fFl)}{\xi m^4 \chi l}$$

$$\tau_2 = -\frac{e (gF^* f)}{m^4 \chi l} = -\frac{e (gF^* l)}{m^4 \chi l} = \frac{e (fF^* l)}{m^4 \chi l}$$

$$-\infty < \rho_{1,2}, \tau_{1,2} < +\infty$$

Also, notice that

$$(q F^2 p) = -a^2 k p k q$$

Expr contains contractions of momenta in pairs

$(pq), (pf), (pg), (qf), (qg), (fg)$

Let us express all such contractions through ρ_1 ,

τ_1, ρ_2, τ_2 and the χ parameters

The key idea is projecting a given 4 – momentum on a basis spanned by the field tensor and another 4 – momentum :

$$q^\mu = C_1 p^\mu + C_2 (Fp)^\mu + C_3 (F^* p)^\mu + C_4 (FFp)^\mu$$

$$C_1 = \frac{\chi_q}{\chi_p}$$

$$C_2 = -\frac{e^2 (qFp)}{m^6 \chi_p^2}$$

$$C_3 = -\frac{e^2 (qF^* p)}{m^6 \chi_p^2}$$

$$C_4 = \frac{e^2}{2 m^4 \chi_p \chi_q} \left(1 - \frac{\chi_q^2}{\chi_p^2} + \frac{e^2 (qFp)^2}{m^8 \chi_p^2} + \frac{e^2 (qF^* p)^2}{m^8 \chi_p^2} \right)$$

This projection is specific to a crossed field,

as in such field $[F^{n>2}]^{\mu\nu} = [F^* F]^{\mu\nu} = 0$

(a non – crossed field may have higher – order terms in powers of F)

Then we substitute the projected vector to

the initial scalar product. Terms such as (qFp) are re – expressed via invariant parameters

Note that expressing combinations such as (qg) via

the invariant parameters ρ_n, τ_n requires an intermediate step. The issue is that q and g enter different vertices,

and their projection results in appearance of combinations such as (pFg) , which do not enter the definitions of ρ_n , τ_n . We use that the vertices are connected by l .

Command **ExpandVector**[**p, q, μ**] is defined in definitions.nb. It allows projecting 4 – vector p on the basis spanned by q and F , and the resulting vector carries index μ ;
See the illustrating example below

Each projection is stored in a function of the Lorentz index μ , given by command

qpexpansion[μ]

where q is the vector being projected and p – the momentum spanning the basis

When defining a projection, we also introduce related substitutions of scalar combinations such as (qFp) , that reexpress them via invariant ρ_n , τ_n ;

These substitutions are stored in arrays, named, for example, as $\text{subs}_{\rho l \tau l qp}$, so that the last two letters are attributed to the momenta in the same order as in 'qpexpansion'

In[40]:= (**example**)

ExpandVector[**p, q, μ**]

$$\text{Out[40]} = \frac{e^2 FFp^\mu (e^2 (FDp \cdot q)^2 + e^2 (Fp \cdot q)^2 + m^8 \chi p^2 - m^8 \chi q^2)}{2 m^{12} \chi p^3 \chi q} - \frac{e^2 FDp^\mu (FDp \cdot q)}{m^6 \chi p^2} - \frac{e^2 Fp^\mu (Fp \cdot q)}{m^6 \chi p^2} + \frac{\chi q p^\mu}{\chi p}$$

(pq)

– project q on p

– contract with p , reexpress via ρ_n , τ_n

```
In[41]:= subs $\rho$ 1 $\tau$ 1qp = {Pair[Momentum[Fp, D], Momentum[q, D]]  $\rightarrow$   $\rho$ 1  $\xi$  m^4  $\chi$ l / e,
  Pair[Momentum[FDp, D], Momentum[q, D]]  $\rightarrow$  - $\tau$ 1 m^4  $\chi$ l / e}
qpexpansion[ $\mu$ _] = Collect[ExpandVector[p, q,  $\mu$ ] /. subs $\rho$ 1 $\tau$ 1qp,
  {pv[ $\mu$ ], Fpv[ $\mu$ ], FDPv[ $\mu$ ], FFpv[ $\mu$ ]}, Simplify]
pq = Contract[pv[ $\mu$ ]  $\times$  qv[ $\mu$ ] /. {qv[ $\mu$ ]  $\rightarrow$  qpexpansion[ $\mu$ ]}
```

$$\text{Out[41]} = \left\{ \text{Fp} \cdot q \rightarrow \frac{m^4 \xi \rho^1 \chi^1}{e}, \text{FDp} \cdot q \rightarrow -\frac{m^4 \tau^1 \chi^1}{e} \right\}$$

$$\text{Out[42]} = \frac{e^2 \text{FFp}^\mu (\xi^2 \rho^1 \chi^1 + \tau^1 \chi^1 + \chi p^2 - \chi q^2)}{2 m^4 \chi p^3 \chi q} + \frac{e \tau^1 \chi^1 \text{FDp}^\mu}{m^2 \chi p^2} - \frac{e \xi \rho^1 \chi^1 \text{Fp}^\mu}{m^2 \chi p^2} + \frac{\chi q p^\mu}{\chi p}$$

$$\text{Out[43]} = \frac{m^2 (\xi^2 \rho^1 \chi^1 + \tau^1 \chi^1 + \chi p^2 - \chi q^2)}{2 \chi p \chi q} + \frac{m^2 \chi q}{\chi p}$$

(lp), required later as an intermediate step
 – project l on p
 – contract with p, reexpress via ρ_n , τ_n

```
In[44]:= subs $\rho$ 1 $\tau$ 1lp = {Pair[Momentum[Fp, D], Momentum[l, D]]  $\rightarrow$  - $\rho$ 1  $\xi$  m^4  $\chi$ l / e,
  Pair[Momentum[FDp, D], Momentum[l, D]]  $\rightarrow$   $\tau$ 1 m^4  $\chi$ l / e,
  Pair[Momentum[FFp, D], Momentum[l, D]]  $\rightarrow$  -av2 kl kp} /.
  ktochiSubs[p, q, f, g, l] /. {av2  $\rightarrow$  -m^2  $\xi$ ^2 / e^2}
lpexpansion[ $\mu$ _] = Collect[ExpandVector[p, l,  $\mu$ ] /. subs $\rho$ 1 $\tau$ 1lp,
  {lv[ $\mu$ ], Flv[ $\mu$ ], FDlv[ $\mu$ ], FFlv[ $\mu$ ]}, Simplify]
lp = Contract[pv[ $\mu$ ]  $\times$  lv[ $\mu$ ] /. {lv[ $\mu$ ]  $\rightarrow$  lpexpansion[ $\mu$ ]}/. subs $\rho$ 1 $\tau$ 1lp
```

$$\text{Out[44]} = \left\{ \text{Fp} \cdot l \rightarrow -\frac{m^4 \xi \rho^1 \chi^1}{e}, \text{FDp} \cdot l \rightarrow \frac{m^4 \tau^1 \chi^1}{e}, \text{FFp} \cdot l \rightarrow \frac{m^6 \chi^1 \chi p}{e^2} \right\}$$

$$\text{Out[45]} = \frac{e^2 \text{FFp}^\mu (\chi^1 (\xi^2 \rho^1 + \tau^1 - 1) + \chi p^2) - 2 e m^2 \tau^1 \chi^1 \chi p \text{FDp}^\mu + 2 m^2 \chi^1 \chi p (e \xi \rho^1 \text{Fp}^\mu + m^2 \chi p p^\mu)}{2 m^4 \chi^1 \chi p^3}$$

$$\text{Out[46]} = \frac{m^6 \chi p^2 (\chi^1 (\xi^2 \rho^1 + \tau^1 - 1) + \chi p^2) + 2 m^6 \chi^1 \chi p^2}{2 m^4 \chi^1 \chi p^3}$$

Projection of p on l,
 required later as an intermediate step (particularly, subs ρ 1 τ 1pl)
 Note that subs ρ 1 τ 1pl is not exactly
 the same as subs ρ 1 τ 1lp! We use both further

```
In[47]:= subsp1τ1pl = {Pair[Momentum[Fl, D], Momentum[p, D]] → ρ1 ξ m^4 χ l / e,
  Pair[Momentum[FDl, D], Momentum[p, D]] → -τ1 m^4 χ l / e,
  Pair[Momentum[FFl, D], Momentum[p, D]] → -av2 kl kp} /.
  ktochiSubs[p, q, f, g, l] /. {av2 → -m^2 ξ^2 / e^2}
plexpansion[μ_] = Collect[ExpandVector[l, p, μ] /. subsp1τ1pl,
  {lv[μ], Flv[μ], FDlv[μ], FFlv[μ]}, Simplify]
```

$$\text{Out[47]} = \left\{ \text{Fl} \cdot p \rightarrow \frac{m^4 \xi \rho_1 \chi l}{e}, \text{FDl} \cdot p \rightarrow -\frac{m^4 \tau_1 \chi l}{e}, \text{FFl} \cdot p \rightarrow \frac{m^6 \chi l \chi p}{e^2} \right\}$$

$$\text{Out[48]} = \frac{e^2 \text{FFl}^\mu (\chi l^2 (\xi^2 \rho_1^2 + \tau_1^2 + 1) - \chi p^2)}{2 m^4 \chi l^3 \chi p} + \frac{e \tau_1 \text{FDl}^\mu}{m^2 \chi l} - \frac{e \xi \rho_1 \text{Fl}^\mu}{m^2 \chi l} + \frac{\chi p l^\mu}{\chi l}$$

(fg)
 -project g on f
 -contract with f, reexpress via ρ_n , τ_n

```
In[49]:= subsp2τ2gf = {Pair[Momentum[Ff, D], Momentum[g, D]] → -ρ2 ξ m^4 χ l / e,
  Pair[Momentum[FDf, D], Momentum[g, D]] → -τ2 m^4 χ l / e}
gfexpansion[μ_] = Collect[ExpandVector[f, g, μ] /. subsp2τ2gf,
  {fv[μ], Ffv[μ], FDfv[μ], FFFv[μ]}, Simplify]
fg = Contract[gfexpansion[μ] × fv[μ]]
```

$$\text{Out[49]} = \left\{ \text{Ff} \cdot g \rightarrow -\frac{m^4 \xi \rho_2 \chi l}{e}, \text{FDf} \cdot g \rightarrow -\frac{m^4 \tau_2 \chi l}{e} \right\}$$

$$\text{Out[50]} = \frac{e^2 \text{FFF}^\mu (\chi l^2 (\xi^2 \rho_2^2 + \tau_2^2) + \chi f^2 - \chi g^2)}{2 m^4 \chi f^3 \chi g} + \frac{e \tau_2 \chi l \text{FDf}^\mu}{m^2 \chi f^2} + \frac{e \xi \rho_2 \chi l \text{Ff}^\mu}{m^2 \chi f^2} + \frac{\chi g f^\mu}{\chi f}$$

$$\text{Out[51]} = \frac{m^2 (\chi l^2 (\xi^2 \rho_2^2 + \tau_2^2) + \chi f^2 - \chi g^2)}{2 \chi f \chi g} + \frac{m^2 \chi g}{\chi f}$$

Projection of g on l, required later as an intermediate step

```
In[52]:= subsp2τ2gl = {Pair[Momentum[Fl, D], Momentum[g, D]] → -ρ2 ξ m^4 χ l / e,
  Pair[Momentum[FDl, D], Momentum[g, D]] → -τ2 m^4 χ l / e}
glexpansion[μ_] = Collect[ExpandVector[l, g, μ] /. subsp2τ2gl,
  {lv[μ], Flv[μ], FDlv[μ], FFlv[μ]}, Simplify]
```

$$\text{Out[52]} = \left\{ \text{Fl} \cdot g \rightarrow -\frac{m^4 \xi \rho_2 \chi l}{e}, \text{FDl} \cdot g \rightarrow -\frac{m^4 \tau_2 \chi l}{e} \right\}$$

$$\text{Out[53]} = -\frac{e^2 \text{FFl}^\mu (\chi g^2 - \chi l^2 (\xi^2 \rho_2^2 + \tau_2^2 + 1))}{2 m^4 \chi g \chi l^3} + \frac{e \tau_2 \text{FDl}^\mu}{m^2 \chi l} + \frac{e \xi \rho_2 \text{Fl}^\mu}{m^2 \chi l} + \frac{\chi g l^\mu}{\chi l}$$

(pg)
 -project g on l
 -contract with p, reexpress via ρ_n ,
 τ_n using also substitutions subsp1τ1pl

and expression for (lp) calculated above

```
In[54]:= gpexpansion[μ_] =
  Collect[ExpandVector[p, g, μ], {pv[μ], Fpv[μ], FDpv[μ], FFpv[μ]}, Simplify]
pg = Simplify[Contract[pv[μ] × gv[μ] /. {gv[μ] → glexpansion[μ]}] /. subsρ1τ1pl /.
  {Contract[pv[μ] × lv[μ]] → lp}]
gp = pg;
```

$$\text{Out[54]} = \frac{\text{FFp}^\mu (e^4 (\text{FDp} \cdot g)^2 + e^4 (\text{Fp} \cdot g)^2 + e^2 m^8 (\chi p^2 - \chi g^2))}{2 m^{12} \chi g \chi p^3} - \frac{e^2 \text{FDp}^\mu (\text{FDp} \cdot g)}{m^6 \chi p^2} - \frac{e^2 \text{Fp}^\mu (\text{Fp} \cdot g)}{m^6 \chi p^2} + \frac{\chi g p^\mu}{\chi p}$$

$$\text{Out[55]} = \frac{m^2 (\chi g^2 (\xi^2 \rho_1^2 + \tau_1^2 + 1) + 2 \chi g \chi p (\xi^2 \rho_1 \rho_2 - \tau_1 \tau_2) + \chi p^2 (\xi^2 \rho_2^2 + \tau_2^2 + 1))}{2 \chi g \chi p}$$

(qg)
 – project q on p (calculated above)
 – project g on l (calculated above)
 – contract, reexpress via ρ_n, τ_n
 – the result will contain (lp),
 which we calculated above and substitute here;
 – In the last step,
 we remove all terms containing $(F)^{n>2}$ and $(F^* F)$ as they are zero

```
In[57]:= qg = Simplify[
  Contract[qv[μ] × gv[μ] /. {qv[μ] → qpexpansion[μ], gv[μ] → glexpansion[μ]}] /.
    subsρ1τ1qp /. subsρ1τ1pl /. subsρ1τ1lp /. {Contract[pv[μ] × lv[μ]] →
      lp} /. {Pair[Momentum[FFp, D], Momentum[FFl, D]] → 0,
      Pair[Momentum[FFp, D], Momentum[FDl, D]] → 0,
      Pair[Momentum[FDp, D], Momentum[FFl, D]] → 0,
      Pair[Momentum[Fp, D], Momentum[FFl, D]] → 0,
      Pair[Momentum[FFp, D], Momentum[Ffl, D]] → 0,
      Pair[Momentum[Fp, D], Momentum[FDl, D]] → 0,
      Pair[Momentum[FDp, D], Momentum[Ffl, D]] → 0,
      Pair[Momentum[Fp, D], Momentum[Ffl, D]] → av2 kp kl,
      Pair[Momentum[FDp, D], Momentum[FDl, D]] → av2 kp kl} /.
    ktouchSubs[p, q, f, g, l] /. {av2 → -m^2 ξ^2 / e^2}]
gq = qg;
```

$$\text{Out[57]} = \frac{m^2 (\xi^2 (\rho_1 \chi g (\chi l + \chi q) + \rho_2 \chi p \chi q)^2 + \tau_1^2 \chi g^2 (\chi l + \chi q)^2 - 2 \tau_1 \tau_2 \chi g \chi p \chi q (\chi l + \chi q) + \chi p^2 ((\tau_2^2 + 1) \chi q^2 + \chi g^2))}{2 \chi g \chi p^2 \chi q}$$

Projection of f on l, required later as an intermediate step

```
In[59]:= subsp2τ2fl = {Pair[Momentum[Fl, D], Momentum[f, D]] → ρ2 ξ m^4 χl / e,
  Pair[Momentum[FDl, D], Momentum[f, D]] → τ2 m^4 χl / e,
  Pair[Momentum[FFl, D], Momentum[f, D]] → -av2 kl kf} /.
  ktochiSubs[p, q, f, g, l] /. {av2 → -m^2 ξ^2 / e^2}
flexpansion[μ_] = Collect[ExpandVector[l, f, μ] /. subsp2τ2fl,
  {lv[μ], Flv[μ], Fdlv[μ], FFlv[μ]}, Simplify]
```

$$\text{Out[59]} = \left\{ f \cdot Fl \rightarrow \frac{m^4 \xi \rho^2 \chi^l}{e}, f \cdot FDI \rightarrow \frac{m^4 \tau^2 \chi^l}{e}, f \cdot FFI \rightarrow \frac{m^6 \chi^f \chi^l}{e^2} \right\}$$

$$\text{Out[60]} = -\frac{e^2 FFI^\mu (\chi^{f^2} - \chi^{l^2} (\xi^2 \rho^2 + \tau^2 + 1))}{2 m^4 \chi^f \chi^{l^3}} - \frac{e \tau^2 FDI^\mu}{m^2 \chi^l} - \frac{e \xi \rho^2 FI^\mu}{m^2 \chi^l} + \frac{\chi^f l^\mu}{\chi^l}$$

(pf)
 -project f on l
 -contract, reexpress via ρ_n, τ_n ,
 also using subsp1τ1pl and scalar product (lp) calculated before

```
In[61]:= fpexpansion[μ_] =
  Collect[ExpandVector[p, f, μ], {pv[μ], Fpv[μ], FDpv[μ], FFpv[μ]}, Simplify]
pf = Simplify[Contract[pv[μ] × fv[μ] /. {fv[μ] → flexpansion[μ]}] /. subsp1τ1pl /.
  {Contract[pv[μ] × lv[μ]] → lp}]
fp = pf;
```

$$\text{Out[61]} = \frac{FFp^\mu (e^4 (f \cdot FDp)^2 + e^4 (f \cdot Fp)^2 + e^2 m^8 (\chi p^2 - \chi^f))}{2 m^{12} \chi^f \chi p^3} - \frac{e^2 FDp^\mu (f \cdot FDp)}{m^6 \chi p^2} - \frac{e^2 Fp^\mu (f \cdot Fp)}{m^6 \chi p^2} + \frac{\chi^f p^\mu}{\chi p}$$

$$\text{Out[62]} = \frac{m^2 (\chi^{f^2} (\xi^2 \rho^2 + \tau^2 + 1) + \chi^f (2 \tau^1 \tau^2 \chi p - 2 \xi^2 \rho^1 \rho^2 \chi p) + \chi p^2 (\xi^2 \rho^2 + \tau^2 + 1))}{2 \chi^f \chi p}$$

(qf)
 -project q on p (calculated above)
 -project f on l (calculated above)
 -contract, reexpress via ρ_n, τ_n ,
 -the result will contain (lp),
 which we calculated above and substitute here;
 -In the last step,
 we remove all terms containing $(F)^{n>2}$ and $(F^* F)$ as they are zero

```

In[64]:= qf = Simplify[
  Contract[qv[μ] × fv[μ] /. {qv[μ] → qpexpansion[μ], fv[μ] → flexpansion[μ]}] /.
    subsρ1τ1qp /. subsρ1τ1pl /. subsρ1τ1lp /. {Contract[pv[μ] × lv[μ] →
      lp} /. {Pair[Momentum[FFp, D], Momentum[FFl, D]] → 0,
    Pair[Momentum[FFp, D], Momentum[FDl, D]] → 0,
    Pair[Momentum[FDp, D], Momentum[FFl, D]] → 0,
    Pair[Momentum[Fp, D], Momentum[FFl, D]] → 0,
    Pair[Momentum[FFp, D], Momentum[Ff, D]] → 0,
    Pair[Momentum[Fp, D], Momentum[FDl, D]] → 0,
    Pair[Momentum[FDp, D], Momentum[Ff, D]] → 0,
    Pair[Momentum[Fp, D], Momentum[Ff, D]] → av2 kp kl,
    Pair[Momentum[FDp, D], Momentum[FDl, D]] → av2 kp kl} /.
    ktochiSubs[p, q, f, g, l] /. {av2 → -m^2 ξ^2 / e^2}]
fq = qf;
Out[64]= 
$$\frac{m^2 (\xi^2 (\rho^2 \chi p \chi q - \rho^1 \chi f (\chi^1 + \chi q))^2 + \tau^1{}^2 \chi f^2 (\chi^1 + \chi q)^2 + 2 \tau^1 \tau^2 \chi f \chi p \chi q (\chi^1 + \chi q) + \chi p^2 ((\tau^2{}^2 + 1) \chi q^2 + \chi f^2))}{2 \chi f \chi p^2 \chi q}$$


```

All scalar products of the form

(aq), (al), ...

can be reduced to only two : (ap) or (af) plus invariant terms,

This is achieved again by projecting 4 –

momenta on the basis spanned by F and p or f

$$q^\mu = C_1 p^\mu + C_2 (Fp)^\mu + C_3 (F^* p)^\mu + C_4 (FFp)^\mu$$

If done correctly,

all terms containing (ap) and (af) in variable Expr will cancel ,
and the expression will depend only on gauge – invariant parameters

The result is stored in Expr4

```

In[66]:= aqsubs = {Pair[Momentum[a, D], Momentum[q, D]] → Contract[av[μ] × qpexpansion[μ]] /.
    ktochiSubs[p, q, f, g, l] /. {av2 → -m^2 ξ^2 / e^2}}
alsubs = {Pair[Momentum[a, D], Momentum[l, D]] → Contract[av[μ] × lpexpansion[μ]] /.
    ktochiSubs[p, q, f, g, l] /. {av2 → -m^2 ξ^2 / e^2}}
agsubs = {Pair[Momentum[a, D], Momentum[g, D]] → Contract[av[μ] × glexpansion[μ]] /.
    alsubs /. ktochiSubs[p, q, f, g, l] /. {av2 → -m^2 ξ^2 / e^2}}
afsubs = {Pair[Momentum[a, D], Momentum[f, D]] → Contract[av[μ] × flxpansion[μ]] /.
    alsubs /. ktochiSubs[p, q, f, g, l] /. {av2 → -m^2 ξ^2 / e^2}}

```

$$\text{Out[66]} = \left\{ a \cdot q \rightarrow \frac{\chi q (a \cdot p)}{\chi p} - \frac{m^2 \xi^2 \rho 1 \chi 1}{e \chi p} \right\}$$

$$\text{Out[67]} = \left\{ a \cdot l \rightarrow \frac{\chi 1 \left(m^2 \chi p (a \cdot p) + \frac{m^4 \xi^2 \rho 1 \chi p}{e} \right)}{m^2 \chi p^2} \right\}$$

$$\text{Out[68]} = \left\{ a \cdot g \rightarrow \frac{\chi g \left(m^2 \chi p (a \cdot p) + \frac{m^4 \xi^2 \rho 1 \chi p}{e} \right)}{m^2 \chi p^2} + \frac{m^2 \xi^2 \rho 2}{e} \right\}$$

$$\text{Out[69]} = \left\{ a \cdot f \rightarrow \frac{\chi f \left(m^2 \chi p (a \cdot p) + \frac{m^4 \xi^2 \rho 1 \chi p}{e} \right)}{m^2 \chi p^2} - \frac{m^2 \xi^2 \rho 2}{e} \right\}$$

```

In[70]:= Expr3 /. aqsubs /. agsubs /. afsubs /. {Pair[Momentum[p, D], Momentum[q, D]] → pq,
    Pair[Momentum[p, D], Momentum[f, D]] → pf,
    Pair[Momentum[p, D], Momentum[g, D]] → pg,
    Pair[Momentum[q, D], Momentum[f, D]] → qf,
    Pair[Momentum[q, D], Momentum[g, D]] → qg,
    Pair[Momentum[f, D], Momentum[g, D]] → fg};
Expand[%];
Expr4 = SortTermsSimplify[Contract[%] /. ktochiSubs[p, q, f, g] /.
    {av2 → -m^2 ξ^2 / e^2} (* /. {χq→χp-χl, χg→χl-χg} *) ];
% /. {D → 4}

```

$$\begin{aligned} \text{Out[73]} = & \frac{16 A2 B2 C0 D0 m^4 \chi f \chi g \xi^4}{\chi p \chi q} + \\ & A1 B2 \left(\frac{8 C1 D0 m^4 (\chi f - \chi g) (\chi p + \chi q) \xi^4}{\chi p \chi q} + \frac{8 C0 D0 m^4 (\rho 2 (\chi f - \chi g) \chi p (\chi p + \chi q) + 2 \rho 1 \chi f \chi g (\chi 1 + \chi p + \chi q)) \xi^4}{\chi p^2 \chi q} \right) + \\ & A2 B1 \left(\frac{8 C0 D1 m^4 (\chi f - \chi g) (\chi p + \chi q) \xi^4}{\chi p \chi q} + \frac{8 C0 D0 m^4 (\rho 2 (\chi f - \chi g) \chi p (\chi p + \chi q) + 2 \rho 1 \chi f \chi g (\chi 1 + \chi p + \chi q)) \xi^4}{\chi p^2 \chi q} \right) + \\ & A0 B2 \left(-16 C2 D0 \xi^4 m^4 + \frac{8 C1 D0 \xi^4 (\rho 1 (\chi f - \chi g) (\chi 1 + 2 \chi q) - 4 \rho 2 \chi p \chi q) m^4}{\chi p \chi q} + \frac{1}{\chi f \chi g \chi p^2 \chi q} \right. \\ & \quad \left. 8 C0 D0 \xi^2 ((2 \rho 1^2 \chi f^2 (\chi 1 + \chi q) \chi g^2 + \rho 1 \rho 2 \chi f (\chi f - \chi g) \chi p (\chi 1 + 2 \chi q) \chi g + \rho 2^2 (\chi f^2 + \chi g^2 - \chi 1^2) \chi p^2 \chi q) \xi^2 + \right. \\ & \quad \left. 2 \tau 1^2 \chi f^2 \chi g^2 (\chi 1 + \chi q) - \tau 1 \tau 2 \chi f (\chi f - \chi g) \chi g \chi p (\chi 1 + 2 \chi q) + \right. \\ & \quad \left. \chi p ((\chi g^2 - \chi 1^2) \chi p \chi q \tau 2^2 - 2 \chi f \chi g \chi p \chi q + \chi f^2 (\chi p \chi q \tau 2^2 + 2 \chi g^2))) m^4 \right) + \end{aligned}$$

$$\begin{aligned}
& \text{A2 B0} \left(-16 \text{C0 D2 } \xi^4 m^4 + \frac{8 \text{C0 D1 } \xi^4 (\rho 1 (\chi f - \chi g) (\chi l + 2 \chi q) - 4 \rho 2 \chi p \chi q) m^4}{\chi p \chi q} + \frac{1}{\chi f \chi g \chi p^2 \chi q} \right. \\
& \quad 8 \text{C0 D0 } \xi^2 ((2 \rho 1^2 \chi f^2 (\chi l + \chi q) \chi g^2 + \rho 1 \rho 2 \chi f (\chi f - \chi g) \chi p (\chi l + 2 \chi q) \chi g + \rho 2^2 (\chi f^2 + \chi g^2 - \chi l^2) \chi p^2 \chi q) \xi^2 + \\
& \quad 2 \tau 1^2 \chi f^2 \chi g^2 (\chi l + \chi q) - \tau 1 \tau 2 \chi f (\chi f - \chi g) \chi g \chi p (\chi l + 2 \chi q) + \\
& \quad \left. \chi p ((\chi g^2 - \chi l^2) \chi p \chi q \tau 2^2 - 2 \chi f \chi g \chi p \chi q + \chi f^2 (\chi p \chi q \tau 2^2 + 2 \chi g^2))) m^4 \right) + \\
& \text{A1 B1} \left(\frac{4 \text{C1 D1 } \xi^4 ((2 \chi p^2 + 2 \chi q^2) \chi f^2 - 8 \chi g \chi p \chi q \chi f + \chi g^2 (2 \chi p^2 + 2 \chi q^2)) m^4}{\chi f \chi g \chi p \chi q} + \right. \\
& \quad \frac{1}{\chi f \chi g \chi p^2 \chi q} 4 \text{C1 D0 } \xi^4 (2 \rho 1 \chi f (\chi f - \chi g) \chi g (\chi p + \chi q) (\chi l + \chi p + \chi q) + \\
& \quad \rho 2 \chi p ((2 \chi p^2 + 2 \chi q^2) \chi f^2 - 8 \chi g \chi p \chi q \chi f + \chi g^2 (2 \chi p^2 + 2 \chi q^2))) m^4 + \\
& \quad \frac{1}{\chi f \chi g \chi p^2 \chi q} 4 \text{C0 D1 } \xi^4 (2 \rho 1 \chi f (\chi f - \chi g) \chi g (\chi p + \chi q) (\chi l + \chi p + \chi q) + \\
& \quad \rho 2 \chi p ((2 \chi p^2 + 2 \chi q^2) \chi f^2 - 8 \chi g \chi p \chi q \chi f + \chi g^2 (2 \chi p^2 + 2 \chi q^2))) m^4 + \\
& \quad \frac{1}{\chi f \chi g \chi p^3 \chi q} 4 \text{C0 D0 } \xi^2 ((\rho 2^2 (4 \chi p \chi q \chi l^2 + 2 \chi f^2 (\chi p - \chi q)^2 + 2 \chi g^2 (\chi p - \chi q)^2 - 16 \chi f \chi g \chi p \chi q) \chi p^2 + \\
& \quad 4 \rho 1 \rho 2 \chi f (\chi f - \chi g) \chi g (\chi p + \chi q) (\chi l + \chi p + \chi q) \chi p + 4 \rho 1^2 \chi f^2 \chi g^2 (\chi l + \chi p + \chi q)^2) \xi^2 + \\
& \quad 4 \tau 1^2 \chi f^2 \chi g^2 (\chi l - \chi p + \chi q)^2 + 4 \tau 1 \tau 2 \chi f (\chi f - \chi g) \chi g \chi p (\chi p - \chi q) (\chi l - \chi p + \chi q) + \\
& \quad \chi p^2 ((-4 \chi p \chi q \tau 2^2 + (2 \tau 2^2 + 2) \chi p^2 + (2 \tau 2^2 + 2) \chi q^2) \chi f^2 + 2 \chi g (2 \chi p^2 + 2 \chi q^2) \chi f + \\
& \quad \left. 4 \tau 2^2 \chi l^2 \chi p \chi q + \chi g^2 (-4 \chi p \chi q \tau 2^2 + (2 \tau 2^2 + 2) \chi p^2 + (2 \tau 2^2 + 2) \chi q^2))) m^4 \right) + \\
& \text{A1 B0} \left(-\frac{8 \text{D2 } \xi^4 (\text{C1 } (\chi f - \chi g) \chi p (\chi p + \chi q) + \text{C0 } \rho 2 (\chi f - \chi g) \chi p (\chi p + \chi q) + 2 \text{C0 } \rho 1 \chi f \chi g (\chi l + \chi p + \chi q)) m^4}{\chi f \chi g \chi p} + \right. \\
& \quad \frac{1}{\chi f \chi g \chi p \chi q} 4 \text{C1 D1 } \xi^4 \\
& \quad (\rho 1 ((2 \chi l \chi p + 2 \chi q (\chi p + \chi q)) \chi f^2 + 2 \chi g (-2 \chi l \chi q - 2 (\chi p + \chi q) \chi q) \chi f + \chi g^2 (2 \chi l \chi p + 2 \chi q (\chi p + \chi q))) - \\
& \quad 4 \rho 2 (\chi f - \chi g) \chi p \chi q (\chi p + \chi q)) m^4 + \frac{1}{\chi f^2 \chi g^2 \chi p^2 \chi q} \\
& \quad 4 \text{C0 D1 } \xi^2 ((2 \rho 1^2 \chi f^2 (\chi f - \chi g) (\chi l + \chi p + \chi q) (\chi l + 2 \chi q) \chi g^2 + \rho 1 \rho 2 \chi f \chi p ((2 \chi l \chi p + 2 \chi q (\chi p + \chi q)) \chi f^2 + \\
& \quad 2 \chi g (-6 \chi l \chi q - 6 (\chi p + \chi q) \chi q) \chi f + \chi g^2 (2 \chi l \chi p + 2 \chi q (\chi p + \chi q))) \chi g - \\
& \quad \rho 2^2 (\chi f - \chi g) (\chi f^2 + 6 \chi g \chi f + \chi g^2 - \chi l^2) \chi p^2 \chi q (\chi p + \chi q) \xi^2 + 2 \tau 1^2 \chi f^2 (\chi f - \chi g) \\
& \quad \chi g^2 \chi l (\chi l - \chi p + \chi q) - \tau 2^2 (\chi f - \chi g) (\chi f^2 + 2 \chi g \chi f + \chi g^2 - \chi l^2) \chi p^2 \chi q (\chi p + \chi q) + 2 \tau 1 \tau 2 \\
& \quad \chi f \chi g \chi p ((\chi l \chi p + \chi q (\chi p + \chi q)) \chi f^2 + 2 \chi g \chi q (\chi l + \chi p + \chi q) \chi f + \chi g^2 (\chi l \chi p + \chi q (\chi p + \chi q)))) m^4 + \\
& \quad \frac{1}{\chi f^2 \chi g^2 \chi p^2 \chi q} 4 \text{C1 D0 } \xi^2 ((2 \rho 1^2 \chi f^2 (\chi f - \chi g) (\chi l + \chi q) (\chi p + \chi q) \chi g^2 + \rho 1 \rho 2 \chi f \chi p ((2 \chi l \chi p + 2 \chi q (\chi p + \chi q)) \\
& \quad \chi f^2 + 2 \chi g (-2 \chi l \chi q - 2 (\chi p + \chi q) \chi q) \chi f + \chi g^2 (2 \chi l \chi p + 2 \chi q (\chi p + \chi q))) \chi g + \\
& \quad \rho 2^2 (\chi f - \chi g) (\chi f^2 + \chi g^2 - \chi l^2) \chi p^2 \chi q (\chi p + \chi q) \xi^2 + 2 \tau 1^2 \chi f^2 (\chi f - \chi g) \chi g^2 (\chi l + \chi q) (\chi p + \chi q) + \\
& \quad (\chi f - \chi g) \chi p (\chi p + \chi q) ((\chi g^2 - \chi l^2) \chi p \chi q \tau 2^2 - 2 \chi f \chi g \chi p \chi q + \chi f^2 (\chi p \chi q \tau 2^2 + 2 \chi g^2)) - 2 \tau 1 \\
& \quad \tau 2 \chi f \chi g \chi p ((\chi l \chi p + \chi q (\chi p + \chi q)) \chi f^2 - 2 \chi g \chi q (\chi l + \chi p + \chi q) \chi f + \chi g^2 (\chi l \chi p + \chi q (\chi p + \chi q)))) m^4 + \\
& \quad \frac{1}{\chi f \chi g \chi p^3 \chi q} 4 \text{C0 D0 } \xi^2 ((4 \chi f^2 \chi g^2 (\chi l + \chi q) (\chi l + \chi p + \chi q) \rho 1^3 + 2 \rho 2 \chi f (\chi f - \chi g) \chi g \chi p \\
& \quad (\chi l^2 + 2 (\chi p + 2 \chi q) \chi l + 3 \chi q (\chi p + \chi q)) \rho 1^2 + \rho 2^2 \chi p^2
\end{aligned}$$

$$\begin{aligned}
& (2(\chi^1 \chi p + \chi q(\chi p + \chi q))\chi f^2 - 8\chi g \chi q(\chi^1 + \chi p + \chi q)\chi f + 2\chi g^2(\chi^1 \chi p + \chi q(\chi p + \chi q)))\rho^1 - \\
& 2\rho^2^3(\chi f - \chi g)\chi p^3 \chi q(\chi p + \chi q)\xi^2 + 2\rho^2 \chi p(\chi f(\chi f - \chi g)\chi g(\chi^1^2 + 2\chi q \chi^1 + \chi q(\chi p + \chi q))\tau^1^2 + \\
& 4\tau^2 \chi f \chi g \chi p \chi q(\chi^1 + \chi p + \chi q)\tau^1 + (\chi f - \chi g)\chi p(\chi p + \chi q)(\chi f \chi g - (\tau^2^2 + 1)\chi p \chi q)) + \\
& \rho^1(4\tau^1^2 \chi f^2(\chi^1 + \chi q)(\chi^1 + \chi p + \chi q)\chi g^2 - 4\tau^1 \tau^2 \chi f(\chi f - \chi g)\chi p(\chi^1 + \chi q)(\chi p + \chi q)\chi g + \\
& \chi p((4(\chi^1 + \chi p + \chi q)\chi g^2 + \chi p((2\tau^2^2 + 2)\chi^1 \chi p + 2(\tau^2^2 + 1)\chi q(\chi p + \chi q)))\chi f^2 + \\
& 4\chi g \chi^1 \chi p(\chi p - \chi q)\chi f + \chi g^2 \chi p((2\tau^2^2 + 2)\chi^1 \chi p + 2(\tau^2^2 + 1)\chi q(\chi p + \chi q))))m^4 \Big) + \\
& \text{A0 B1} \left(-\frac{8 \text{C2 D1} \xi^4 (\chi f - \chi g)(\chi p + \chi q)m^4}{\chi f \chi g} - \frac{8 \text{C2 D0} \xi^4 (\rho^2 (\chi f - \chi g)\chi p(\chi p + \chi q) + 2\rho^1 \chi f \chi g(\chi^1 + \chi p + \chi q))m^4}{\chi f \chi g \chi p} + \right. \\
& \frac{1}{\chi f \chi g \chi p \chi q} 4 \text{C1 D1} \xi^4 \\
& (\rho^1((2\chi^1 \chi p + 2\chi q(\chi p + \chi q))\chi f^2 + 2\chi g(-2\chi^1 \chi q - 2(\chi p + \chi q)\chi q)\chi f + \chi g^2(2\chi^1 \chi p + 2\chi q(\chi p + \chi q))) - \\
& 4\rho^2(\chi f - \chi g)\chi p \chi q(\chi p + \chi q))m^4 + \frac{1}{\chi f^2 \chi g^2 \chi p^2 \chi q} \\
& 4 \text{C1 D0} \xi^2 ((2\rho^1^2 \chi f^2(\chi f - \chi g)(\chi^1 + \chi p + \chi q)(\chi^1 + 2\chi q)\chi g^2 + \rho^1 \rho^2 \chi f \chi p((2\chi^1 \chi p + 2\chi q(\chi p + \chi q))\chi f^2 + \\
& 2\chi g(-6\chi^1 \chi q - 6(\chi p + \chi q)\chi q)\chi f + \chi g^2(2\chi^1 \chi p + 2\chi q(\chi p + \chi q)))\chi g - \\
& \rho^2^2(\chi f - \chi g)(\chi f^2 + 6\chi g \chi f + \chi g^2 - \chi^1^2)\chi p^2 \chi q(\chi p + \chi q))\xi^2 + 2\tau^1^2 \chi f^2(\chi f - \chi g) \\
& \chi g^2 \chi^1(\chi^1 - \chi p + \chi q) - \tau^2^2(\chi f - \chi g)(\chi f^2 + 2\chi g \chi f + \chi g^2 - \chi^1^2)\chi p^2 \chi q(\chi p + \chi q) + 2\tau^1 \tau^2 \\
& \chi f \chi g \chi p((\chi^1 \chi p + \chi q(\chi p + \chi q))\chi f^2 + 2\chi g \chi q(\chi^1 + \chi p + \chi q)\chi f + \chi g^2(\chi^1 \chi p + \chi q(\chi p + \chi q))))m^4 + \\
& \frac{1}{\chi f^2 \chi g^2 \chi p^2 \chi q} 4 \text{C0 D1} \xi^2 ((2\rho^1^2 \chi f^2(\chi f - \chi g)(\chi^1 + \chi q)(\chi p + \chi q)\chi g^2 + \rho^1 \rho^2 \chi f \chi p((2\chi^1 \chi p + 2\chi q(\chi p + \chi q)) \\
& \chi f^2 + 2\chi g(-2\chi^1 \chi q - 2(\chi p + \chi q)\chi q)\chi f + \chi g^2(2\chi^1 \chi p + 2\chi q(\chi p + \chi q)))\chi g + \\
& \rho^2^2(\chi f - \chi g)(\chi f^2 + \chi g^2 - \chi^1^2)\chi p^2 \chi q(\chi p + \chi q))\xi^2 + 2\tau^1^2 \chi f^2(\chi f - \chi g)\chi g^2(\chi^1 + \chi q)(\chi p + \chi q) + \\
& (\chi f - \chi g)\chi p(\chi p + \chi q)((\chi g^2 - \chi^1^2)\chi p \chi q \tau^2^2 - 2\chi f \chi g \chi p \chi q + \chi f^2(\chi p \chi q \tau^2^2 + 2\chi g^2)) - 2\tau^1 \\
& \tau^2 \chi f \chi g \chi p((\chi^1 \chi p + \chi q(\chi p + \chi q))\chi f^2 - 2\chi g \chi q(\chi^1 + \chi p + \chi q)\chi f + \chi g^2(\chi^1 \chi p + \chi q(\chi p + \chi q))))m^4 + \\
& \frac{1}{\chi f \chi g \chi p^3 \chi q} 4 \text{C0 D0} \xi^2 ((4\chi f^2 \chi g^2(\chi^1 + \chi q)(\chi^1 + \chi p + \chi q)\rho^1^3 + 2\rho^2 \chi f(\chi f - \chi g)\chi g \chi p \\
& (\chi^1^2 + 2(\chi p + 2\chi q)\chi^1 + 3\chi q(\chi p + \chi q))\rho^1^2 + \rho^2^2 \chi p^2 \\
& (2(\chi^1 \chi p + \chi q(\chi p + \chi q))\chi f^2 - 8\chi g \chi q(\chi^1 + \chi p + \chi q)\chi f + 2\chi g^2(\chi^1 \chi p + \chi q(\chi p + \chi q)))\rho^1 - \\
& 2\rho^2^3(\chi f - \chi g)\chi p^3 \chi q(\chi p + \chi q)\xi^2 + 2\rho^2 \chi p(\chi f(\chi f - \chi g)\chi g(\chi^1^2 + 2\chi q \chi^1 + \chi q(\chi p + \chi q))\tau^1^2 + \\
& 4\tau^2 \chi f \chi g \chi p \chi q(\chi^1 + \chi p + \chi q)\tau^1 + (\chi f - \chi g)\chi p(\chi p + \chi q)(\chi f \chi g - (\tau^2^2 + 1)\chi p \chi q)) + \\
& \rho^1(4\tau^1^2 \chi f^2(\chi^1 + \chi q)(\chi^1 + \chi p + \chi q)\chi g^2 - 4\tau^1 \tau^2 \chi f(\chi f - \chi g)\chi p(\chi^1 + \chi q)(\chi p + \chi q)\chi g + \\
& \chi p((4(\chi^1 + \chi p + \chi q)\chi g^2 + \chi p((2\tau^2^2 + 2)\chi^1 \chi p + 2(\tau^2^2 + 1)\chi q(\chi p + \chi q)))\chi f^2 + \\
& 4\chi g \chi^1 \chi p(\chi p - \chi q)\chi f + \chi g^2 \chi p((2\tau^2^2 + 2)\chi^1 \chi p + 2(\tau^2^2 + 1)\chi q(\chi p + \chi q))))m^4 \Big) + \\
& \text{A0 B0} \left(\frac{16 \text{C2 D2} \xi^4 \chi p \chi q m^4}{\chi f \chi g} - \frac{8 \text{C2 D1} \xi^4 (\rho^1(\chi f - \chi g)(\chi^1 + 2\chi q) - 4\rho^2 \chi p \chi q)m^4}{\chi f \chi g} - \right. \\
& \frac{1}{\chi f^2 \chi g^2 \chi p} \\
& 8 \text{C2 D0} \xi^2 ((2\rho^1^2 \chi f^2(\chi^1 + \chi q)\chi g^2 + \rho^1 \rho^2 \chi f(\chi f - \chi g)\chi p(\chi^1 + 2\chi q)\chi g + \rho^2^2(\chi f^2 + \chi g^2 - \chi^1^2)\chi p^2 \chi q)\xi^2 + \\
& 2\tau^1^2 \chi f^2 \chi g^2(\chi^1 + \chi q) - \tau^1 \tau^2 \chi f(\chi f - \chi g)\chi g \chi p(\chi^1 + 2\chi q) + \\
& \chi p((\chi g^2 - \chi^1^2)\chi p \chi q \tau^2^2 - 2\chi f \chi g \chi p \chi q + \chi f^2(\chi p \chi q \tau^2^2 + 2\chi g^2)))m^4 -
\end{aligned}$$

$$\begin{aligned}
& \frac{1}{\chi f^2 \chi g^2 \chi p} 8 D2 \xi^2 (C1 \chi f \chi g \chi p (\rho 1 (\chi f - \chi g) (\chi l^2 + 2 \chi q) - 4 \rho 2 \chi p \chi q) \xi^2 + \\
& \quad C0 ((2 \rho 1^2 \chi f^2 (\chi l + \chi q) \chi g^2 + \rho 1 \rho 2 \chi f (\chi f - \chi g) \chi p (\chi l + 2 \chi q) \chi g + \rho 2^2 (\chi f^2 + \chi g^2 - \chi l^2) \chi p^2 \chi q) \xi^2 + \\
& \quad 2 \tau 1^2 \chi f^2 \chi g^2 (\chi l + \chi q) - \tau 1 \tau 2 \chi f (\chi f - \chi g) \chi g \chi p (\chi l + 2 \chi q) + \\
& \quad \chi p ((\chi g^2 - \chi l^2) \chi p \chi q \tau 2^2 - 2 \chi f \chi g \chi p \chi q + \chi f^2 (\chi p \chi q \tau 2^2 + 2 \chi g^2)))) m^4 + \\
& \frac{1}{\chi f^2 \chi g^2 \chi p \chi q} 4 C1 D1 \xi^2 (2 \chi g \chi p^2 \chi f^3 + 2 \chi g \chi q^2 \chi f^3 - 4 \chi g \chi p \chi q \chi f^3 + 4 \tau 2^2 \chi p^2 \chi q^2 \chi f^2 + \\
& \quad 2 \chi g^3 \chi p^2 \chi f + 2 \chi g^3 \chi q^2 \chi f + 8 \tau 2^2 \chi g \chi p^2 \chi q^2 \chi f - 4 \chi g^3 \chi p \chi q \chi f + \\
& \quad \tau 1^2 \chi g ((2 \chi l^2 + 4 \chi q \chi l + 4 \chi q^2) \chi f^2 + 2 \chi g (4 \chi q^2 + 4 \chi l \chi q) \chi f + \chi g^2 (2 \chi l^2 + 4 \chi q \chi l + 4 \chi q^2)) \chi f + \\
& \quad 4 \tau 2^2 \chi g^2 \chi p^2 \chi q^2 - 4 \tau 2^2 \chi l^2 \chi p^2 \chi q^2 + \\
& \quad \xi^2 (\chi f \chi g ((2 \chi l^2 + 4 \chi q \chi l + 4 \chi q^2) \chi f^2 + 2 \chi g (-4 \chi q^2 - 4 \chi l \chi q) \chi f + \chi g^2 (2 \chi l^2 + 4 \chi q \chi l + 4 \chi q^2)) \rho 1^2 - \\
& \quad 8 \rho 2 \chi f (\chi f - \chi g) \chi g \chi p \chi q (\chi l + 2 \chi q) \rho 1 + 4 \rho 2^2 (\chi f^2 + 6 \chi g \chi f + \chi g^2 - \chi l^2) \chi p^2 \chi q^2)) m^4 + \\
& \frac{1}{\chi f \chi g \chi p^2 \chi q} 4 C1 D0 \xi^2 ((2 \chi f (\chi f - \chi g) \chi g (\chi l^2 + 3 \chi q \chi l + 2 \chi q^2) \rho 1^3 + \rho 2 \chi p \\
& \quad ((2 \chi l^2 + 4 \chi q \chi l + 4 \chi q^2) \chi f^2 + 2 \chi g (-8 \chi q^2 - 8 \chi l \chi q) \chi f + \chi g^2 (2 \chi l^2 + 4 \chi q \chi l + 4 \chi q^2)) \rho 1^2 - \\
& \quad 6 \rho 2^2 (\chi f - \chi g) \chi p^2 \chi q (\chi l + 2 \chi q) \rho 1 + 8 \rho 2^3 \chi p^3 \chi q^2) \xi^2 + \\
& \quad \rho 2 \chi p (((2 \chi l^2 + 4 \chi q \chi l + 4 \chi q^2) \chi f^2 + \chi g^2 (2 \chi l^2 + 4 \chi q \chi l + 4 \chi q^2)) \tau 1^2 + \\
& \quad 4 \tau 2 (\chi f - \chi g) \chi p \chi q (\chi l + 2 \chi q) \tau 1 + 2 \chi g^2 \chi p^2 + 2 \chi f^2 (\chi p - \chi q)^2 + \\
& \quad 2 \chi g^2 \chi q^2 + 8 \tau 2^2 \chi p^2 \chi q^2 + 8 \chi p^2 \chi q^2 - 4 \chi g^2 \chi p \chi q - 8 \chi f \chi g \chi p \chi q) + \\
& \quad 2 \rho 1 (\chi f (\chi f - \chi g) \chi g (\chi l^2 + 3 \chi q \chi l + 2 \chi q^2) \tau 1^2 + 8 \tau 2 \chi f \chi g \chi p \chi q (\chi l + \chi q) \tau 1 + \\
& \quad (\chi f - \chi g) (\chi f \chi g (\chi p^2 + \chi q (\chi l + \chi q)) - (\tau 2^2 + 1) \chi p^2 \chi q (\chi l + 2 \chi q)))) m^4 + \\
& \frac{1}{\chi f \chi g \chi p^2 \chi q} 4 C0 D1 \xi^2 ((2 \chi f (\chi f - \chi g) \chi g (\chi l^2 + 3 \chi q \chi l + 2 \chi q^2) \rho 1^3 + \rho 2 \chi p \\
& \quad ((2 \chi l^2 + 4 \chi q \chi l + 4 \chi q^2) \chi f^2 + 2 \chi g (-8 \chi q^2 - 8 \chi l \chi q) \chi f + \chi g^2 (2 \chi l^2 + 4 \chi q \chi l + 4 \chi q^2)) \rho 1^2 - \\
& \quad 6 \rho 2^2 (\chi f - \chi g) \chi p^2 \chi q (\chi l + 2 \chi q) \rho 1 + 8 \rho 2^3 \chi p^3 \chi q^2) \xi^2 + \\
& \quad \rho 2 \chi p (((2 \chi l^2 + 4 \chi q \chi l + 4 \chi q^2) \chi f^2 + \chi g^2 (2 \chi l^2 + 4 \chi q \chi l + 4 \chi q^2)) \tau 1^2 + \\
& \quad 4 \tau 2 (\chi f - \chi g) \chi p \chi q (\chi l + 2 \chi q) \tau 1 + 2 \chi g^2 \chi p^2 + 2 \chi f^2 (\chi p - \chi q)^2 + \\
& \quad 2 \chi g^2 \chi q^2 + 8 \tau 2^2 \chi p^2 \chi q^2 + 8 \chi p^2 \chi q^2 - 4 \chi g^2 \chi p \chi q - 8 \chi f \chi g \chi p \chi q) + \\
& \quad 2 \rho 1 (\chi f (\chi f - \chi g) \chi g (\chi l^2 + 3 \chi q \chi l + 2 \chi q^2) \tau 1^2 + 8 \tau 2 \chi f \chi g \chi p \chi q (\chi l + \chi q) \tau 1 + \\
& \quad (\chi f - \chi g) (\chi f \chi g (\chi p^2 + \chi q (\chi l + \chi q)) - (\tau 2^2 + 1) \chi p^2 \chi q (\chi l + 2 \chi q)))) m^4 + \\
& \frac{1}{\chi f \chi g \chi p^3 \chi q} 4 C0 D0 ((4 \chi f^2 \chi g^2 (\chi l + \chi q)^2 \rho 1^4 + 4 \rho 2 \chi f (\chi f - \chi g) \chi g \chi p (\chi l^2 + 3 \chi q \chi l + 2 \chi q^2) \rho 1^3 + \\
& \quad \rho 2^2 \chi p^2 (2 (\chi l^2 + 2 \chi q \chi l + 2 \chi q^2) \chi f^2 - 16 \chi g \chi q (\chi l + \chi q) \chi f + 2 \chi g^2 (\chi l^2 + 2 \chi q \chi l + 2 \chi q^2)) \rho 1^2 - \\
& \quad 4 \rho 2^3 (\chi f - \chi g) \chi p^3 \chi q (\chi l + 2 \chi q) \rho 1 + 4 \rho 2^4 \chi p^4 \chi q^2) \xi^4 + \\
& \quad (((4 ((2 \tau 1^2 + 1) \chi l^2 + 2 (2 \chi q \tau 1^2 + \chi q) \chi l + \chi p^2 + 2 \tau 1^2 \chi q^2 + \chi q^2) \chi g^2 - 4 \tau 1 \tau 2 \chi p (\chi l^2 + \\
& \quad 3 \chi q \chi l + 2 \chi q^2) \chi g + \chi p^2 ((2 \tau 2^2 + 2) \chi l^2 + 4 (\tau 2^2 + 1) \chi q \chi l + 4 (\tau 2^2 + 1) \chi q^2)) \\
& \quad \chi f^2 + 2 \chi g \chi p (2 \chi p \chi l^2 + 2 \tau 1 \tau 2 \chi g (\chi l^2 + 3 \chi q \chi l + 2 \chi q^2)) \chi f + \\
& \quad \chi g^2 \chi p^2 ((2 \tau 2^2 + 2) \chi l^2 + 4 (\tau 2^2 + 1) \chi q \chi l + 4 (\tau 2^2 + 1) \chi q^2)) \rho 1^2 + \\
& \quad 4 \rho 2 \chi p (\chi f (\chi f - \chi g) \chi g (\chi l^2 + 3 \chi q \chi l + 2 \chi q^2) \tau 1^2 + 8 \tau 2 \chi f \chi g \chi p \chi q (\chi l + \chi q) \tau 1 + \\
& \quad (\chi f - \chi g) (\chi f \chi g (\chi p^2 + \chi q (\chi l + \chi q)) - (\tau 2^2 + 1) \chi p^2 \chi q (\chi l + 2 \chi q)) \rho 1 + \\
& \quad \rho 2^2 \chi p^2 ((2 (\chi l^2 + 2 \chi q \chi l + 2 \chi q^2) \chi f^2 + 2 \chi g^2 (\chi l^2 + 2 \chi q \chi l + 2 \chi q^2)) \tau 1^2 + \\
& \quad 4 \tau 2 (\chi f - \chi g) \chi p \chi q (\chi l + 2 \chi q) \tau 1 + 2 \chi f^2 \chi p^2 + 2 \chi g^2 \chi p^2 + \\
& \quad 2 \chi f^2 \chi q^2 + 2 \chi g^2 \chi q^2 + 8 \tau 2^2 \chi p^2 \chi q^2 + 8 \chi p^2 \chi q^2 - 4 \chi l^2 \chi p \chi q)) \xi^2 + \\
& \quad 4 \tau 1^4 \chi f^2 \chi g^2 (\chi l + \chi q)^2 - 4 \tau 1^3 \tau 2 \chi f (\chi f - \chi g) \chi g \chi p (\chi l^2 + 3 \chi q \chi l + 2 \chi q^2) +
\end{aligned}$$

$$\begin{aligned} & \chi p^2 (4 \chi p^2 \chi q^2 \tau^2 + 8 \chi p^2 \chi q^2 \tau^2 - 4 \chi l^2 \chi p \chi q \tau^2 + 4 \chi p^2 \chi q^2 + \\ & 2 \chi f \chi g (2 \chi p^2 - 8 \chi q \chi p + 2 \chi q^2) + \chi g^2 ((2 \tau^2 + 2) \chi p^2 - 4 \chi q \chi p + (2 \tau^2 + 2) \chi q^2) + \\ & \chi f^2 (4 \chi g^2 + (2 \tau^2 + 2) \chi p^2 + (2 \tau^2 + 2) \chi q^2 - 4 \chi p \chi q)) + \\ & 4 \tau l \tau^2 (\chi f - \chi g) \chi p ((\tau^2 + 1) \chi p^2 \chi q (\chi l + 2 \chi q) - \chi f \chi g (\chi p^2 + \chi q (\chi l + \chi q))) + \\ & \tau l^2 ((4 (\chi l^2 + 2 \chi q \chi l + \chi p^2 + \chi q^2) \chi g^2 + \chi p^2 ((2 \tau^2 + 2) \chi l^2 + 4 (\tau^2 + 1) \chi q \chi l + 4 (\tau^2 + 1) \chi q^2)) \chi f^2 + \\ & 2 \chi g \chi p^2 (-8 \chi q^2 \tau^2 - 8 \chi l \chi q \tau^2 + 2 \chi l^2) \chi f + \\ & \chi g^2 \chi p^2 ((2 \tau^2 + 2) \chi l^2 + 4 (\tau^2 + 1) \chi q \chi l + 4 (\tau^2 + 1) \chi q^2))) m^4 \Big) \end{aligned}$$

Check

No terms linear in a, as expected

```
In[74]:= Collect[Coefficient[Expr4, Pair[Momentum[a, D], Momentum[p, D]]], e]
```

```
Out[74]= 0
```

Finally, in this step, recall that

$$l = p + s k - q$$

$$l' = p + t k - q$$

Namely,

$$l^2 = 2 m^2 + 2 s [(kp) - (kq)] - 2 (pq)$$

$$(l')^2 = 2 m^2 + 2 t [(kp) - (kq)] - 2 (pq)$$

so we need to substitute (pq) here as well

$$\frac{1}{(l^2 + i\epsilon)(l'^2 + i\epsilon)} \text{ enters Coeff, therefore we rewrite it}$$

Note that we still did not pass to

new variables in the integral measure ,

we will introduce the Jacobian determinant in the following steps

```
In[75]:= Diffs4 = Diffs3;
```

```
Coeff4 = Simplify[
```

```
  Coeff3 /. {l1v2 -> 2 m^2 + 2 s (kp - kq) - 2 pq, l2v2 -> 2 m^2 + 2 t (kp - kq) - 2 pq} /. 
```

```
  ktochiSubs[p, q, f, g, l]]
```

```
Out[76]= (a^2 \xi^3 \chi p^2 \chi q^2) / (64 \pi^2 g0 L \phi m^2 p0 q0 \chi f (m^2 (\xi^3 \rho l^2 \chi l^2 + \xi (\tau l^2 \chi l^2 + (\chi p - \chi q)^2) + 2 s \chi p \chi q (\chi q - \chi p)) - i \epsilon \xi \chi p \chi q) \\ (i \epsilon \xi \chi p \chi q + m^2 (\xi^3 \rho l^2 \chi l^2 + \xi (\tau l^2 \chi l^2 + (\chi p - \chi q)^2) + 2 t \chi p \chi q (\chi q - \chi p))))
```

Step 6: Recovering Airy functions

In this step, we substitute the explicit form of An,

Bn, Cn, Dn and pass to invariant variables in the process

Recall some definitions introduced at the beginning before Step 1

$$A_0(s, q, p) = \int \frac{d\phi}{2\pi} e^{is\phi} e^{-i\left(\frac{\alpha}{2}\phi^2 - \frac{4\beta}{3}\phi^3\right)} = \frac{\text{Exp}\left[i\left(\frac{\alpha s}{8\beta} - \frac{\alpha^3}{192\beta^2}\right)\right]}{(4\beta)^{1/3}} \text{Ai}(y)$$

$$A_1(s, q, p) = -i \frac{dA_0}{ds} = \int \frac{d\phi}{2\pi} e^{is\phi} \phi e^{-i\left(\frac{\alpha}{2}\phi^2 - \frac{4\beta}{3}\phi^3\right)} = \frac{\text{Exp}\left[i\left(\frac{\alpha s}{8\beta} - \frac{\alpha^3}{192\beta^2}\right)\right]}{(4\beta)^{1/3}} \left\{ \frac{\alpha}{8\beta} \text{Ai}(y) - i \frac{1}{(4\beta)^{1/3}} \text{Ai}'(y) \right\}$$

$$A_2(s, q, p) = (-i)^2 \frac{d^2 A_0}{ds^2} = \int \frac{d\phi}{2\pi} e^{is\phi} \phi^2 e^{-i\left(\frac{\alpha}{2}\phi^2 - \frac{4\beta}{3}\phi^3\right)} = \frac{\text{Exp}\left[i\left(\frac{\alpha s}{8\beta} - \frac{\alpha^3}{192\beta^2}\right)\right]}{(4\beta)^{1/3}} \left\{ \left[2\left(\frac{\alpha}{8\beta}\right)^2 - \frac{s}{4\beta} \right] \text{Ai}(y) - i \frac{\alpha}{(4\beta)^{4/3}} \text{Ai}'(y) \right\}$$

$$y = y(s, q, p) = \frac{1}{[4\beta(q, p)]^{1/3}} \left[s - \frac{\alpha^2(q, p)}{16\beta(q, p)} \right]$$

$$B_n = A_n^*$$

Note that each term in Expr4 is

proportional to multiplication of four symbols

$A_k B_l C_m D_n$,

where $k, l, m, n = 0, 1$,

2. Each represents a function A_n or B_n , in particular,

$$A_n \rightarrow A_n[s, q, p]$$

$$B_n \rightarrow B_n[t, q, p]$$

$$C_n \rightarrow A_n[\tilde{s} - s, f, -g]$$

$$D_n \rightarrow B_n[\tilde{s} - t, f, -g]$$

In effect, after writing functions A_n and B_n in Expr explicitly, we will obtain a linear combination of Ai and Ai' function tetrads, each having a different argument

We perform the substitution in the following order

1. Substitute A_n, B_n in place of A_n, B_n, C_n, D_n in Expr and Coeff

2. Functions A_n, B_n depend on momenta through $y[s, q, p]$,

$\alpha[q, p]$, and $\beta[q, p]$. Therefore,

we rewrite all these functions explicitly in terms of $\rho_{1,2}$, $\tau_{1,2}$, χ

3. Substitute y , α , β functions

4. Simplify the resulting expression and collect terms

In what follows, we first (Step 6.1) define

some symbols that contain a list of substitutions,
and then (Step 6.2) apply these substitutions to
Expr4 and Coeff4 obtained in the previous step

Step 6.1: Preparations: define substitutions $A_n \rightarrow A_i(y)$, $A_i'(y)$

We recover the explicit form of A_n and B_n by using pre -
defined substitutions

As mentioned (see the comment right after title ' Step 6 : ... '),
each A_n takes the form

$$A_n(s, q, p) = \frac{\text{Exp}\left[i \left(\frac{\alpha s}{8\beta} - \frac{\alpha^3}{192\beta^2} \right)\right]}{(4\beta)^{1/3}} * R_n[A_i(y), A_i'(y)],$$

where R_n is a linear combination
of the Airy functions (similarly for B_n)

As the prefactor here does not depend on n ,
it is convenient to keep the overall factor in Coeff. We substitute A_n ,
 B_n , C_n , D_n , appearing in Expr, by the corresponding expressions for R_n

SubsAnBack is the list of substitutions of symbols A_n , B_n , C_n ,
 D_n with A_n and B_n in the form of functions of their arguments

AnsubsNoFactor is introduced in definitions.nb

It contains the substitutions of A_n and B_n in a functional form with
corresponding explicit expressions for $R_n[A_i(y), A_i'(y)]$,
namely, linear combinations of Airy functions

Four factors $1/(4\beta)^{1/3}$ (one per each of A_n , B_n , C_n , and D_n)
are collected in a common factor named **AnFactor**

The sum of four exponentials, $i \left(\frac{\alpha s}{8\beta} - \frac{\alpha^3}{192\beta^2} \right)$, etc,

will be introduced in a separate variable in what follows

Note that Airy functions are expressed using Mathematica's `AiryAi` and `AiryAiPrime`, which obey usual Airy function properties

```
In[77]:= SubsAnBack = {A0 -> A0[s, q, p], A1 -> A1[s, q, p], A2 -> A2[s, q, p],
  B0 -> B0[t, q, p], B1 -> B1[t, q, p], B2 -> B2[t, q, p],
  C0 -> A0[-s + stilde, f, -g],
  C1 -> A1[-s + stilde, f, -g], C2 -> A2[-s + stilde, f, -g],
  D0 -> B0[stilde - t, f, -g], D1 -> B1[stilde - t, f, -g], D2 -> B2[stilde - t, f, -g]}
```

```
Out[77]:= {A0 -> A0(s, q, p), A1 -> A1(s, q, p), A2 -> A2(s, q, p), B0 -> B0(t, q, p), B1 -> B1(t, q, p),
  B2 -> B2(t, q, p), C0 -> A0(stilde - s, f, -g), C1 -> A1(stilde - s, f, -g), C2 -> A2(stilde - s, f, -g),
  D0 -> B0(stilde - t, f, -g), D1 -> B1(stilde - t, f, -g), D2 -> B2(stilde - t, f, -g)}
```

```
In[78]:= AnSubsNoFactor (*print the list of substitutions, we use them later*)
AnFactor = 1 / (4 β[q, p] 4 β[f, -g] * 4 β[q, p] 4 β[f, -g]) ^ (1 / 3)
```

```
Out[78]:= {A0(s_, q_, p_) -> Ai(y(s, q, p)), B0(s_, q_, p_) -> Ai(y(s, q, p)),
  A1(s_, q_, p_) -> \frac{\alpha(q, p) Ai(y(s, q, p))}{8 \beta(q, p)} - \frac{i Ai'(y(s, q, p))}{2^{2/3} \sqrt[3]{\beta(q, p)}}, B1(s_, q_, p_) -> \frac{\alpha(q, p) Ai(y(s, q, p))}{8 \beta(q, p)} + \frac{i Ai'(y(s, q, p))}{2^{2/3} \sqrt[3]{\beta(q, p)}},
  A2(s_, q_, p_) -> Ai(y(s, q, p)) \left( \frac{\alpha(q, p)^2}{64 \beta(q, p)^2} - \frac{y(s, q, p)}{2 \sqrt[3]{2} \beta(q, p)^{2/3}} \right) - \frac{i \alpha(q, p) Ai'(y(s, q, p))}{4 \times 2^{2/3} \beta(q, p)^{4/3}},
  B2(s_, q_, p_) -> Ai(y(s, q, p)) \left( \frac{\alpha(q, p)^2}{64 \beta(q, p)^2} - \frac{y(s, q, p)}{2 \sqrt[3]{2} \beta(q, p)^{2/3}} \right) + \frac{i \alpha(q, p) Ai'(y(s, q, p))}{4 \times 2^{2/3} \beta(q, p)^{4/3}} \}
```

```
Out[79]:= \frac{1}{4 \times 2^{2/3} \sqrt[3]{\beta(f, -g)^2 \beta(q, p)^2}}
```

Explicit expressions for y , α , β in terms of invariant variables

Recall the definitions

$$y[s, q, p] = (4 \beta[q, p])^{2/3} \left[\frac{s}{4 \beta[q, p]} - \left(\frac{\alpha[q, p]}{8 \beta[q, p]} \right)^2 \right]$$

Similarly, we can write

$$y[t, q, p]$$

$$y[\tilde{s} - s, f, -g]$$

$$y[\tilde{s} - t, f, -g]$$

$$\alpha(q, p) = -e \left(\frac{ap}{kp} - \frac{aq}{kq} \right)$$

$$\beta(q, p) = \frac{e^2 a^2}{8} \left(\frac{1}{kp} - \frac{1}{kq} \right)$$

and analogously for $\alpha(f, -g)$, $\beta(f, -g)$

Let us rewrite $\tilde{s}$ (entering C_n and D_n), which we defined above as

$$\tilde{s} = \frac{\xi}{m^2 \chi_f} (pq + pg - qg - m^2)$$

in terms of the invariants. We already calculated scalar products (pq) , (pg) , (qg) , so we reuse these results

```
In[80]:= (*The definition of  $\tilde{s}$ *)
Simplify[( $\xi / m^2 / \chi_f$ ) (pq + pg - qg - m^2)];
Expand[% /. { $\chi_l \rightarrow \chi_p - \chi_q$ }];
Collect[%, {1 /  $\chi_f / \chi_g$ , 1 /  $\chi_f / \chi_p / \chi_q$ ,  $\rho_1$ ,  $\tau_1$ ,  $\rho_2$ ,  $\tau_2$ }, Simplify]
stildeDef = Simplify[% /. { $\chi_p - \chi_q \rightarrow \chi_l$ ,  $\chi_g - \chi_p + \chi_q \rightarrow -\chi_f$ }]
```

$$\begin{aligned} \text{Out[82]} = & -\frac{\xi^3 \rho_1^2 (\chi_p - \chi_q) (\chi_g - \chi_p + \chi_q)}{2 \chi_f \chi_p \chi_q} + \frac{\frac{1}{2} \xi^3 \rho_2^2 (\chi_p - \chi_q) + \frac{1}{2} \xi \tau_2^2 (\chi_p - \chi_q) + \frac{1}{2} \xi (\chi_p - \chi_q)}{\chi_f \chi_g} - \\ & \frac{\xi \tau_1^2 (\chi_p - \chi_q) (\chi_g - \chi_p + \chi_q)}{2 \chi_f \chi_p \chi_q} - \frac{\xi (\chi_p - \chi_q) (\chi_g - \chi_p + \chi_q)}{2 \chi_f \chi_p \chi_q} \\ \text{Out[83]} = & \frac{\xi \chi_l (\chi_f \chi_g (\xi^2 \rho_1^2 + \tau_1^2 + 1) + \chi_p \chi_q (\xi^2 \rho_2^2 + \tau_2^2 + 1))}{2 \chi_f \chi_g \chi_p \chi_q} \end{aligned}$$

Now, we find the arguments for the Airy functions

Commands **yArg**, **αdef**, and **βdef** are introduced in definitions.nb
They are used to write the Airy function arguments explicitly

```
In[84]:= (*example*)
yArg[s, q, p]
Expand[%]
```

$$\begin{aligned} \text{Out[84]} = & 2 \sqrt[3]{2} \beta(q, p)^{2/3} \left(\frac{s}{4 \beta(q, p)} - \frac{\alpha(q, p)^2}{64 \beta(q, p)^2} \right) \\ \text{Out[85]} = & \frac{s}{2^{2/3} \sqrt[3]{\beta(q, p)}} - \frac{\alpha(q, p)^2}{16 \times 2^{2/3} \beta(q, p)^{4/3}} \end{aligned}$$

Now, we rewrite α and β in terms of invariants by replacing scalar products (ap) , (af) , etc, which we identified earlier

```

In[86]:=  $\alpha$ def[q, p] /. {Pair[Momentum[a, D], Momentum[q, D]]  $\rightarrow$  av[ $\mu$ ]  $\times$  qv[ $\mu$ ]}
 $\alpha$ qpInv =
  Simplify[Contract[% /. {qv[ $\mu$ ]  $\rightarrow$  qpexpansion[ $\mu$ ]}] /. ktochiSubs[p, q, f, g, l] /.
    {av2  $\rightarrow$  -m^2  $\xi$ ^2 / e^2}] /. {xp - xq  $\rightarrow$  x $l$ , xq - xp  $\rightarrow$  -x $l$ }
 $\beta$ def[q, p] /. {Pair[Momentum[a, D], Momentum[q, D]]  $\rightarrow$  av[ $\mu$ ]  $\times$  qv[ $\mu$ ]}
 $\beta$ qpInv =
  Simplify[Contract[% /. {qv[ $\mu$ ]  $\rightarrow$  qpexpansion[ $\mu$ ]}] /. ktochiSubs[p, q, f, g, l] /.
    {av2  $\rightarrow$  -m^2  $\xi$ ^2 / e^2}] /. {xp - xq  $\rightarrow$  x $l$ , xq - xp  $\rightarrow$  -x $l$ }

```

$$\text{Out[86]} = -e \left(\frac{a \cdot p}{k \cdot p} - \frac{a^\mu q^\mu}{k \cdot q} \right)$$

$$\text{Out[87]} = -\frac{\xi^3 \rho_1 \chi^1}{\chi p \chi q}$$

$$\text{Out[88]} = \frac{1}{8} a^2 e^2 \left(\frac{1}{k \cdot p} - \frac{1}{k \cdot q} \right)$$

$$\text{Out[89]} = \frac{\xi^3 \chi^1}{8 \chi p \chi q}$$

```

In[90]:=  $\alpha$ def[f, -g] /. {Pair[Momentum[a, D], Momentum[g, D]]  $\rightarrow$  av[ $\mu$ ]  $\times$  gv[ $\mu$ ]}
 $\alpha$ fgInv =
  Simplify[Contract[% /. {gv[ $\mu$ ]  $\rightarrow$  gfexpansion[ $\mu$ ]}] /. ktochiSubs[p, q, f, g, l] /.
    {av2  $\rightarrow$  -m^2  $\xi$ ^2 / e^2}] /. {xf + xg  $\rightarrow$  x $l$ }
 $\beta$ def[f, -g] /. {Pair[Momentum[a, D], Momentum[g, D]]  $\rightarrow$  av[ $\mu$ ]  $\times$  gv[ $\mu$ ]}
 $\beta$ fgInv =
  Simplify[Contract[% /. {gv[ $\mu$ ]  $\rightarrow$  gfexpansion[ $\mu$ ]}] /. ktochiSubs[p, q, f, g, l] /.
    {av2  $\rightarrow$  -m^2  $\xi$ ^2 / e^2}] /. {xf + xg  $\rightarrow$  x $l$ }

```

$$\text{Out[90]} = -e \left(\frac{a^\mu g^\mu}{g \cdot k} - \frac{a \cdot f}{f \cdot k} \right)$$

$$\text{Out[91]} = -\frac{\xi^3 \rho_2 \chi^1}{\chi f \chi g}$$

$$\text{Out[92]} = \frac{1}{8} a^2 e^2 \left(-\frac{1}{f \cdot k} - \frac{1}{g \cdot k} \right)$$

$$\text{Out[93]} = \frac{\xi^3 \chi^1}{8 \chi f \chi g}$$

With this, we can write the Airy function arguments explicitly

Notations

ysqp == y (s, q, p)

ytqp == y (t, q, p)

In[94]:= `yArg[s, q, p] /. {α[q, p] → αqpInv, β[q, p] → βqpInv};`

$$ysqp = \frac{1}{\xi} \left(\frac{\chi^l}{2 \chi p \chi q} \right)^{-1/3} \text{Simplify} \left[\text{Expand}[\%] \left(\frac{\xi^3 \chi^l}{2 \chi p \chi q} \right)^{1/3} \right]$$

`yArg[t, q, p] /. {α[q, p] → αqpInv, β[q, p] → βqpInv};`

$$ytqp = \frac{1}{\xi} \left(\frac{\chi^l}{2 \chi p \chi q} \right)^{-1/3} \text{Simplify} \left[\text{Expand}[\%] \left(\frac{\xi^3 \chi^l}{2 \chi p \chi q} \right)^{1/3} \right]$$

Out[95]=

$$\frac{\sqrt[3]{2} \left(s - \frac{\xi^3 \rho^2 \chi^l}{2 \chi p \chi q} \right)}{\xi \sqrt[3]{\frac{\chi^l}{\chi p \chi q}}}$$

Out[97]=

$$\frac{\sqrt[3]{2} \left(t - \frac{\xi^3 \rho^2 \chi^l}{2 \chi p \chi q} \right)}{\xi \sqrt[3]{\frac{\chi^l}{\chi p \chi q}}}$$

Notations

`ysfg` == $y(s, f, -g)$ (used only in an intermediate step)

`ystsfg` == $y(\tilde{s} - s, f, -g)$

`ytfg` == $y(t, f, -g)$ (used only in an intermediate step)

`ysttfg` == $y(\tilde{s} - t, f, -g)$

In[98]:= $\text{yArg}[s, f, -g] /. \{\alpha[f, -g] \rightarrow \alpha\text{fgInv}, \beta[f, -g] \rightarrow \beta\text{fgInv}\};$

$$\text{ysfg} = \frac{1}{\xi} \left(\frac{\chi^l}{2 \chi^f \chi^g} \right)^{-1/3} \text{Expand} \left[\% \left(\frac{\xi^3 \chi^l}{2 \chi^f \chi^g} \right)^{1/3} \right]$$

$\text{ystsfg} =$

$$\frac{1}{\xi} \left(\frac{\chi^l}{2 \chi^f \chi^g} \right)^{-1/3} \text{Collect} \left[\text{Expand} \left[\left(\text{ysfg} / \left(\frac{1}{\xi} \left(\frac{\chi^l}{2 \chi^f \chi^g} \right)^{-1/3} \right) \right) /. \{s \rightarrow \text{stildeDef} - s\} \right], \{\chi^p, \chi^q, \chi^f, \chi^g\}, \text{Simplify} \right]$$

$\text{yArg}[t, f, -g] /. \{\alpha[f, -g] \rightarrow \alpha\text{fgInv}, \beta[f, -g] \rightarrow \beta\text{fgInv}\};$

$$\text{ytfg} = \frac{1}{\xi} \left(\frac{\chi^l}{2 \chi^f \chi^g} \right)^{-1/3} \text{Expand} \left[\% \left(\frac{\xi^3 \chi^l}{2 \chi^f \chi^g} \right)^{1/3} \right]$$

$\text{ysttfg} =$

$$\frac{1}{\xi} \left(\frac{\chi^l}{2 \chi^f \chi^g} \right)^{-1/3} \text{Collect} \left[\text{Expand} \left[\left(\text{ytfg} / \left(\frac{1}{\xi} \left(\frac{\chi^l}{2 \chi^f \chi^g} \right)^{-1/3} \right) \right) /. \{t \rightarrow \text{stildeDef} - t\} \right], \{\chi^p, \chi^q, \chi^f, \chi^g\}, \text{Simplify} \right]$$

Out[99]=
$$\frac{\sqrt[3]{2} \left(s - \frac{\xi^3 \rho^2 \chi^l}{2 \chi^f \chi^g} \right)}{\xi \sqrt[3]{\frac{\chi^l}{\chi^f \chi^g}}}$$

Out[100]=
$$\frac{\sqrt[3]{2} \left(\frac{\xi \chi^l (\xi^2 \rho^2 + \tau^2 + 1)}{2 \chi^p \chi^q} + \frac{\xi (\tau^2 + 1) \chi^l}{2 \chi^f \chi^g} - s \right)}{\xi \sqrt[3]{\frac{\chi^l}{\chi^f \chi^g}}}$$

Out[102]=
$$\frac{\sqrt[3]{2} \left(t - \frac{\xi^3 \rho^2 \chi^l}{2 \chi^f \chi^g} \right)}{\xi \sqrt[3]{\frac{\chi^l}{\chi^f \chi^g}}}$$

Out[103]=
$$\frac{\sqrt[3]{2} \left(\frac{\xi \chi^l (\xi^2 \rho^2 + \tau^2 + 1)}{2 \chi^p \chi^q} + \frac{\xi (\tau^2 + 1) \chi^l}{2 \chi^f \chi^g} - t \right)}{\xi \sqrt[3]{\frac{\chi^l}{\chi^f \chi^g}}}$$

The resulting list for α ,
 β and y variables : **Subs $\alpha\beta$** , **SubsyArgExplicit**

SubsyArgsInAi is used to make the output expressions more compact
 by replacing the Airy function arguments with shorthands

$\text{ys1} == \text{y}[s, q, p]$

$\text{yt1} == \text{y}[t, q, p]$

$\text{ys2} == \text{y}[-s + \text{stilde}, f, -g]$

$\text{yt2} == \text{y}[\text{stilde} - t, f, -g]$

In[104]:= **Subs** $\alpha\beta = \{\alpha[q, p] \rightarrow \alpha qpInv, \beta[q, p] \rightarrow \beta qpInv, \alpha[f, -g] \rightarrow \alpha fgInv, \beta[f, -g] \rightarrow \beta fgInv\}$

SubsyArgExplicit = $\{y[s, q, p] \rightarrow ysqp, y[t, q, p] \rightarrow ytpq,$

$y[-s + stilde, f, -g] \rightarrow ystsfg, y[stilde - t, f, -g] \rightarrow ysttfg\}$

SubsyArgsInAi =

$\{AiryAi[y[s, q, p]] \rightarrow AiryAi[ys1], AiryAi[y[t, q, p]] \rightarrow AiryAi[yt1],$

$AiryAi[y[-s + stilde, f, -g]] \rightarrow AiryAi[ys2],$

$AiryAi[y[stilde - t, f, -g]] \rightarrow AiryAi[yt2],$

$AiryAiPrime[y[s, q, p]] \rightarrow AiryAiPrime[ys1],$

$AiryAiPrime[y[t, q, p]] \rightarrow AiryAiPrime[yt1],$

$AiryAiPrime[y[-s + stilde, f, -g]] \rightarrow AiryAiPrime[ys2],$

$AiryAiPrime[y[stilde - t, f, -g]] \rightarrow AiryAiPrime[yt2]\}$

$$\text{Out[104]} = \left\{ \alpha(q, p) \rightarrow -\frac{\xi^3 \rho^1 \chi^1}{\chi p \chi q}, \beta(q, p) \rightarrow \frac{\xi^3 \chi^1}{8 \chi p \chi q}, \alpha(f, -g) \rightarrow -\frac{\xi^3 \rho^2 \chi^1}{\chi f \chi g}, \beta(f, -g) \rightarrow \frac{\xi^3 \chi^1}{8 \chi f \chi g} \right\}$$

$$\text{Out[105]} = \left\{ y(s, q, p) \rightarrow \frac{\sqrt[3]{2} \left(s - \frac{\xi^3 \rho^1 \chi^1}{2 \chi p \chi q} \right)}{\xi \sqrt[3]{\frac{\chi^1}{\chi p \chi q}}}, y(t, q, p) \rightarrow \frac{\sqrt[3]{2} \left(t - \frac{\xi^3 \rho^1 \chi^1}{2 \chi p \chi q} \right)}{\xi \sqrt[3]{\frac{\chi^1}{\chi p \chi q}}}, \right.$$

$$\left. y(stilde - s, f, -g) \rightarrow \frac{\sqrt[3]{2} \left(\frac{\xi \chi^1 (\xi^2 \rho^1 \chi^1 + \tau^1 \chi^1)}{2 \chi p \chi q} + \frac{\xi (\tau^2 + 1) \chi^1}{2 \chi f \chi g} - s \right)}{\xi \sqrt[3]{\frac{\chi^1}{\chi f \chi g}}}, y(stilde - t, f, -g) \rightarrow \frac{\sqrt[3]{2} \left(\frac{\xi \chi^1 (\xi^2 \rho^1 \chi^1 + \tau^1 \chi^1)}{2 \chi p \chi q} + \frac{\xi (\tau^2 + 1) \chi^1}{2 \chi f \chi g} - t \right)}{\xi \sqrt[3]{\frac{\chi^1}{\chi f \chi g}}} \right\}$$

Out[106]= **Ai**($y(s, q, p)$) $\rightarrow$ **Ai**(ys1), **Ai**($y(t, q, p)$) $\rightarrow$ **Ai**(yt1), **Ai**($y(stilde - s, f, -g)$) $\rightarrow$ **Ai**(ys2),

Ai($y(stilde - t, f, -g)$) $\rightarrow$ **Ai**(yt2), **Ai'**($y(s, q, p)$) $\rightarrow$ **Ai'**(ys1), **Ai'**($y(t, q, p)$) $\rightarrow$ **Ai'**(yt1),

Ai'($y(stilde - s, f, -g)$) $\rightarrow$ **Ai'**(ys2), **Ai'**($y(stilde - t, f, -g)$) $\rightarrow$ **Ai'**(yt2)}

Some default assumptions used in command **Simplify** below

In[107]:= **assump** $\xi\chi = \{\xi > 0, \chi p > 0, \chi q > 0, \chi f > 0, \chi g > 0\}$

assump**All** = $\{u > 0, v > 4, \chi p > 0, \chi q > 0, \chi g > 0, \chi f > 0, \chi f > 0, \xi > 0\}$

\$Assumptions = **assumpAll**;

Out[107]= $\{\xi > 0, \chi p > 0, \chi q > 0, \chi f > 0, \chi g > 0\}$

Out[108]= $\{u > 0, v > 4, \chi p > 0, \chi q > 0, \chi g > 0, \chi f > 0, \chi f > 0, \xi > 0\}$

```
In[110]:= (*a small test, all the substitutions defined above are applied one by one*)
A2 B2 C2 D2 /. SubsAnBack /. AnsubsNoFactor /. SubsyArgsInAi
% /. Subsaβ /. SubsyArgExplicit
```

$$\begin{aligned} \text{Out[110]} = & \left(\text{Ai}(ys1) \left(\frac{\alpha(q, p)^2}{64 \beta(q, p)^2} - \frac{y(s, q, p)}{2 \sqrt[3]{2} \beta(q, p)^{2/3}} \right) - \frac{i \text{Ai}'(ys1) \alpha(q, p)}{4 \times 2^{2/3} \beta(q, p)^{4/3}} \right) \\ & \left(\text{Ai}(yt1) \left(\frac{\alpha(q, p)^2}{64 \beta(q, p)^2} - \frac{y(t, q, p)}{2 \sqrt[3]{2} \beta(q, p)^{2/3}} \right) + \frac{i \text{Ai}'(yt1) \alpha(q, p)}{4 \times 2^{2/3} \beta(q, p)^{4/3}} \right) \\ & \left(\text{Ai}(ys2) \left(\frac{\alpha(f, -g)^2}{64 \beta(f, -g)^2} - \frac{y(\text{stilde} - s, f, -g)}{2 \sqrt[3]{2} \beta(f, -g)^{2/3}} \right) - \frac{i \text{Ai}'(ys2) \alpha(f, -g)}{4 \times 2^{2/3} \beta(f, -g)^{4/3}} \right) \\ & \left(\text{Ai}(yt2) \left(\frac{\alpha(f, -g)^2}{64 \beta(f, -g)^2} - \frac{y(\text{stilde} - t, f, -g)}{2 \sqrt[3]{2} \beta(f, -g)^{2/3}} \right) + \frac{i \text{Ai}'(yt2) \alpha(f, -g)}{4 \times 2^{2/3} \beta(f, -g)^{4/3}} \right) \\ \text{Out[111]} = & \left(\text{Ai}(ys1) \left(\rho 1^2 - \frac{2 \left(s - \frac{\xi^3 \rho 1^2 \chi 1}{2 \chi p \chi q} \right)}{\xi \sqrt[3]{\frac{\chi 1}{\chi p \chi q} \left(\frac{\xi^3 \chi 1}{\chi p \chi q} \right)^{2/3}}} \right) + \frac{2 i \sqrt[3]{2} \xi^3 \rho 1 \chi 1 \text{Ai}'(ys1)}{\chi p \chi q \left(\frac{\xi^3 \chi 1}{\chi p \chi q} \right)^{4/3}} \right) \\ & \left(\text{Ai}(yt1) \left(\rho 1^2 - \frac{2 \left(t - \frac{\xi^3 \rho 1^2 \chi 1}{2 \chi p \chi q} \right)}{\xi \sqrt[3]{\frac{\chi 1}{\chi p \chi q} \left(\frac{\xi^3 \chi 1}{\chi p \chi q} \right)^{2/3}}} \right) - \frac{2 i \sqrt[3]{2} \xi^3 \rho 1 \chi 1 \text{Ai}'(yt1)}{\chi p \chi q \left(\frac{\xi^3 \chi 1}{\chi p \chi q} \right)^{4/3}} \right) \\ & \left(\text{Ai}(ys2) \left(\rho 2^2 - \frac{2 \left(\frac{\xi \chi 1 (\xi^2 \rho 1^2 + \tau 1^2 + 1)}{2 \chi p \chi q} + \frac{\xi (\tau 2^2 + 1) \chi 1}{2 \chi f \chi g} - s \right)}{\xi \sqrt[3]{\frac{\chi 1}{\chi f \chi g} \left(\frac{\xi^3 \chi 1}{\chi f \chi g} \right)^{2/3}}} \right) + \frac{2 i \sqrt[3]{2} \xi^3 \rho 2 \chi 1 \text{Ai}'(ys2)}{\chi f \chi g \left(\frac{\xi^3 \chi 1}{\chi f \chi g} \right)^{4/3}} \right) \\ & \left(\text{Ai}(yt2) \left(\rho 2^2 - \frac{2 \left(\frac{\xi \chi 1 (\xi^2 \rho 1^2 + \tau 1^2 + 1)}{2 \chi p \chi q} + \frac{\xi (\tau 2^2 + 1) \chi 1}{2 \chi f \chi g} - t \right)}{\xi \sqrt[3]{\frac{\chi 1}{\chi f \chi g} \left(\frac{\xi^3 \chi 1}{\chi f \chi g} \right)^{2/3}}} \right) - \frac{2 i \sqrt[3]{2} \xi^3 \rho 2 \chi 1 \text{Ai}'(yt2)}{\chi f \chi g \left(\frac{\xi^3 \chi 1}{\chi f \chi g} \right)^{4/3}} \right) \end{aligned}$$

Step 6.2: Substitutions and simplification of resulting expressions

Now we use the introduced substitutions to express An, Bn, Cn, Dn explicitly through invariant variables

First, let us calculate

Phase5 –

the sum of all exponentials (with i omitted) of the form $\left(\frac{\alpha s}{8 \beta} - \frac{\alpha^3}{192 \beta^2} \right)$

```
In[112]:= Simplify[(AnPhase[s, q, p] +
  AnPhase[st - s, f, -g] - AnPhase[t, q, p] - AnPhase[st - t, f, -g])]
Phase5 = Simplify[% /. {st → stildeDef} /. Subsaβ, Assumptions → assumptξχ]
```

$$\text{Out[112]} = \frac{(s-t)(\beta(f, -g)\alpha(q, p) - \alpha(f, -g)\beta(q, p))}{8\beta(f, -g)\beta(q, p)}$$

$$\text{Out[113]} = -((\rho_1 - \rho_2)(s - t))$$

Performing the actual substitutions as described, to obtain Expr5

Note that AnFactor is introduced in variable Expr and not in Coeff

Expr5 is lengthy and is not printed for this reason

```
In[114]:= Diffs5 = Diffs4;
Coeff5 = Coeff4;
Expr4 * AnFactor;
Expand[%] /. SubsAnBack /. AnsubsNoFactor /. SubsyArgsInAi /. Subsaβ /.
  SubsyArgExplicit;
Expr5 = Expand[%];
```

In order to simplify the result,

let us now pass to new variables in the integral measure

$$u = \frac{\chi_l}{\chi_q}, \quad 0 < u < \infty$$

$$v^2 = \frac{\chi_l^2}{\chi_g \chi_f}, \quad 4 < v < \infty$$

$$\chi_g = \frac{\chi_l}{2} \left(1 + \sigma \sqrt{1 - \frac{4}{v}} \right), \quad \chi_f = \frac{\chi_l}{2} \left(1 - \sigma \sqrt{1 - \frac{4}{v}} \right),$$

$\sigma = \pm 1$ are two branches, arising in this variable change;

The final result is given by summation over σ

In effect,

combined with variable changes introduced above $(\rho_{1,2}, \tau_{1,2})$,

the overall integral measure transforms as

$$d^3g d^3q \rightarrow |J_1^{-1}| |J_2^{-1}| d\rho_1 d\tau_1 du d\rho_2 d\tau_2 dv,$$

with Jacobian determinants given by

$$|J_1^{-1}| = q_0 m^2 \xi \frac{u^2}{(1+u)^3}$$

$$|J_2^{-1}| = g_0 m^2 \xi \frac{\chi_l}{\chi_g} \frac{1}{v \sqrt{v(v-4)}}$$

$$\text{In[119]:= } \text{jac1} = q_0 m^2 \xi \frac{u^2}{(1+u)^3}$$

$$\text{jac2} = g_0 m^2 \xi \frac{\chi^l}{\chi g} \frac{1}{v \sqrt{v-4}}$$

$$\text{Out[119]= } \frac{m^2 \xi q_0 u^2}{(u+1)^3}$$

$$\text{Out[120]= } \frac{g_0 m^2 \xi \chi^l}{v \sqrt{v-4} v \chi g}$$

$$\begin{aligned} \text{In[121]:= } \text{subsu} &= \{\chi q \rightarrow \chi p / (1+u), \chi^l \rightarrow \chi p u / (1+u)\} \\ \text{subsv}\sigma &= \{\chi g \rightarrow (1/2) \chi^l (1 + \sigma \sqrt{1-4/v}), \chi^f \rightarrow (1/2) \chi^l (1 - \sigma \sqrt{1-4/v})\} \end{aligned}$$

$$\text{Out[121]= } \left\{ \chi q \rightarrow \frac{\chi p}{u+1}, \chi^l \rightarrow \frac{u \chi p}{u+1} \right\}$$

$$\text{Out[122]= } \left\{ \chi g \rightarrow \frac{1}{2} \chi^l \left(\sigma \sqrt{1 - \frac{4}{v}} + 1 \right), \chi^f \rightarrow \frac{1}{2} \chi^l \left(1 - \sigma \sqrt{1 - \frac{4}{v}} \right) \right\}$$

$$\text{In[123]:= } (*\text{It is straightforward to see that the following result depends only on } \sigma^2, \text{ so we can just set } \sigma \rightarrow 1*)$$

$$\text{Coeff5} * \text{jac1} * \text{jac2};$$

$$\% /. \{\sigma \rightarrow 1\};$$

$$\text{Coeff6} = \text{Simplify}[\% /. \text{subsv}\sigma /. \text{subsu}]$$

$$\text{Diffs6} = \text{Diffs5} /. \{d3q \rightarrow d\rho_1 d\tau_1 du, d3g \rightarrow d\rho_2 d\tau_2 dv\}$$

$$\text{Phase6} = \text{Phase5}$$

$$\begin{aligned} \text{Out[125]= } & (a^2 m^2 \xi^5 u) / \left(16 \pi^2 L \phi p_0 v \sqrt{v-4} v \chi p \left(\sigma \sqrt{\frac{v-4}{v}} - 1 \right) \left(\sigma \sqrt{\frac{v-4}{v}} + 1 \right) \right. \\ & \left. (i \epsilon \xi (u+1) - m^2 u (\xi^2 \rho_1^2 + \tau_1^2 + 1) - 2 s \chi p) (i \epsilon \xi (u+1) + m^2 u (\xi^2 \rho_1^2 + \tau_1^2 + 1) - 2 t \chi p) \right) \end{aligned}$$

$$\text{Out[126]= } d\rho_1 d\rho_2 ds dt d\tau_1 d\tau_2 du dv$$

$$\text{Out[127]= } -((\rho_1 - \rho_2)(s - t))$$

Summation over two σ branches

$$\text{In[128]:= } \text{Expr6}\sigma = \text{Expr5} /. \text{subsv}\sigma /. \text{subsu};$$

$$\text{Expr6} = (\text{Expr6}\sigma /. \{\sigma \rightarrow 1\}) + (\text{Expr6}\sigma /. \{\sigma \rightarrow -1\});$$

In the next step, we collect terms in Expr -- all possible combinations of four Airy functions and their derivatives -- and simplify them

```

In[130]:= Collect4Airy = {
    AiryAi[ys1] AiryAi[ys2] AiryAi[yt1] AiryAi[yt2],
    AiryAi'[ys1] AiryAi[ys2] AiryAi[yt1] AiryAi[yt2],
    AiryAi[ys1] AiryAi'[ys2] AiryAi[yt1] AiryAi[yt2],
    AiryAi[ys1] AiryAi[ys2] AiryAi'[yt1] AiryAi[yt2],
    AiryAi[ys1] AiryAi[ys2] AiryAi[yt1] AiryAi'[yt2],

    AiryAi'[ys1] AiryAi'[ys2] AiryAi[yt1] AiryAi[yt2],
    AiryAi'[ys1] AiryAi[ys2] AiryAi'[yt1] AiryAi[yt2],
    AiryAi'[ys1] AiryAi[ys2] AiryAi[yt1] AiryAi'[yt2],
    AiryAi[ys1] AiryAi'[ys2] AiryAi'[yt1] AiryAi[yt2],
    AiryAi[ys1] AiryAi'[ys2] AiryAi[yt1] AiryAi'[yt2],
    AiryAi[ys1] AiryAi[ys2] AiryAi'[yt1] AiryAi'[yt2],

    AiryAi'[ys1] AiryAi'[ys2] AiryAi'[yt1] AiryAi[yt2],
    AiryAi'[ys1] AiryAi'[ys2] AiryAi[yt1] AiryAi'[yt2],
    AiryAi'[ys1] AiryAi[ys2] AiryAi'[yt1] AiryAi'[yt2],
    AiryAi[ys1] AiryAi'[ys2] AiryAi'[yt1] AiryAi'[yt2],
    AiryAi'[ys1] AiryAi'[ys2] AiryAi'[yt1] AiryAi'[yt2]
}

Out[130]:= {Ai(ys1) Ai(ys2) Ai(yt1) Ai(yt2), Ai'(ys1) Ai(ys2) Ai(yt1) Ai(yt2),
    Ai(ys1) Ai'(ys2) Ai(yt1) Ai(yt2), Ai(ys1) Ai(ys2) Ai'(yt1) Ai(yt2), Ai(ys1) Ai(ys2) Ai(yt1) Ai'(yt2),
    Ai'(ys1) Ai'(ys2) Ai(yt1) Ai(yt2), Ai'(ys1) Ai(ys2) Ai'(yt1) Ai(yt2), Ai'(ys1) Ai(ys2) Ai(yt1) Ai'(yt2),
    Ai(ys1) Ai'(ys2) Ai'(yt1) Ai(yt2), Ai(ys1) Ai'(ys2) Ai(yt1) Ai'(yt2), Ai(ys1) Ai(ys2) Ai'(yt1) Ai'(yt2),
    Ai'(ys1) Ai'(ys2) Ai'(yt1) Ai(yt2), Ai'(ys1) Ai'(ys2) Ai(yt1) Ai'(yt2), Ai'(ys1) Ai(ys2) Ai'(yt1) Ai'(yt2),
    Ai(ys1) Ai'(ys2) Ai'(yt1) Ai'(yt2), Ai'(ys1) Ai'(ys2) Ai'(yt1) Ai'(yt2)}

```

In[131]:= Simplify[

Expr6 /. {D → 4},

Assumptions → {u > 0, v > 4, xp > 0, xq > 0, xg > 0, xf > 0, χf > 0, ξ > 0}];

Expr7 = PowerExpand[Collect[Expand[%], Collect4Airy, Simplify]];

Diffs7 = Diffs6

Coeff7 = Coeff6

Phase7 = Phase6

Expr7

Out[133]:= dp1 dp2 ds dt dτ1 dτ2 du dv

$$\text{Out[134]} = (\alpha^2 m^2 \xi^5 u) \left/ \left(16 \pi^2 L \phi p_0 v \sqrt{(v-4)v} \chi p \left(\sigma \sqrt{\frac{v-4}{v}} - 1 \right) \left(\sigma \sqrt{\frac{v-4}{v}} + 1 \right) \right. \right. \right. \\ \left. \left. \left. (\text{i} \epsilon \xi (u+1) - m^2 u (\xi u (\xi^2 \rho_1^2 + \tau_1^2 + 1) - 2 s \chi p)) (\text{i} \epsilon \xi (u+1) + m^2 u (\xi u (\xi^2 \rho_1^2 + \tau_1^2 + 1) - 2 t \chi p)) \right) \right)$$

Out[135]:= -((ρ1 - ρ2) (s - t))

$$\text{Out[136]} = \frac{1}{(u+1)^{5/3} v^{5/3} \xi^6} 32 \sqrt[3]{2} \chi p^{4/3} \\ \left(-8 v \tau_1^2 \tau_2^2 \xi^2 + 2 v^2 \tau_1^2 (\tau_2^2 + 1) \xi^2 + u^2 ((\tau_1^2 + 1) (\tau_2^2 + 1) v^2 - 2 (\tau_1^2 + 1) \tau_2^2 v + 8 (\xi^2 \rho_1^2 + \tau_1^2 + 1)^2) \xi^2 - \right. \\ \left. 2 u (4 v \xi \tau_2^2 \tau_1^2 - v^2 \xi (\tau_2^2 + 1) \tau_1^2 + 8 (s+t) (\xi^2 \rho_1^2 + \tau_1^2 + 1) \chi p) \xi + 32 s t \chi p^2 \right) \text{Ai}(ys_1) \text{Ai}(ys_2) \text{Ai}(yt_1) \\ \text{Ai}(yt_2) m^4 + \frac{1}{u^{2/3} (u+1)^{8/3} v^{5/3} \xi^4} 64 \tau_1 \tau_2 (u^{5/3} (u+1)^{2/3} (u+2) \chi p^{2/3} v^{5/3} + 2 u^{2/3} (u+1)^{2/3} \chi p^{2/3} v^{5/3} - \\ 2 ((u+1)^{2/3} v^{2/3} \chi p^{2/3} u^{8/3} + 4 (u+1)^{2/3} v^{2/3} \chi p^{2/3} u^{5/3} + 4 (u+1)^{2/3} v^{2/3} \chi p^{2/3} u^{2/3})) \chi p^{4/3} \text{Ai}(yt_1) \text{Ai}(yt_2) \\ \text{Ai}'(ys_1) \text{Ai}'(ys_2) m^4 + \frac{64 ((u^2 + 2 u + 2) v (\tau_2^2 + 1) - 2 u^2 \tau_2^2) \chi p^2 \text{Ai}(ys_2) \text{Ai}(yt_2) \text{Ai}'(ys_1) \text{Ai}'(yt_1) m^4}{u^{2/3} (u+1)^{5/3} v^{2/3} \xi^4} + \\ \frac{64 ((v-2) u^2 + 2 v u + 2 v) \tau_1 \tau_2 \chi p^2 \text{Ai}(ys_1) \text{Ai}(yt_2) \text{Ai}'(ys_2) \text{Ai}'(yt_1) m^4}{(u+1)^2 v \xi^4} + \\ \frac{64 ((v-2) u^2 + 2 v u + 2 v) \tau_1 \tau_2 \chi p^2 \text{Ai}(ys_2) \text{Ai}(yt_1) \text{Ai}'(ys_1) \text{Ai}'(yt_2) m^4}{(u+1)^2 v \xi^4} - \\ \frac{1}{u^2 (u+1)^{8/3} v^{8/3} \xi^4} 64 (-\sqrt[3]{u+1} v^{7/3} \chi p^{2/3} u^{14/3} + 2 \sqrt[3]{u+1} v^{4/3} \chi p^{2/3} u^{14/3} + \tau_1^2 \\ (-\sqrt[3]{u+1} v^{7/3} \chi p^{2/3} u^{14/3} + 2 \sqrt[3]{u+1} v^{4/3} \chi p^{2/3} u^{14/3} - 2 \sqrt[3]{u+1} v^{7/3} \chi p^{2/3} u^{11/3} - 2 \sqrt[3]{u+1} v^{7/3} \chi p^{2/3} u^{8/3})) \\ \chi p^{4/3} \text{Ai}(ys_1) \text{Ai}(yt_1) \text{Ai}'(ys_2) \text{Ai}'(yt_2) m^4 + \\ \frac{64 ((v-2) u^2 + 2 (v-4) u + 2 (v-4)) \tau_1 \tau_2 \chi p^2 \text{Ai}(ys_1) \text{Ai}(ys_2) \text{Ai}'(yt_1) \text{Ai}'(yt_2) m^4}{(u+1)^2 v \xi^4} - \\ \frac{1}{u^{2/3} (u+1)^{8/3} v^{8/3} \xi^4} 64 \times 2^{2/3} \chi p^{4/3} \\ (-\sqrt[3]{u+1} v^{7/3} \chi p^{4/3} u^{8/3} + 2 \sqrt[3]{u+1} v^{4/3} \chi p^{4/3} u^{8/3} - 2 \sqrt[3]{u+1} v^{7/3} \chi p^{4/3} u^{5/3} + 8 \sqrt[3]{u+1} v^{4/3} \chi p^{4/3} u^{5/3} - \\ 2 \sqrt[3]{u+1} v^{7/3} \chi p^{4/3} u^{2/3} + 8 \sqrt[3]{u+1} v^{4/3} \chi p^{4/3} u^{2/3}) \text{Ai}'(ys_1) \text{Ai}'(ys_2) \text{Ai}'(yt_1) \text{Ai}'(yt_2) m^4$$

At this stage, we observe that Expr7 contains a number of terms that are proportional to ρ_1^n , $n = 1, 2, 3, 4$. In what follows,

we will carry out integrals over $\rho_{1,2}$. Apparently, these terms can be removed by changing variables s and t , namely, by shifting them. This simplifies the integration greatly.

As another observation, we do not get any such terms proportional to powers of ρ_2

```
In[137]:= Coefficient[Expr7, {ρ1, ρ1^2, ρ1^3, ρ1^4}];
Coefficient[%, AiryAi[ys1] AiryAi[ys2] AiryAi[yt1] AiryAi[yt2]]
Out[138]= {0, - $\frac{512 \sqrt{2} m^4 s u \chi p^{7/3}}{\xi^3 (u+1)^{5/3} v^{5/3}} - \frac{512 \sqrt{2} m^4 t u \chi p^{7/3}}{\xi^3 (u+1)^{5/3} v^{5/3}} + \frac{512 \sqrt{2} m^4 \tau l^2 u^2 \chi p^{4/3}}{\xi^2 (u+1)^{5/3} v^{5/3}} + \frac{512 \sqrt{2} m^4 u^2 \chi p^{4/3}}{\xi^2 (u+1)^{5/3} v^{5/3}},$ 
0,  $\frac{256 \sqrt{2} m^4 u^2 \chi p^{4/3}}{(u+1)^{5/3} v^{5/3}}$ }
```

```
In[139]:= Coefficient[Expr7, {ρ2, ρ2^2, ρ2^3, ρ2^4}]
Out[139]= {0, 0, 0, 0}
```

Step 7: Shift of s and t and intermediate results

We shift s and t variables to simplify Coeff7 and remove the explicit dependence on τi and ρi in the denominator

Note that we use the property

$$i \in * X = i \in \text{for } X > 0$$

```
In[140]:= shift = {s → s +  $\frac{u \xi}{2 \chi p} (1 + \tau l^2 + \xi^2 \rho l^2)$ , t → t +  $\frac{u \xi}{2 \chi p} (1 + \tau l^2 + \xi^2 \rho l^2)$ };
factor8 = (8 m^2 u χ p / ξ^3 / (1 + u))^2;
Coeff8 = Simplify[Coeff7 * factor8 /. {ie → 0} /. shift] /.
{1 / s → 1 / (s + iε), 1 / t → 1 / (t - iε)}
Diffs8 = Diffs7;
Phase8 = Phase7;
```

$$\text{Out[142]} = -\frac{\alpha^2 m^2 u}{\pi^2 L \phi \xi p_0 (u+1)^2 \sqrt{(v-4)v} \chi p (i\epsilon + s) (t - i\epsilon) ((\sigma^2 - 1)v - 4\sigma^2)}$$

In[145]:= **Expand**[**Simplify**[**Expr7** / **factor8** /. **shift**] /. { $\left(\sqrt{\frac{v-4}{v}} - 1 \right) \left(\sqrt{\frac{v-4}{v}} + 1 \right) \rightarrow -\frac{4}{v} \}$];

Simplify[**PowerExpand**[%],

Assumptions → {**u** > 0, **v** > 4, **χp** > 0, **χq** > 0, **χg** > 0, **χf** > 0, **χf** > 0, **ξ** > 0}];

Expr8 = **Collect**[**Expand**[**PowerExpand**[%]], **Collect4Airy**, **Simplify**]

$$\begin{aligned} \text{Out[147]} = & \frac{1}{2^{2/3} u^2 v^{5/3} \chi p^{2/3}} \sqrt[3]{u+1} \text{Ai}(ys1) \text{Ai}(ys2) \text{Ai}(yt1) \text{Ai}(yt2) (32 s t \chi p^2 + \\ & \xi^2 (\tau_1^2 + 1) u^2 v (-2 \tau_2^2 + \tau_2^2 v + v) + 2 \xi^2 \tau_1^2 u v (-4 \tau_2^2 + \tau_2^2 v + v) + 2 \xi^2 \tau_1^2 (\tau_2^2 + 1) v^2 - 8 \xi^2 \tau_1^2 \tau_2^2 v) + \\ & \frac{2^{2/3} \xi^2 \chi p^{2/3} (u^2 (v-2) + 2 u (v-4) + 2 (v-4)) \text{Ai}'(ys1) \text{Ai}'(ys2) \text{Ai}'(yt1) \text{Ai}'(yt2)}{u^2 \sqrt[3]{(u+1) v^4}} + \\ & \frac{\xi^2 \tau_1 \tau_2 (u^2 (v-2) + 2 u (v-4) + 2 (v-4)) \text{Ai}'(ys1) \text{Ai}'(ys2) \text{Ai}(yt1) \text{Ai}(yt2)}{u^2 v} + \\ & \frac{\xi^2 \tau_1 \tau_2 (u^2 (v-2) + 2 u v + 2 v) \text{Ai}(ys1) \text{Ai}'(ys2) \text{Ai}'(yt1) \text{Ai}(yt2)}{u^2 v} + \\ & \frac{\xi^2 \tau_1 \tau_2 (u^2 (v-2) + 2 u v + 2 v) \text{Ai}'(ys1) \text{Ai}(ys2) \text{Ai}(yt1) \text{Ai}'(yt2)}{u^2 v} + \\ & \frac{\xi^2 \tau_1 \tau_2 (u^2 (v-2) + 2 u (v-4) + 2 (v-4)) \text{Ai}(ys1) \text{Ai}(ys2) \text{Ai}'(yt1) \text{Ai}'(yt2)}{u^2 v} + \\ & \frac{\xi^2 \sqrt[3]{u+1} \text{Ai}'(ys1) \text{Ai}(ys2) \text{Ai}'(yt1) \text{Ai}(yt2) ((\tau_2^2 + 1) (u^2 + 2 u + 2) v - 2 \tau_2^2 u^2)}{u^{8/3} v^{2/3}} + \\ & \frac{\xi^2 \text{Ai}(ys1) \text{Ai}'(ys2) \text{Ai}(yt1) \text{Ai}'(yt2) ((\tau_1^2 + 1) u^2 (v-2) + 2 \tau_1^2 u v + 2 \tau_1^2 v)}{\sqrt[3]{u^4 (u+1) v^4}} \end{aligned}$$

In[148]:= **(*must be all zeros*)**

Coefficient[**Expr8**, {**ρ1**, **ρ1**^2, **ρ1**^3, **ρ1**^4}]

Out[148]:= {0, 0, 0, 0}

Let us collect the intermediate result for Expr, expressed term by term (collected for different Airy function combinations)

In[149]:= **Coefficient**[**Expr8**, **AiryAi**[**ys1**] **AiryAi**[**ys2**] **AiryAi**[**yt1**] **AiryAi**[**yt2**]] /.
{**τ1** → **τ₁**, **τ2** → **τ₂**}

$$\text{Out[149]} = \frac{\sqrt[3]{u+1} (32 s t \chi p^2 + \xi^2 (\tau_1^2 + 1) u^2 v (-2 \tau_2^2 + \tau_2^2 v + v) + 2 \xi^2 \tau_1^2 u v (-4 \tau_2^2 + \tau_2^2 v + v) + 2 \xi^2 \tau_1^2 (\tau_2^2 + 1) v^2 - 8 \xi^2 \tau_1^2 \tau_2^2 v)}{2^{2/3} u^2 v^{5/3} \chi p^{2/3}}$$

In[150]:= **(Coefficient**[**Expr8**, **AiryAi**[**ys1**] **AiryAi**[**ys2**] **AiryAiPrime**[**yt1**] **AiryAiPrime**[**yt2**]] /.
{**τ1** → **τ₁**, **τ2** → **τ₂**}) * **AiryAi**[**ys1**] **AiryAi**[**ys2**] **AiryAiPrime**[**yt1**] **AiryAiPrime**[**yt2**]

$$\text{Out[150]} = \frac{\xi^2 \tau_1 \tau_2 (u^2 (v-2) + 2 u (v-4) + 2 (v-4)) \text{Ai}(ys1) \text{Ai}(ys2) \text{Ai}'(yt1) \text{Ai}'(yt2)}{u^2 v}$$

In[151]:= **Coefficient**[Expr8, **AiryAiPrime**[ys1] **AiryAiPrime**[ys2] **AiryAi**[yt1] **AiryAi**[yt2]] /.
 $\{\tau_1 \rightarrow \tau_1, \tau_2 \rightarrow \tau_2\}$

$$\text{Out[151]} = \frac{\xi^2 \tau_1 \tau_2 (u^2 (v-2) + 2 u (v-4) + 2 (v-4))}{u^2 v}$$

In[152]:= **Coefficient**[Expr8, **AiryAiPrime**[ys1] **AiryAi**[ys2] **AiryAiPrime**[yt1] **AiryAi**[yt2]] /.
 $\{\tau_1 \rightarrow \tau_1, \tau_2 \rightarrow \tau_2\}$

$$\text{Out[152]} = \frac{\xi^2 \sqrt[3]{u+1} ((\tau_2^2 + 1)(u^2 + 2 u + 2)v - 2 \tau_2^2 u^2)}{u^{8/3} v^{2/3}}$$

In[153]:= **Coefficient**[Expr8, **AiryAi**[ys1] **AiryAiPrime**[ys2] **AiryAi**[yt1] **AiryAiPrime**[yt2]] /.
 $\{\tau_1 \rightarrow \tau_1, \tau_2 \rightarrow \tau_2\}$

$$\text{Out[153]} = \frac{\xi^2 ((\tau_1^2 + 1)u^2 (v-2) + 2 \tau_1^2 u v + 2 \tau_1^2 v)}{\sqrt[3]{u^4 (u+1) v^4}}$$

In[154]:= **Coefficient**[Expr8, **AiryAiPrime**[ys1] **AiryAi**[ys2] **AiryAi**[yt1] **AiryAiPrime**[yt2]] /.
 $\{\tau_1 \rightarrow \tau_1, \tau_2 \rightarrow \tau_2\}$

$$\text{Out[154]} = \frac{\xi^2 \tau_1 \tau_2 (u^2 (v-2) + 2 u v + 2 v)}{u^2 v}$$

In[155]:= **Coefficient**[Expr8, **AiryAi**[ys1] **AiryAiPrime**[ys2] **AiryAiPrime**[yt1] **AiryAi**[yt2]] /.
 $\{\tau_1 \rightarrow \tau_1, \tau_2 \rightarrow \tau_2\}$

$$\text{Out[155]} = \frac{\xi^2 \tau_1 \tau_2 (u^2 (v-2) + 2 u v + 2 v)}{u^2 v}$$

In[156]:= **Coefficient**[Expr8, **AiryAiPrime**[ys1] **AiryAiPrime**[ys2]
AiryAiPrime[yt1] **AiryAiPrime**[yt2]] /. $\{\tau_1 \rightarrow \tau_1, \tau_2 \rightarrow \tau_2\}$

$$\text{Out[156]} = \frac{2^{2/3} \xi^2 \chi p^{2/3} (u^2 (v-2) + 2 u (v-4) + 2 (v-4))}{u^2 \sqrt[3]{(u+1) v^4}}$$

In[157]:= **Coefficient**[Expr8, **AiryAiPrime**[ys1] **AiryAiPrime**[ys2]
AiryAiPrime[yt1] **AiryAiPrime**[yt2]] /. $\{\tau_1 \rightarrow \tau_1, \tau_2 \rightarrow \tau_2\}$

$$\text{Out[157]} = \frac{2^{2/3} \xi^2 \chi p^{2/3} (u^2 (v-2) + 2 u (v-4) + 2 (v-4))}{u^2 \sqrt[3]{(u+1) v^4}}$$

Explicit expressions for the y -
arguments of the Airy functions after the shift
They depend on τ_1^2 , but are independent of ρ_1

```
In[158]:= ys1new = Simplify[ysqp /. subsvσ /. subsu /. shift] /. {σ^2 → 1}
ys2new = Simplify[ystsfg /. subsvσ /. subsu /. shift] /. {σ^2 → 1}
yt1new = Simplify[ytqp /. subsvσ /. subsu /. shift] /. {σ^2 → 1}
yt2new = Simplify[ysttfg /. subsvσ /. subsu /. shift] /. {σ^2 → 1}
```

$$\text{Out[158]} = \frac{2 s \chi p + \xi (\tau^2 + 1) u}{2^{2/3} \xi \sqrt[3]{u \chi p^2}}$$

$$\text{Out[159]} = -\frac{4 s u \chi p - 2 \xi (\tau^2 + 1) (u + 1) v}{2 \times 2^{2/3} \xi u \chi p \sqrt[3]{\frac{u v + v}{u \chi p}}}$$

$$\text{Out[160]} = \frac{2 t \chi p + \xi (\tau^2 + 1) u}{2^{2/3} \xi \sqrt[3]{u \chi p^2}}$$

$$\text{Out[161]} = -\frac{4 t u \chi p - 2 \xi (\tau^2 + 1) (u + 1) v}{2 \times 2^{2/3} \xi u \chi p \sqrt[3]{\frac{u v + v}{u \chi p}}}$$

The integral over τ_1 is symmetric from $-\infty$ to $+\infty$;
The Airy functions are even functions of τ_1 , hence,
all terms proportional to τ_1 in an odd power give zero upon integration;

We remove such terms in this step

```
In[162]:= Expr9 = Simplify[(Expr8 + (Expr8 /. {τ1 → -τ1})) / 2]
Diffs9 = Diffs8
Coeff9 = Coeff8
Phase9 = Phase8
```

$$\begin{aligned} \text{Out[162]} = & \frac{1}{2^{2/3} u^2 v^{5/3} \chi p^{2/3}} \sqrt[3]{u+1} \text{Ai}(ys1) \text{Ai}(ys2) \text{Ai}(yt1) \text{Ai}(yt2) (32 s t \chi p^2 + \\ & \xi^2 (\tau^2 + 1) u^2 v (-2 \tau^2 + \tau^2 v + v) + 2 \xi^2 \tau^2 u v (-4 \tau^2 + \tau^2 v + v) + 2 \xi^2 \tau^2 (\tau^2 + 1) v^2 - 8 \xi^2 \tau^2 \tau^2 v) + \\ & \frac{2^{2/3} \xi^2 \chi p^{2/3} (u^2 (v-2) + 2 u (v-4) + 2 (v-4)) \text{Ai}'(ys1) \text{Ai}'(ys2) \text{Ai}'(yt1) \text{Ai}'(yt2)}{u^2 \sqrt[3]{(u+1) v^4}} + \\ & \frac{\xi^2 \sqrt[3]{u+1} \text{Ai}'(ys1) \text{Ai}(ys2) \text{Ai}'(yt1) \text{Ai}(yt2) ((\tau^2 + 1) (u^2 + 2 u + 2) v - 2 \tau^2 u^2)}{u^{8/3} v^{2/3}} + \\ & \frac{\xi^2 \text{Ai}(ys1) \text{Ai}'(ys2) \text{Ai}(yt1) \text{Ai}'(yt2) ((\tau^2 + 1) u^2 (v-2) + 2 \tau^2 u v + 2 \tau^2 v)}{\sqrt[3]{u^4 (u+1) v^4}} \end{aligned}$$

$$\text{Out[163]} = d\rho_1 d\rho_2 ds dt d\tau_1 d\tau_2 du dv$$

$$\text{Out[164]} = -\frac{\alpha^2 m^2 u}{\pi^2 L \phi \xi p_0 (u+1)^2 \sqrt{(v-4) v} \chi p (i\epsilon + s) (t - i\epsilon) ((\sigma^2 - 1) v - 4 \sigma^2)}$$

$$\text{Out[165]} = -((\rho_1 - \rho_2) (s - t))$$

Step 8: Integration over ρ_1, ρ_2, s and t ; one- and two-step contributions

Integrals over $d\rho_1 d\rho_2 ds dt$ are considered together and are regularised as described in section 6.1 in [arXiv 2605.12479] (eqs. (6.36) , (6.37))

The procedure amounts to

$$\int d\rho_1 d\rho_2 ds dt \text{Coeff}_9 * \text{Expr}_9 * \text{Exp}[-i (\rho_1 - \rho_2) (s - t)] \rightarrow I_{1st} + I_{2st}$$

where

$$I_{1st} = 2 \pi L_\phi \int \frac{ds}{s^2} \text{Coeff}_{1st} * (G[s] - G[-s] - 2 G[0])$$

$$I_{2st} = 2 \pi^2 L_\phi^2 \text{Coeff}_{2st} * G[0]$$

correspond to one - and two - step contributions, respectively

factor10 is an overall factor which we transfer from the Expr to Coeff

We introduce G_s and split Expr into two contributions, Expr_{1st} and Expr_{2st} , which we track separately

Phase is zero in this step, so we drop the corresponding variable

```
In[166]:= factor10 = 8 m^4;
Gs = Expr9 / factor10 /. {t -> s} /. {yt1 -> ys1, yt2 -> ys2}
Phase10 = Phase9 /. {t -> s}

Out[167]= 
$$\frac{1}{8 m^4} \left( \frac{1}{2^{2/3} u^2 v^{5/3} \chi p^{2/3}} \sqrt[3]{u+1} \text{Ai}(ys1)^2 \text{Ai}(ys2)^2 (32 s^2 \chi p^2 + \xi^2 (\tau^2 + 1) u^2 v (-2 \tau^2 + \tau^2 v + v) + \right.$$


$$2 \xi^2 \tau^2 u v (-4 \tau^2 + \tau^2 v + v) + 2 \xi^2 \tau^2 (\tau^2 + 1) v^2 - 8 \xi^2 \tau^2 \tau^2 v) +$$


$$\frac{2^{2/3} \xi^2 \chi p^{2/3} (u^2 (v - 2) + 2 u (v - 4) + 2 (v - 4)) \text{Ai}'(ys1)^2 \text{Ai}'(ys2)^2}{u^2 \sqrt[3]{(u+1) v^4}} +$$


$$\frac{\xi^2 \sqrt[3]{u+1} \text{Ai}'(ys1)^2 \text{Ai}(ys2)^2 ((\tau^2 + 1) (u^2 + 2 u + 2) v - 2 \tau^2 u^2)}{u^{8/3} v^{2/3}} +$$


$$\left. \frac{\xi^2 \text{Ai}(ys1)^2 \text{Ai}'(ys2)^2 ((\tau^2 + 1) u^2 (v - 2) + 2 \tau^2 u v + 2 \tau^2 v)}{\sqrt[3]{u^4 (u+1) v^4}} \right)$$

```

Out[168]= 0

One - step contribution

$ys1$ and $ys2$ are $y_{1,2} (s)$ arguments, introduced above
 $yms1$ and $yms1$ are the same arguments with $s \rightarrow -s$

y01 and y02 are given by $y_{1,2}(0)$

```
In[169]:= DiffS101st = DiffS9 /. {dt → 1, dρ1 → 1, dρ2 → 1}
Coeff101st = Simplify[Coeff9 * 2 π L ϕ * factor10 /. {t → s} /. {ie → 0}]
Expr101st =
  Gs + (Gs /. {s → -s, ys1 → yms1, ys2 → yms2}) - 2 (Gs /. {s → 0, ys1 → y01, ys2 → y02})
Out[169]= ds dτ1 dτ2 du dv
```

$$\text{Out[170]= } -\frac{16 \alpha^2 m^6 u}{\pi \xi p_0 s^2 (u+1)^2 \sqrt{(v-4)v} \chi p ((\sigma^2-1)v-4\sigma^2)}$$

$$\text{Out[171]= } -\frac{1}{4 m^4} \left(\frac{1}{2^{2/3} u^2 v^{5/3} \chi p^{2/3}} \sqrt[3]{u+1} \right. \\
\begin{aligned} & (-8 v \tau_1^2 \tau_2^2 \xi^2 + 2 v^2 \tau_1^2 (\tau_2^2 + 1) \xi^2 + 2 u v \tau_1^2 (v \tau_2^2 - 4 \tau_2^2 + v) \xi^2 + u^2 v (\tau_1^2 + 1) (v \tau_2^2 - 2 \tau_2^2 + v) \xi^2) \\ & \text{Ai}(y_{01})^2 \text{Ai}(y_{02})^2 + \frac{\sqrt[3]{u+1} \xi^2 ((u^2 + 2 u + 2) v (\tau_2^2 + 1) - 2 u^2 \tau_2^2) \text{Ai}'(y_{01})^2 \text{Ai}(y_{02})^2}{u^{8/3} v^{2/3}} + \\ & \frac{\xi^2 ((v-2) (\tau_1^2 + 1) u^2 + 2 v \tau_1^2 u + 2 v \tau_1^2) \text{Ai}(y_{01})^2 \text{Ai}'(y_{02})^2}{\sqrt[3]{u^4 (u+1) v^4}} + \\ & \left. \frac{2^{2/3} ((v-2) u^2 + 2 (v-4) u + 2 (v-4)) \xi^2 \chi p^{2/3} \text{Ai}'(y_{01})^2 \text{Ai}'(y_{02})^2}{u^2 \sqrt[3]{(u+1) v^4}} \right) + \\ & \frac{1}{8 m^4} \left(\frac{1}{2^{2/3} u^2 v^{5/3} \chi p^{2/3}} \sqrt[3]{u+1} \right. \\ & (-8 v \tau_1^2 \tau_2^2 \xi^2 + 2 v^2 \tau_1^2 (\tau_2^2 + 1) \xi^2 + 2 u v \tau_1^2 (v \tau_2^2 - 4 \tau_2^2 + v) \xi^2 + \\ & u^2 v (\tau_1^2 + 1) (v \tau_2^2 - 2 \tau_2^2 + v) \xi^2 + 32 s^2 \chi p^2) \text{Ai}(y_{ms1})^2 \text{Ai}(y_{ms2})^2 + \\ & \frac{\sqrt[3]{u+1} \xi^2 ((u^2 + 2 u + 2) v (\tau_2^2 + 1) - 2 u^2 \tau_2^2) \text{Ai}'(y_{ms1})^2 \text{Ai}(y_{ms2})^2}{u^{8/3} v^{2/3}} + \\ & \frac{\xi^2 ((v-2) (\tau_1^2 + 1) u^2 + 2 v \tau_1^2 u + 2 v \tau_1^2) \text{Ai}(y_{ms1})^2 \text{Ai}'(y_{ms2})^2}{\sqrt[3]{u^4 (u+1) v^4}} + \\ & \left. \frac{2^{2/3} ((v-2) u^2 + 2 (v-4) u + 2 (v-4)) \xi^2 \chi p^{2/3} \text{Ai}'(y_{ms1})^2 \text{Ai}'(y_{ms2})^2}{u^2 \sqrt[3]{(u+1) v^4}} \right) + \\ & \frac{1}{8 m^4} \left(\frac{1}{2^{2/3} u^2 v^{5/3} \chi p^{2/3}} \sqrt[3]{u+1} \right. \\ & (-8 v \tau_1^2 \tau_2^2 \xi^2 + 2 v^2 \tau_1^2 (\tau_2^2 + 1) \xi^2 + 2 u v \tau_1^2 (v \tau_2^2 - 4 \tau_2^2 + v) \xi^2 + \\ & u^2 v (\tau_1^2 + 1) (v \tau_2^2 - 2 \tau_2^2 + v) \xi^2 + 32 s^2 \chi p^2) \text{Ai}(y_{s1})^2 \text{Ai}(y_{s2})^2 + \\ & \frac{\sqrt[3]{u+1} \xi^2 ((u^2 + 2 u + 2) v (\tau_2^2 + 1) - 2 u^2 \tau_2^2) \text{Ai}'(y_{s1})^2 \text{Ai}(y_{s2})^2}{u^{8/3} v^{2/3}} + \\ & \frac{\xi^2 ((v-2) (\tau_1^2 + 1) u^2 + 2 v \tau_1^2 u + 2 v \tau_1^2) \text{Ai}(y_{s1})^2 \text{Ai}'(y_{s2})^2}{\sqrt[3]{u^4 (u+1) v^4}} + \\ & \left. \frac{2^{2/3} ((v-2) u^2 + 2 (v-4) u + 2 (v-4)) \xi^2 \chi p^{2/3} \text{Ai}'(y_{s1})^2 \text{Ai}'(y_{s2})^2}{u^2 \sqrt[3]{(u+1) v^4}} \right) \end{aligned}$$

Two – step contribution

Note that the integral over s is carried out in this term

```
In[172]:= Diffsl02 st = Diffs9 /. {ds → 1, dt → 1, dp1 → 1, dp2 → 1}
Coeffl02 st = Simplify[Coeff9 * 2 π^2 Lφ^2 * factor10 /. {t → 1, s → 1} /. {ie → 0}]
Exprl02 st = Gs /. {s → 0, ys1 → y01, ys2 → y02}
```

```
Out[172]= dr1 dτ2 du dv
```

$$\text{Out[173]} = -\frac{16 \alpha^2 L \phi m^6 u}{\xi p_0 (u+1)^2 \sqrt{(v-4)v} \chi p ((\sigma^2-1)v - 4\sigma^2)}$$

$$\begin{aligned} \text{Out[174]} = & \frac{1}{8 m^4} \left(\frac{2^{2/3} \xi^2 \chi p^{2/3} (u^2 (v-2) + 2u(v-4) + 2(v-4)) \text{Ai}'(y_{01})^2 \text{Ai}'(y_{02})^2}{u^2 \sqrt[3]{(u+1)v^4}} + \frac{1}{2^{2/3} u^2 v^{5/3} \chi p^{2/3}} \sqrt[3]{u+1} \text{Ai}(y_{01})^2 \text{Ai}(y_{02})^2 \right. \\ & \frac{(\xi^2 (\tau^2 + 1) u^2 v (-2\tau^2 + \tau^2 v + v) + 2\xi^2 \tau^2 u v (-4\tau^2 + \tau^2 v + v) + 2\xi^2 \tau^2 (\tau^2 + 1) v^2 - 8\xi^2 \tau^2 \tau^2 v) + \xi^2 \sqrt[3]{u+1} \text{Ai}'(y_{01})^2 \text{Ai}(y_{02})^2 ((\tau^2 + 1)(u^2 + 2u + 2)v - 2\tau^2 u^2)}{u^{8/3} v^{2/3}} + \\ & \left. \frac{\xi^2 \text{Ai}(y_{01})^2 \text{Ai}'(y_{02})^2 ((\tau^2 + 1) u^2 (v-2) + 2\tau^2 u v + 2\tau^2 v)}{\sqrt[3]{u^4 (u+1)v^4}} \right) \end{aligned}$$

Step 9: Integration over τ

Integrate over τ

$$\int d\tau \tau^{2k} \text{Ai}^2(a + x \tau^2) = \frac{\Gamma(k+1/2)}{2^{2k/3+1} \pi^{1/2} x^{k+1/2}} \text{Ai}_{k+1}(2^{2/3} a)$$

$$\int d\tau \text{Ai}^2(a + x \tau^2) = \frac{1}{2 x^{1/2}} \text{Ai}_1(2^{2/3} a)$$

$$\int d\tau \tau^2 \text{Ai}^2(a + x \tau^2) =$$

$$\frac{1}{4 \times 2^{2/3} x^{3/2}} \text{Ai}_2(2^{2/3} a) = \frac{1}{4 \times 2^{2/3} x^{3/2}} [-\text{Ai}'(2^{2/3} a) - 2^{2/3} a \text{Ai}_1(2^{2/3} a)]$$

$$\int d\tau \text{Ai}'^2(a + x \tau^2) = \frac{1}{4 \times 2^{2/3} x^{1/2}} [-3 \text{Ai}'(2^{2/3} a) - 2^{2/3} a \text{Ai}_1(2^{2/3} a)]$$

$$\text{Ai}_2(y) = -\text{Ai}'(y) - y \text{Ai}_1(y)$$

$ys_i = a_i + x_i \tau_i$, see a_i and x_i below

We will perform the integration for G_s

and express the overall result via integrated G_s

Explicit representation for the Airy function arguments

```
In[175]:= Expand[PowerExpand[ys1new]]
a1expl = (u / 2 / xp) ^ (2 / 3) Expand[Simplify[(% / (u / 2 / xp) ^ (2 / 3) /. {τ1 → 0})]]
x1expl = Coefficient[%%, τ1^2]
```

$$\text{Out[175]} = \frac{\sqrt[3]{2} s \sqrt[3]{xp}}{\xi \sqrt[3]{u}} + \frac{\tau^2 u^{2/3}}{2^{2/3} xp^{2/3}} + \frac{u^{2/3}}{2^{2/3} xp^{2/3}}$$

$$\text{Out[176]} = \frac{\left(\frac{u}{xp}\right)^{2/3} \left(\frac{2 s xp}{\xi u} + 1\right)}{2^{2/3}}$$

$$\text{Out[177]} = \frac{u^{2/3}}{2^{2/3} xp^{2/3}}$$

```
In[178]:= 2 ^ (2 / 3) \left( \frac{(u + 1) v}{4 u xp} \right)^{2/3} *
```

```
Collect[Expand[PowerExpand[ys2new / 2 ^ (2 / 3) / \left( \frac{(u + 1) v}{4 u xp} \right)^{2/3}]], τ2, Simplify]
```

```
a2expl = % /. {τ2 → 0}
x2expl = Coefficient[%%, τ2^2]
```

$$\text{Out[178]} = \frac{\left(\frac{(u+1)v}{u xp}\right)^{2/3} \left(-\frac{2 s u xp}{\xi u v + \xi v} + \tau^2 + 1\right)}{2^{2/3}}$$

$$\text{Out[179]} = \frac{\left(\frac{(u+1)v}{u xp}\right)^{2/3} \left(1 - \frac{2 s u xp}{\xi u v + \xi v}\right)}{2^{2/3}}$$

$$\text{Out[180]} = \frac{\left(\frac{(u+1)v}{u xp}\right)^{2/3}}{2^{2/3}}$$

```
In[181]:= 2 ^ (2 / 3) Simplify[a1expl, Assumptions → assumpAll]
Simplify[% /. {s → 0}, Assumptions → assumpAll]
2 ^ (2 / 3) \left( \frac{(u + 1) v}{4 u xp} \right)^{2/3} Simplify[a2expl / \left( \frac{(u + 1) v}{4 u xp} \right)^{2/3}, Assumptions → assumpAll]
Simplify[% /. {s → 0}, Assumptions → assumpAll]
```

$$\text{Out[181]} = \left(\frac{u}{xp}\right)^{2/3} \left(\frac{2 s xp}{\xi u} + 1\right)$$

$$\text{Out[182]} = \left(\frac{u}{xp}\right)^{2/3}$$

$$\text{Out[183]} = \left(\frac{(u+1)v}{u xp}\right)^{2/3} \left(1 - \frac{2 s u xp}{\xi u v + \xi v}\right)$$

$$\text{Out[184]} = \left(\frac{(u+1)v}{u xp}\right)^{2/3}$$

The following expression is used later for collecting terms


```
In[185]:= collectAirya = {AiryAiPrime[22/3 a1] AiryAiPrime[22/3 a2],
  Ai1[22/3 a1] Ai1[22/3 a2],
  AiryAiPrime[22/3 a1] Ai1[22/3 a2],
  Ai1[22/3 a1] AiryAiPrime[22/3 a2]};
```

Integration of Gs over $\tau_{1,2}$ is
implemented in this step via substitutions in Gs

Result is stored in GsIntegrated

```
In[186]:= Collect[Gs, {τ1, AiryAi[ys1]2, AiryAiPrime[ys1]2}, Simplify] /.
  {τ12 AiryAi[ys1]2 →
    
$$\frac{1}{4 \times 2^{2/3} x \exp^{3/2}} (-\text{AiryAiPrime}[2^{2/3} a1] - 2^{2/3} a1 \exp \text{Ai}_1[2^{2/3} a1]) \} /.
  \{\text{AiryAi}[ys1]^2 \rightarrow \frac{1}{2 x \exp^{1/2}} \text{Ai}_1[2^{2/3} a1], \text{AiryAiPrime}[ys1]^2 \rightarrow
    \frac{1}{4 \times 2^{2/3} x \exp^{1/2}} (-3 \text{AiryAiPrime}[2^{2/3} a1] - 2^{2/3} a1 \exp \text{Ai}_1[2^{2/3} a1]) \};
Expand[\%];
Collect[\%, {τ2, AiryAi[ys2]2, AiryAiPrime[ys2]2}, Simplify] /.
  {τ22 AiryAi[ys2]2 →
    
$$\frac{1}{4 \times 2^{2/3} x2 \exp^{3/2}} (-\text{AiryAiPrime}[2^{2/3} a2] - 2^{2/3} a2 \exp \text{Ai}_1[2^{2/3} a2]) \} /.
  \{\text{AiryAi}[ys2]^2 \rightarrow \frac{1}{2 x2 \exp^{1/2}} \text{Ai}_1[2^{2/3} a2], \text{AiryAiPrime}[ys2]^2 \rightarrow
    \frac{1}{4 \times 2^{2/3} x2 \exp^{1/2}} (-3 \text{AiryAiPrime}[2^{2/3} a2] - 2^{2/3} a2 \exp \text{Ai}_1[2^{2/3} a2]) \};
PowerExpand[\%];
GsIntegrated = Collect[Expand[\%],
  collectAirya, Simplify]$$$$

```

```
Out[190]= 
$$-\frac{1}{8 m^4 u^3 (u+1) v^2} \text{Ai}_1(2^{2/3} a1) \text{Ai}_1(2^{2/3} a2) (u v (\xi^2 + 2 s^2 \chi p^2 + 4 \xi s \chi p) - 12 s^2 u \chi p^2 +$$


$$u^3 (\xi + s \chi p) (v (\xi + s \chi p) - 2 s \chi p) + 2 u^2 (v (\xi + s \chi p)^2 - s \chi p (\xi + 6 s \chi p)) + 2 \xi s v \chi p) -$$


$$\frac{\xi (u^2 + 2 u + 2) \chi p^{2/3} \text{Ai}'(2^{2/3} a1) \text{Ai}_1(2^{2/3} a2) (v (\xi + s u \chi p + \xi u) - 2 s u \chi p)}{8 m^4 u^{8/3} (u+1) v^2} +$$


$$\frac{\xi (v - 2) \chi p^{2/3} \text{Ai}_1(2^{2/3} a1) \text{Ai}'(2^{2/3} a2) (u^2 (\xi + s \chi p) + u (\xi + 2 s \chi p) + 2 s \chi p)}{8 m^4 \sqrt[3]{u^7} (u+1)^2 v^5} +$$


$$\frac{\xi^2 \chi p^{4/3} \text{Ai}'(2^{2/3} a1) \text{Ai}'(2^{2/3} a2) (u^2 (v - 2) + u (2 v - 5) + 2 v - 5)}{8 m^4 u^2 (u+1)^{2/3} v^{5/3}}$$

```

One step term ;
Expressed via an integral over s

(equivalent of variable T in [arXiv 2605.12479] in Eq. (6.36))

```
In[191]:= Diffs111st = Diffs101st /. {dτ1 → 1, dτ2 → 1}
Coeff111st = Coeff101st
Expr111st = (GsIntegrated /. {a1 → a1expl, a2 → a2expl}) +
  (GsIntegrated /. {a1 → a1expl, a2 → a2expl} /. {s → -s}) -
  2 (GsIntegrated /. {a1 → a1expl, a2 → a2expl} /. {s → 0})
Out[191]= ds du dv
```

$$\text{Out[192]} = -\frac{16 \alpha^2 m^6 u}{\pi \xi p_0 s^2 (u+1)^2 \sqrt{(v-4)v} \chi_P((\sigma^2-1)v-4\sigma^2)}$$

Out[193]=

$$\begin{aligned}
& \frac{((v-2)u^2 + (2v-5)u + 2v-5)\chi p^{4/3} \operatorname{Ai}'\left(\left(\frac{u}{\chi p}\right)^{2/3} \left(\frac{2s\chi p}{u\xi} + 1\right)\right) \operatorname{Ai}'\left(\left(\frac{(u+1)v}{u\chi p}\right)^{2/3} \left(1 - \frac{2su\chi p}{uv\xi + v\xi}\right)\right) \xi^2}{8m^4 u^2 (u+1)^{2/3} v^{5/3}} + \\
& \frac{((v-2)u^2 + (2v-5)u + 2v-5)\chi p^{4/3} \operatorname{Ai}'\left(\left(\frac{u}{\chi p}\right)^{2/3} \left(1 - \frac{2s\chi p}{u\xi}\right)\right) \operatorname{Ai}'\left(\left(\frac{(u+1)v}{u\chi p}\right)^{2/3} \left(\frac{2su\chi p}{uv\xi + v\xi} + 1\right)\right) \xi^2}{8m^4 u^2 (u+1)^{2/3} v^{5/3}} + \\
& \frac{(v-2)\chi p^{2/3} ((\xi - s\chi p)u^2 + (\xi - 2s\chi p)u - 2s\chi p) \operatorname{Ai}'\left(\left(\frac{(u+1)v}{u\chi p}\right)^{2/3} \left(\frac{2su\chi p}{uv\xi + v\xi} + 1\right)\right) \operatorname{Ai}_1\left(\left(\frac{u}{\chi p}\right)^{2/3} \left(1 - \frac{2s\chi p}{u\xi}\right)\right) \xi}{8m^4 \sqrt[3]{u^7} (u+1)^2 v^5} + \\
& \frac{(v-2)\chi p^{2/3} ((\xi + s\chi p)u^2 + (\xi + 2s\chi p)u + 2s\chi p) \operatorname{Ai}'\left(\left(\frac{(u+1)v}{u\chi p}\right)^{2/3} \left(1 - \frac{2su\chi p}{uv\xi + v\xi}\right)\right) \operatorname{Ai}_1\left(\left(\frac{u}{\chi p}\right)^{2/3} \left(\frac{2s\chi p}{u\xi} + 1\right)\right) \xi}{8m^4 \sqrt[3]{u^7} (u+1)^2 v^5} - \\
& \frac{(u^2 + 2u + 2)\chi p^{2/3} (v(u\xi + \xi + s\chi p) - 2su\chi p) \operatorname{Ai}'\left(\left(\frac{u}{\chi p}\right)^{2/3} \left(\frac{2s\chi p}{u\xi} + 1\right)\right) \operatorname{Ai}_1\left(\left(\frac{(u+1)v}{u\chi p}\right)^{2/3} \left(1 - \frac{2su\chi p}{uv\xi + v\xi}\right)\right) \xi}{8m^4 u^{8/3} (u+1)v^2} - \\
& \frac{(u^2 + 2u + 2)\chi p^{2/3} (2su\chi p + v(u\xi + \xi - s\chi p)) \operatorname{Ai}'\left(\left(\frac{u}{\chi p}\right)^{2/3} \left(1 - \frac{2s\chi p}{u\xi}\right)\right) \operatorname{Ai}_1\left(\left(\frac{(u+1)v}{u\chi p}\right)^{2/3} \left(\frac{2su\chi p}{uv\xi + v\xi} + 1\right)\right) \xi}{8m^4 u^{8/3} (u+1)v^2} - \\
& 2 \left(\frac{((v-2)u^2 + (2v-5)u + 2v-5)\chi p^{4/3} \operatorname{Ai}'\left(\left(\frac{u}{\chi p}\right)^{2/3}\right) \operatorname{Ai}'\left(\left(\frac{(u+1)v}{u\chi p}\right)^{2/3}\right) \xi^2}{8m^4 u^2 (u+1)^{2/3} v^{5/3}} + \right. \\
& \frac{(v-2)(\xi u^2 + \xi u)\chi p^{2/3} \operatorname{Ai}'\left(\left(\frac{(u+1)v}{u\chi p}\right)^{2/3}\right) \operatorname{Ai}_1\left(\left(\frac{u}{\chi p}\right)^{2/3}\right) \xi}{8m^4 \sqrt[3]{u^7} (u+1)^2 v^5} - \\
& \frac{(u^2 + 2u + 2)(u\xi + \xi)\chi p^{2/3} \operatorname{Ai}'\left(\left(\frac{u}{\chi p}\right)^{2/3}\right) \operatorname{Ai}_1\left(\left(\frac{(u+1)v}{u\chi p}\right)^{2/3}\right) \xi}{8m^4 u^{8/3} (u+1)v} - \\
& \left. \frac{(v\xi^2 u^3 + 2v\xi^2 u^2 + v\xi^2 u) \operatorname{Ai}_1\left(\left(\frac{u}{\chi p}\right)^{2/3}\right) \operatorname{Ai}_1\left(\left(\frac{(u+1)v}{u\chi p}\right)^{2/3}\right)}{8m^4 u^3 (u+1)v^2} \right) - \\
& \frac{1}{8m^4 u^3 (u+1)v^2} ((\xi + s\chi p)(v(\xi + s\chi p) - 2s\chi p)u^3 + 2(v(\xi + s\chi p)^2 - s\chi p(\xi + 6s\chi p))u^2 - 12s^2\chi p^2 u + \\
& v(\xi^2 + 4s\chi p\xi + 2s^2\chi p^2)u + 2sv\xi\chi p) \operatorname{Ai}_1\left(\left(\frac{u}{\chi p}\right)^{2/3} \left(\frac{2s\chi p}{u\xi} + 1\right)\right) \operatorname{Ai}_1\left(\left(\frac{(u+1)v}{u\chi p}\right)^{2/3} \left(1 - \frac{2su\chi p}{uv\xi + v\xi}\right)\right) - \\
& \frac{1}{8m^4 u^3 (u+1)v^2} ((\xi - s\chi p)(2s\chi p + v(\xi - s\chi p))u^3 + 2(v(\xi - s\chi p)^2 + s\chi p(\xi - 6s\chi p))u^2 - 12s^2\chi p^2 u + \\
& v(\xi^2 - 4s\chi p\xi + 2s^2\chi p^2)u - 2sv\xi\chi p) \operatorname{Ai}_1\left(\left(\frac{u}{\chi p}\right)^{2/3} \left(1 - \frac{2s\chi p}{u\xi}\right)\right) \operatorname{Ai}_1\left(\left(\frac{(u+1)v}{u\chi p}\right)^{2/3} \left(\frac{2su\chi p}{uv\xi + v\xi} + 1\right)\right)
\end{aligned}$$

Two - step term

```

In[194]:= Diffss112 st = Diffss102 st /. {dτ1 → 1, dτ2 → 1}
Coeff112 st = Coeff102 st
Expr112 st = GsIntegrated /. {a1 → a1expl, a2 → a2expl} /. {s → 0}
Out[194]= du dv

```

$$\begin{aligned}
\text{Out[195]} &= -\frac{16 \alpha^2 L \phi m^6 u}{\xi p_0 (u+1)^2 \sqrt{(v-4)v} \chi p ((\sigma^2 - 1)v - 4 \sigma^2)} \\
\text{Out[196]} &= -\frac{\xi (u^2 + 2u + 2) \chi p^{2/3} (\xi + \xi u) \text{Ai}'\left(\left(\frac{u}{\chi p}\right)^{2/3}\right) \text{Ai}_1\left(\left(\frac{(u+1)v}{u \chi p}\right)^{2/3}\right)}{8 m^4 u^{8/3} (u+1)v} + \\
&\quad \frac{\xi (v-2) \chi p^{2/3} (\xi u^2 + \xi u) \text{Ai}_1\left(\left(\frac{u}{\chi p}\right)^{2/3}\right) \text{Ai}'\left(\left(\frac{(u+1)v}{u \chi p}\right)^{2/3}\right) - \text{Ai}_1\left(\left(\frac{u}{\chi p}\right)^{2/3}\right) (\xi^2 u^3 v + 2 \xi^2 u^2 v + \xi^2 u v) \text{Ai}_1\left(\left(\frac{(u+1)v}{u \chi p}\right)^{2/3}\right)}{8 m^4 \sqrt[3]{u^7} (u+1)^2 v^5} - \frac{\text{Ai}_1\left(\left(\frac{u}{\chi p}\right)^{2/3}\right) (\xi^2 u^3 v + 2 \xi^2 u^2 v + \xi^2 u v) \text{Ai}_1\left(\left(\frac{(u+1)v}{u \chi p}\right)^{2/3}\right)}{8 m^4 u^3 (u+1) v^2} + \\
&\quad \frac{\xi^2 \chi p^{4/3} (u^2 (v-2) + u (2v-5) + 2v-5) \text{Ai}'\left(\left(\frac{u}{\chi p}\right)^{2/3}\right) \text{Ai}_1\left(\left(\frac{(u+1)v}{u \chi p}\right)^{2/3}\right)}{8 m^4 u^2 (u+1)^{2/3} v^{5/3}}
\end{aligned}$$

Step 10: Rewriting the expressions to shape them into final result

We pass to variable $\lambda = s \frac{2 m^2 u \chi_p}{\xi (1+u)}$,
which is the virtuality of the intermediate photon

```

In[197]:= a1expl
targ = u ^ (2 / 3) / χp ^ (2 / 3) (1 + (1 + u) / u ^ 2 * λ / m ^ 2)
Simplify[2 ^ (2 / 3) * %% == targ]
stoλ = Solve[% , s] [[1]]
ξarg = Simplify[2 ^ (2 / 3) a2expl /. stoλ]
jacstoλ = D[s /. stoλ, λ]

```

$$\text{Out[197]} = \frac{\left(\frac{u}{\chi p}\right)^{2/3} \left(\frac{2 s \chi p}{\xi u} + 1\right)}{2^{2/3}}$$

$$\text{Out[198]} = \frac{u^{2/3} \left(\frac{\lambda (u+1)}{m^2 u^2} + 1\right)}{\chi p^{2/3}}$$

$$\text{Out[199]} = 2 m s u \chi p = \frac{\lambda \xi (u+1)}{m}$$

$$\text{Out[200]} = \left\{ s \rightarrow \frac{\lambda \xi + \lambda \xi u}{2 m^2 u \chi p} \right\}$$

$$\text{Out[201]} = \frac{(m^2 v - \lambda) \left(\frac{(u+1)v}{u \chi p}\right)^{2/3}}{m^2 v}$$

$$\text{Out[202]} = \frac{\xi + \xi u}{2 m^2 u \chi p}$$

factor12 is an overall factor, which we transfer from Expr to Coeff

$\phi\lambda$ is the new function resulting from
passing from s to λ in G_s and diving it by factor12

The Airy function arguments are now

$$t = 2^{2/3} a_1$$

$$\xi = 2^{2/3} a_2$$

```
In[203]:= factor12 = (u + 1) ξ^2 / (8 u^2 v m^4);
φλ = Collect[GsIntegrated / factor12 /. stoλ, collectAirya, Simplify] /.
{a1 → t / 2^(2/3), a2 → ξ / 2^(2/3)}
```

$$\begin{aligned} \text{Out[204]} = & -\frac{(u^2 + 2u + 2) \text{Ai}_1(\xi) \text{Ai}'_1(t) \left(\frac{\chi p}{u}\right)^{2/3} (2m^2 v + \lambda(v-2))}{2m^2 (u+1)v} + \frac{(v-2) \text{Ai}_1(t) \text{Ai}'_1(\xi) (2m^2 u^2 + \lambda(u^2 + 2u + 2)) \left(\frac{\chi p}{u v + v}\right)^{2/3}}{2m^2 u^{4/3}} + \\ & \frac{\text{Ai}_1(\xi) \text{Ai}_1(t) (-4m^4 u^2 v - 4\lambda m^2 (u^2 (v-1) + u v + v) + \lambda^2 (-u^2 (v-2) - 2u(v-6) - 2(v-6)))}{4m^4 u^2 v} + \\ & \frac{\chi p^{4/3} \text{Ai}'_1(\xi) \text{Ai}'_1(t) (u^2 (v-2) + u(2v-5) + 2v-5)}{(u+1)^{5/3} v^{2/3}} \end{aligned}$$

```
In[205]:= Diffs12_1st = Diffs11_1st /. {ds → dλ}
Coeff12_1st = Simplify[(Coeff11_1st * jacstoλ * factor12) /. stoλ]
Expr12_1st = (φλ /. {t → targ, ξ → ξarg}) +
(φλ /. {t → targ, ξ → ξarg} /. {λ → -λ}) - 2 (φλ /. {t → targ, ξ → ξarg} /. {λ → 0})
```

Out[205]= dλ du dv

$$\text{Out[206]} = -\frac{4\alpha^2 m^4}{\pi \lambda^2 p_0 (u+1)^2 v \sqrt{(v-4)v} ((\sigma^2 - 1)v - 4\sigma^2)}$$

$$\begin{aligned}
& \text{Out[207]=} \frac{((v-2)u^2 + (2v-5)u + 2v-5) \text{Ai}'\left(\frac{(vm^2+\lambda)\left(\frac{(u+1)v}{u\chi p}\right)^{2/3}}{m^2 v}\right) \text{Ai}'\left(\frac{u^{2/3}\left(1-\frac{(u+1)\lambda}{m^2 u^2}\right)}{\chi p^{2/3}}\right) \chi p^{4/3}}{(u+1)^{5/3} v^{2/3}} + \\
& \frac{((v-2)u^2 + (2v-5)u + 2v-5) \text{Ai}'\left(\frac{(m^2 v-\lambda)\left(\frac{(u+1)v}{u\chi p}\right)^{2/3}}{m^2 v}\right) \text{Ai}'\left(\frac{u^{2/3}\left(\frac{(u+1)\lambda}{m^2 u^2}+1\right)}{\chi p^{2/3}}\right) \chi p^{4/3}}{(u+1)^{5/3} v^{2/3}} - \\
& \frac{(u^2 + 2u + 2)(2vm^2 + (v-2)\lambda)\left(\frac{\chi p}{u}\right)^{2/3} \text{Ai}'\left(\frac{u^{2/3}\left(\frac{(u+1)\lambda}{m^2 u^2}+1\right)}{\chi p^{2/3}}\right) \text{Ai}_1\left(\frac{(m^2 v-\lambda)\left(\frac{(u+1)v}{u\chi p}\right)^{2/3}}{m^2 v}\right)}{2m^2(u+1)v} - \\
& \frac{(u^2 + 2u + 2)(2m^2 v - (v-2)\lambda)\left(\frac{\chi p}{u}\right)^{2/3} \text{Ai}'\left(\frac{u^{2/3}\left(1-\frac{(u+1)\lambda}{m^2 u^2}\right)}{\chi p^{2/3}}\right) \text{Ai}_1\left(\frac{(vm^2+\lambda)\left(\frac{(u+1)v}{u\chi p}\right)^{2/3}}{m^2 v}\right)}{2m^2(u+1)v} - \\
& 2 \left(\frac{\left((v-2)u^2 + (2v-5)u + 2v-5\right) \text{Ai}'\left(\left(\frac{(u+1)v}{u\chi p}\right)^{2/3}\right) \text{Ai}'\left(\frac{u^{2/3}}{\chi p^{2/3}}\right) \chi p^{4/3}}{(u+1)^{5/3} v^{2/3}} - \frac{(u^2 + 2u + 2)\left(\frac{\chi p}{u}\right)^{2/3} \text{Ai}'\left(\frac{u^{2/3}}{\chi p^{2/3}}\right) \text{Ai}_1\left(\left(\frac{(u+1)v}{u\chi p}\right)^{2/3}\right)}{u+1} + \right. \\
& \left. u^{2/3}(v-2)\left(\frac{\chi p}{uv+v}\right)^{2/3} \text{Ai}'\left(\left(\frac{(u+1)v}{u\chi p}\right)^{2/3}\right) \text{Ai}_1\left(\frac{u^{2/3}}{\chi p^{2/3}}\right) - \text{Ai}_1\left(\left(\frac{(u+1)v}{u\chi p}\right)^{2/3}\right) \text{Ai}_1\left(\frac{u^{2/3}}{\chi p^{2/3}}\right) \right) + \\
& \frac{(v-2)(2m^2 u^2 - (u^2 + 2u + 2)\lambda)\left(\frac{\chi p}{uv+v}\right)^{2/3} \text{Ai}'\left(\frac{(vm^2+\lambda)\left(\frac{(u+1)v}{u\chi p}\right)^{2/3}}{m^2 v}\right) \text{Ai}_1\left(\frac{u^{2/3}\left(1-\frac{(u+1)\lambda}{m^2 u^2}\right)}{\chi p^{2/3}}\right)}{2m^2 u^{4/3}} + \frac{1}{4m^4 u^2 v} \\
& (-4u^2 v m^4 + 4((v-1)u^2 + vu + v)\lambda m^2 + (-(v-2)u^2) - 2(v-6)u - 2(v-6)\lambda^2) \text{Ai}_1\left(\frac{(vm^2 + \lambda)\left(\frac{(u+1)v}{u\chi p}\right)^{2/3}}{m^2 v}\right) \\
& \text{Ai}_1\left(\frac{u^{2/3}\left(1-\frac{(u+1)\lambda}{m^2 u^2}\right)}{\chi p^{2/3}}\right) + \frac{(v-2)(2m^2 u^2 + (u^2 + 2u + 2)\lambda)\left(\frac{\chi p}{uv+v}\right)^{2/3} \text{Ai}'\left(\frac{(m^2 v-\lambda)\left(\frac{(u+1)v}{u\chi p}\right)^{2/3}}{m^2 v}\right) \text{Ai}_1\left(\frac{u^{2/3}\left(\frac{(u+1)\lambda}{m^2 u^2}+1\right)}{\chi p^{2/3}}\right)}{2m^2 u^{4/3}} + \\
& \frac{1}{4m^4 u^2 v} (-4u^2 v m^4 - 4((v-1)u^2 + vu + v)\lambda m^2 + (-(v-2)u^2) - 2(v-6)u - 2(v-6)\lambda^2) \\
& \text{Ai}_1\left(\frac{(m^2 v - \lambda)\left(\frac{(u+1)v}{u\chi p}\right)^{2/3}}{m^2 v}\right) \text{Ai}_1\left(\frac{u^{2/3}\left(\frac{(u+1)\lambda}{m^2 u^2}+1\right)}{\chi p^{2/3}}\right)
\end{aligned}$$

```
In[208]:= Diff12st = Diff11st
          Coeff12st = Factor[Simplify[(Coeff11st * factor12) /. stoλ]]
          Expr12st = (φλ /. {t → targ, ξ → ξarg} /. {λ → 0})
```

```
Out[208]= du dv
```

$$\text{Out[209]} = -\frac{2 \alpha^2 L \phi m^2 \xi}{p_0 u (u+1) v \sqrt{(v-4)v} \chi p (-4 \sigma^2 + \sigma^2 v - v)}$$

$$\begin{aligned} \text{Out[210]} = & -\text{Ai}_1\left(\frac{u^{2/3}}{\chi p^{2/3}}\right) \text{Ai}_1\left(\left(\frac{(u+1)v}{u \chi p}\right)^{2/3}\right) + u^{2/3} (v-2) \text{Ai}_1\left(\frac{u^{2/3}}{\chi p^{2/3}}\right) \left(\frac{\chi p}{u v + v}\right)^{2/3} \text{Ai}'\left(\left(\frac{(u+1)v}{u \chi p}\right)^{2/3}\right) - \\ & \frac{(u^2 + 2 u + 2) \left(\frac{\chi p}{u}\right)^{2/3} \text{Ai}'\left(\frac{u^{2/3}}{\chi p^{2/3}}\right) \text{Ai}_1\left(\left(\frac{(u+1)v}{u \chi p}\right)^{2/3}\right)}{u+1} + \frac{\chi p^{4/3} (u^2 (v-2) + u (2 v - 5) + 2 v - 5) \text{Ai}'\left(\frac{u^{2/3}}{\chi p^{2/3}}\right) \text{Ai}'\left(\left(\frac{(u+1)v}{u \chi p}\right)^{2/3}\right)}{(u+1)^{5/3} v^{2/3}} \end{aligned}$$

The last two cells are our final result,
which can be found in [arXiv 2605.12479]

For the one – step contribution

($\phi_{\text{tree}}(\lambda)$ is equivalent of $\phi\lambda$ used in this notebook), and $\chi = \chi p$

$$W_{\text{dir}}^{1\text{st}} = \frac{\alpha^2 m^4}{\pi p^0} \int_0^\infty du \int_4^\infty dv \int_0^\infty \frac{d\lambda}{\lambda^2} \frac{\phi_{\text{tree}}(\lambda) + \phi_{\text{tree}}(-\lambda) - 2\phi_{\text{tree}}(0)}{(1+u)^2 v^{3/2} \sqrt{v-4}}, \quad (6.54)$$

with

$$\begin{aligned} \phi_{\text{tree}}(\lambda) = & \chi^{4/3} \frac{u^2(v-2) + (1+u)(2v-5)}{(1+u)^{5/3} v^{2/3}} \text{Ai}'(t) \text{Ai}'(\zeta) \\ & + \left[\frac{\chi u}{(1+u)v} \right]^{2/3} \left(1 + \frac{\lambda}{m^2} \frac{u^2 + 2u + 2}{2u^2} \right) (v-2) \text{Ai}_1(t) \text{Ai}'(\zeta) \\ & - \left(\frac{\chi}{u} \right)^{2/3} \frac{u^2 + 2u + 2}{1+u} \left(1 + \frac{\lambda}{m^2} \frac{v-2}{2v} \right) \text{Ai}'(t) \text{Ai}_1(\zeta) \\ & - \left[1 + \frac{\lambda}{m^2} \left(\frac{u^2 + u + 1}{u^2} - \frac{1}{v} \right) \right. \\ & \quad \left. + \frac{\lambda^2}{m^4} \frac{u^2(v-2) + 2(1+u)(v-6)}{4u^2 v} \right] \text{Ai}_1(t) \text{Ai}_1(\zeta). \end{aligned} \quad (6.55)$$

$$W_{\text{dir}}^{1\text{st}} = \frac{\alpha^2 m^4}{\pi p^0} \int_0^\infty du \int_4^\infty dv \int_0^\infty \frac{d\lambda}{\lambda^2} \frac{\phi_{\text{tree}}(\lambda) + \phi_{\text{tree}}(-\lambda) - 2\phi_{\text{tree}}(0)}{(1+u)^2 v^{3/2} \sqrt{v-4}}, \quad (6.54)$$

with

$$\begin{aligned} \phi_{\text{tree}}(\lambda) = & \chi^{4/3} \frac{u^2(v-2) + (1+u)(2v-5)}{(1+u)^{5/3} v^{2/3}} \text{Ai}'(t) \text{Ai}'(\zeta) \\ & + \left[\frac{\chi u}{(1+u)v} \right]^{2/3} \left(1 + \frac{\lambda}{m^2} \frac{u^2 + 2u + 2}{2u^2} \right) (v-2) \text{Ai}_1(t) \text{Ai}'(\zeta) \\ & - \left(\frac{\chi}{u} \right)^{2/3} \frac{u^2 + 2u + 2}{1+u} \left(1 + \frac{\lambda}{m^2} \frac{v-2}{2v} \right) \text{Ai}'(t) \text{Ai}_1(\zeta) \\ & - \left[1 + \frac{\lambda}{m^2} \left(\frac{u^2 + u + 1}{u^2} - \frac{1}{v} \right) \right. \\ & \quad \left. + \frac{\lambda^2}{m^4} \frac{u^2(v-2) + 2(1+u)(v-6)}{4u^2 v} \right] \text{Ai}_1(t) \text{Ai}_1(\zeta). \end{aligned} \quad (6.55)$$

The two –

step result obtained here is equivalent to the following expression

$$W_{\text{dir},2\text{-loop}}^{2\text{st}}(\chi) = \frac{\alpha^2 m^2 \theta(\chi)}{2p^0 \chi} \xi L_\varphi \int_0^\infty du \int_4^\infty dv \frac{\phi(0)}{u(1+u)v^{3/2} \sqrt{v-4}}, \quad (6.53)$$

$$W_{\text{dir},2\text{-loop}}^{2\text{st}}(\chi) = \frac{\alpha^2 m^2 \theta(\chi)}{2p^0 \chi} \xi L_\varphi \int_0^\infty du \int_4^\infty dv \frac{\phi(0)}{u(1+u)v^{3/2} \sqrt{v-4}}, \quad (6.53)$$